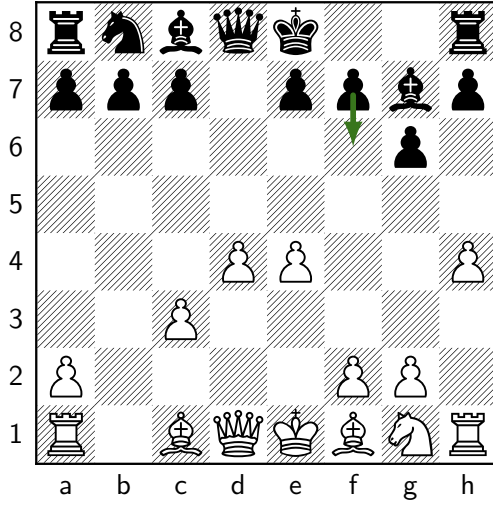


Melanie-ANALOGY — Grunfeld

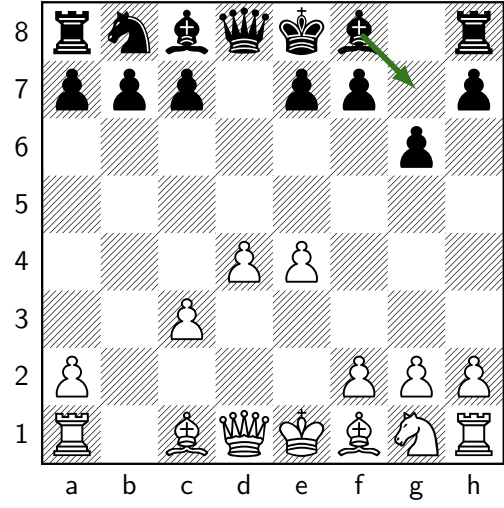
Melanie (Deep_Vision)

Abstract

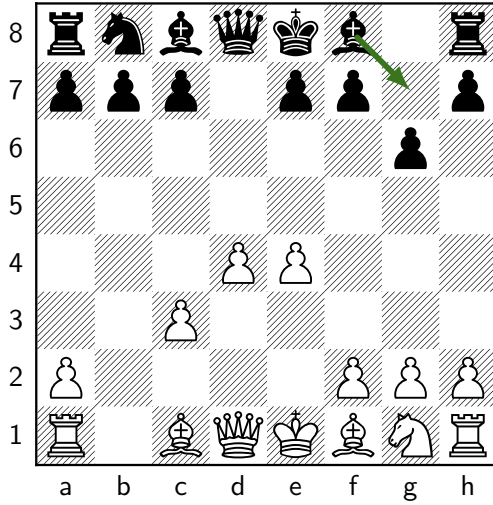
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



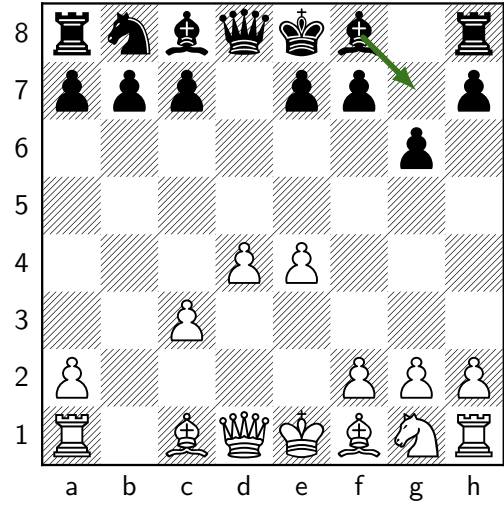
(a)



(b)

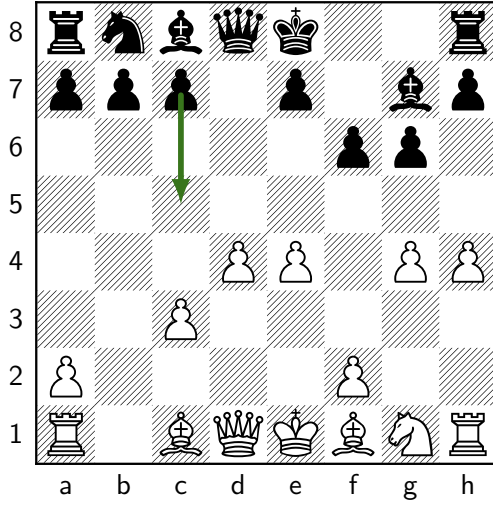


(c)

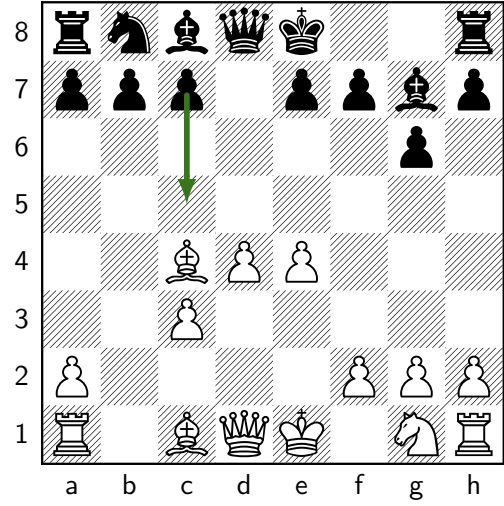


(d)

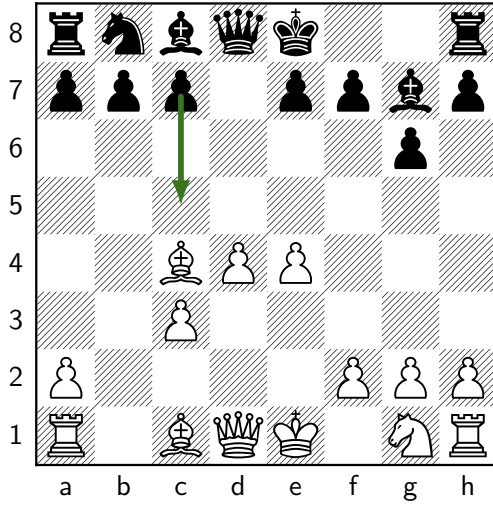
Figure 1: **Move 7, Black: f6**. (a) Our position; Melanie plays f6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#296 (ply 11) [structural distance $d=21$, QAP obj=61]: the master played Bg7; under μ our Pf2 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Ra8, our Rh8 as their Pc7; transferred move f7f6 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#297 (ply 11) [structural distance $d=21$, QAP obj=61]: the master played Bg7; under μ our Pf2 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Ra8, our Rh8 as their Pc7; transferred move f7f6 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#366 (ply 11) [structural distance $d=21$, QAP obj=61]: the master played Bg7; under μ our Pf2 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Ra8, our Rh8 as their Pc7; transferred move f7f6 (**endorses this move**). Synthesis: 40 retrieved analogs transferred to f6 (summed QAP objective 2421, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



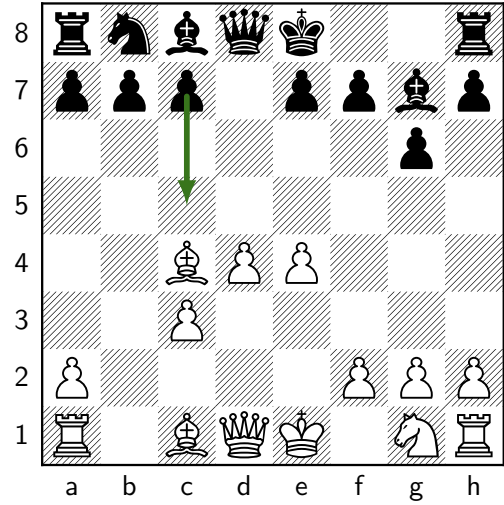
(a)



(b)

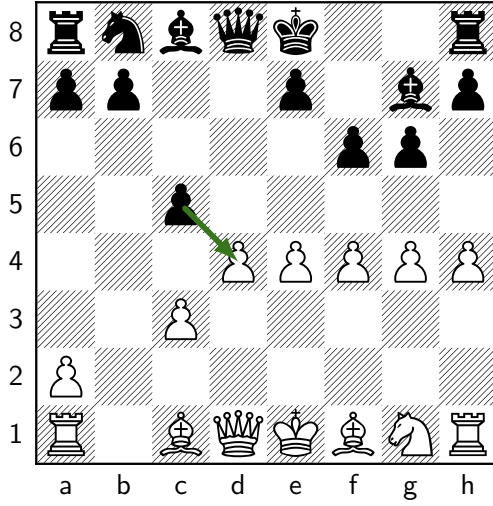


(c)

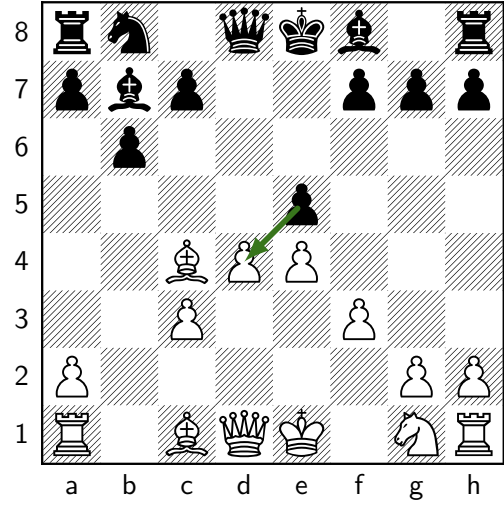


(d)

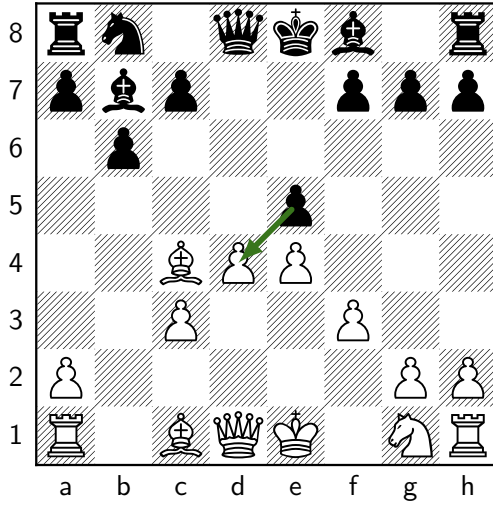
Figure 2: **Move 8, Black: c5.** (a) Our position; Melanie plays c5 (c7c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#366` (ply 13) [structural distance $d=37$, QAP obj=78]: the master played c5; under μ our $Pc2$ plays the role of that game’s $Pc2$; cross-type roles: our $Pe7$ as their $Nb8$, our $Pf6$ as their $Bc8$; transferred move c7c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#367` (ply 13) [structural distance $d=37$, QAP obj=78]: the master played c5; under μ our $Pc2$ plays the role of that game’s $Pc2$; cross-type roles: our $Pe7$ as their $Nb8$, our $Pf6$ as their $Bc8$; transferred move c7c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#630` (ply 13) [structural distance $d=37$, QAP obj=78]: the master played c5; under μ our $Pc2$ plays the role of that game’s $Pc2$; cross-type roles: our $Pe7$ as their $Nb8$, our $Pf6$ as their $Bc8$; transferred move c7c5 (**endorses this move**). Synthesis: 34 retrieved analogs transferred to c5 (summed QAP objective 2576, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



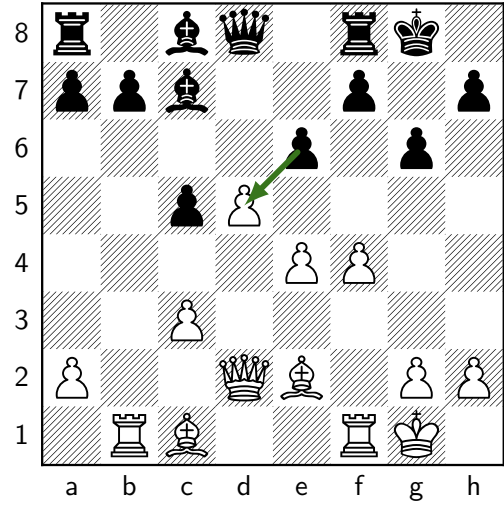
(a)



(b)

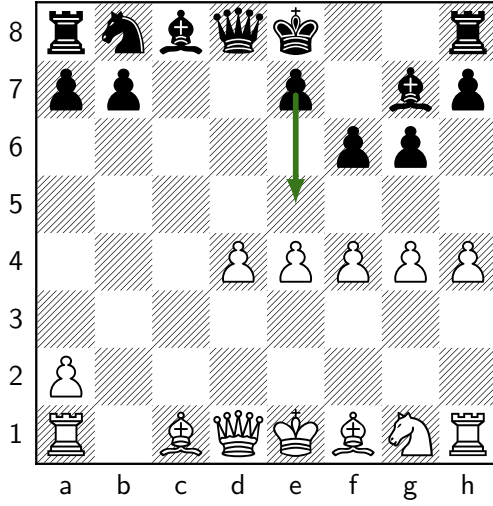


(c)

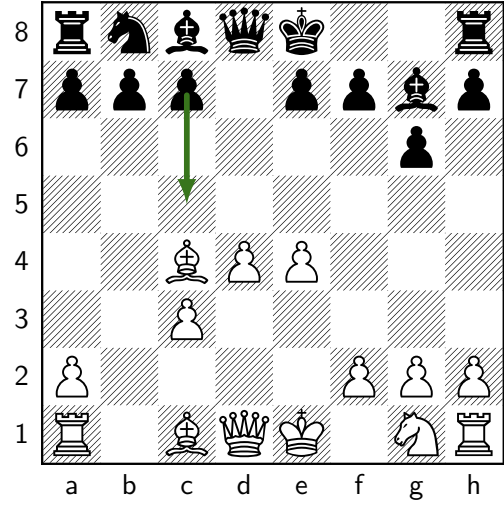


(d)

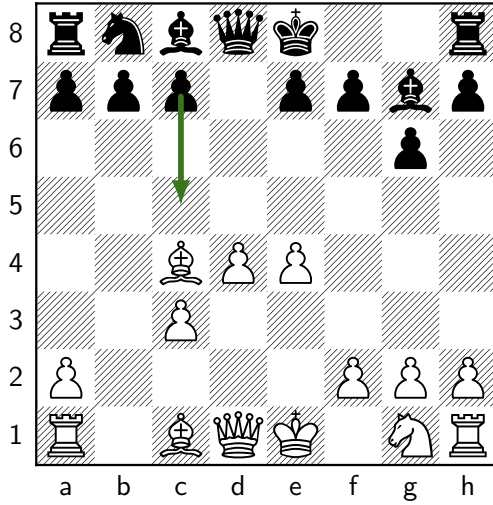
Figure 3: **Move 9, Black: cxd4**. (a) Our position; Melanie plays cxd4 (c5d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2127` (ply 15) [structural distance $d=64$, QAP obj=119]: the master played **exd4**; under μ our $Pc4$ plays the role of that game’s $Pe4$; cross-type roles: our $Pb7$ as their $Ra8$, our $Ph7$ as their $Nb8$; transferred move **c5d4 (endorses this move)**. (c) Retrieved from `ccrl_top_p0000#2128` (ply 15) [structural distance $d=64$, QAP obj=119]: the master played **exd4**; under μ our $Pc4$ plays the role of that game’s $Pe4$; cross-type roles: our $Pb7$ as their $Ra8$, our $Ph7$ as their $Nb8$; transferred move **c5d4 (endorses this move)**. (d) Retrieved from `ccrl_top_p0000#2671` (ply 27) [structural distance $d=71$, QAP obj=101]: the master played **exd5**; under μ our $Pc4$ plays the role of that game’s $Pe3$; cross-type roles: our $Nb8$ as their $Ra8$, our $Pe7$ as their $Bc8$; transferred move **c5d4 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to **cxd4** (summed QAP objective 440, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



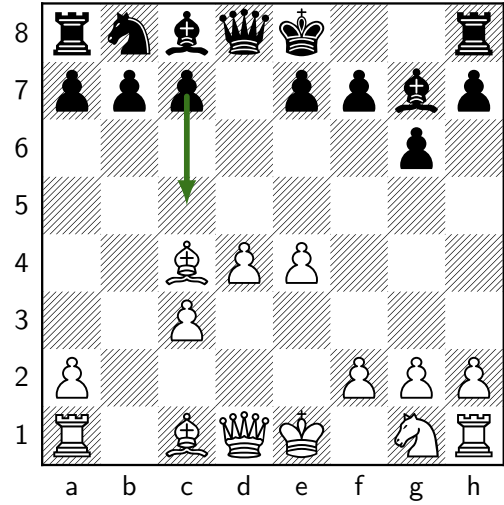
(a)



(b)

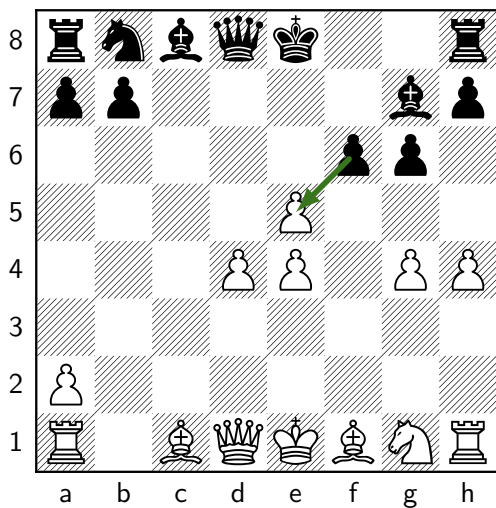


(c)

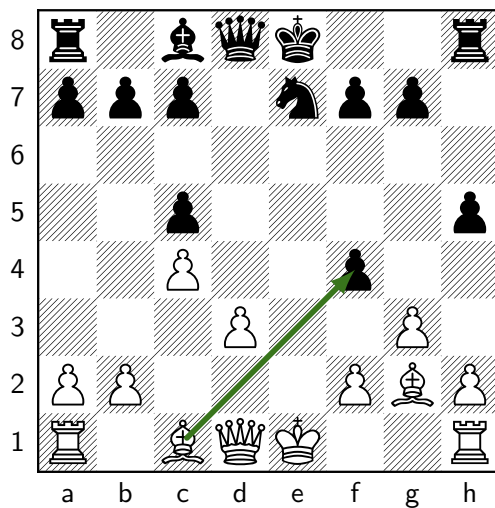


(d)

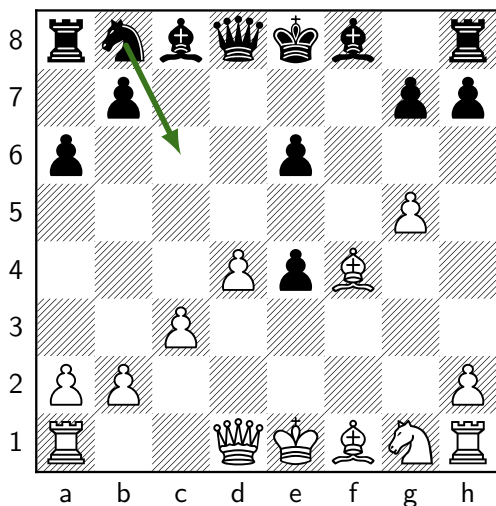
Figure 4: **Move 10, Black: e5 (e7e5).** (a) Our position; Melanie plays e5 (e7e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#366` (ply 13) [structural distance $d=55$, QAP obj=77]: the master played c5; under μ our $Pe2$ plays the role of that game’s $Pc2$; cross-type roles: our $ng1$ as their $Ra8$, our $pf4$ as their $Nb8$; transferred move e7e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#367` (ply 13) [structural distance $d=55$, QAP obj=77]: the master played c5; under μ our $Pe2$ plays the role of that game’s $Pc2$; cross-type roles: our $ng1$ as their $Ra8$, our $pf4$ as their $Nb8$; transferred move e7e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#630` (ply 13) [structural distance $d=55$, QAP obj=77]: the master played c5; under μ our $Pe2$ plays the role of that game’s $Pc2$; cross-type roles: our $ng1$ as their $Ra8$, our $pf4$ as their $Nb8$; transferred move e7e5 (**endorses this move**). Synthesis: 28 retrieved analogs transferred to e5 (summed QAP objective 2148, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



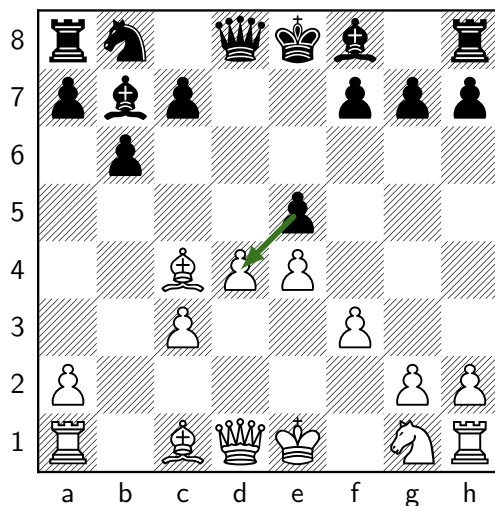
(a)



(b)

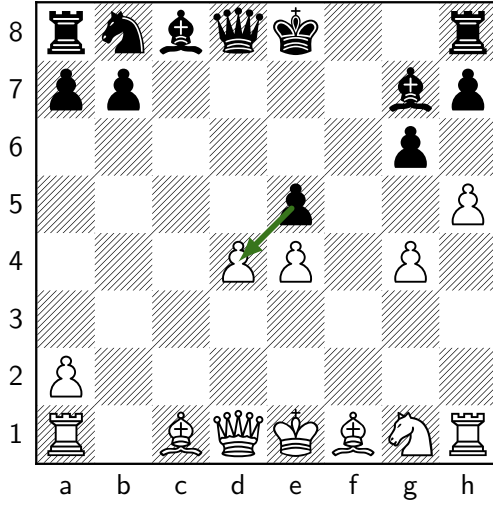


(c)

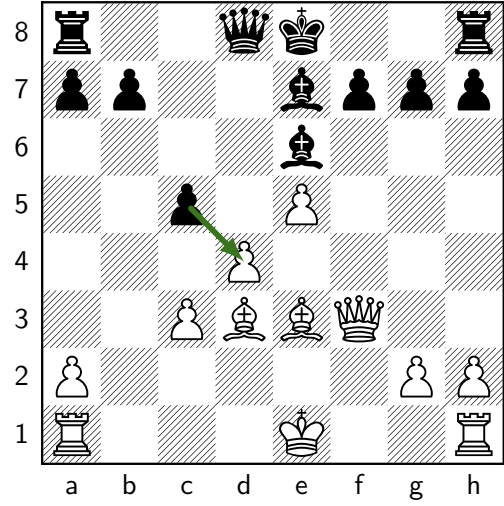


(d)

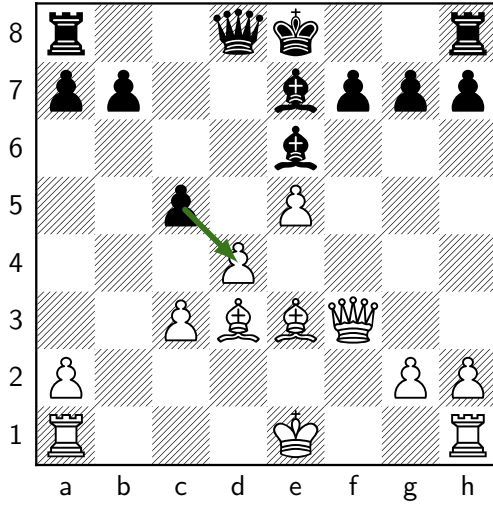
Figure 5: **Move 11, Black: fxe5**. (a) Our position; Melanie plays fxe5 (f6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1519` (ply 20) [structural distance $d=67$, QAP obj=64]: the master played Bxf4; under μ our Pf3 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Qd1, our bfl as their Rh1; transferred move f6e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3211` (ply 21) [structural distance $d=72$, QAP obj=58]: the master played Nc6; under μ our Pf3 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Bg7 as their Ra8, our Pg6 as their Bf8; transferred move f6e5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2127` (ply 15) [structural distance $d=66$, QAP obj=52]: the master played exd4; under μ our Qd1 plays the role of that game’s Pe4 (a **cross-type** correspondence); cross-type roles: our Pb7 as their Ra8, our ph4 as their Bf8; transferred move d8d6. Synthesis: 2 retrieved analogs transferred to fxe5 (summed QAP objective 121, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



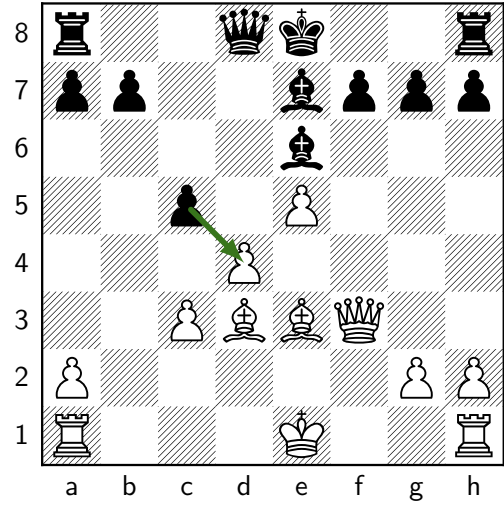
(a)



(b)

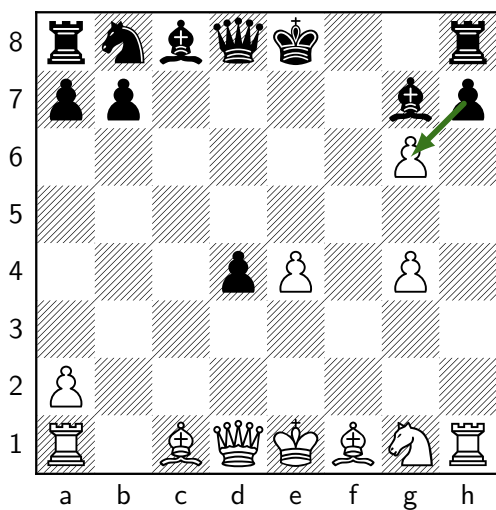


(c)

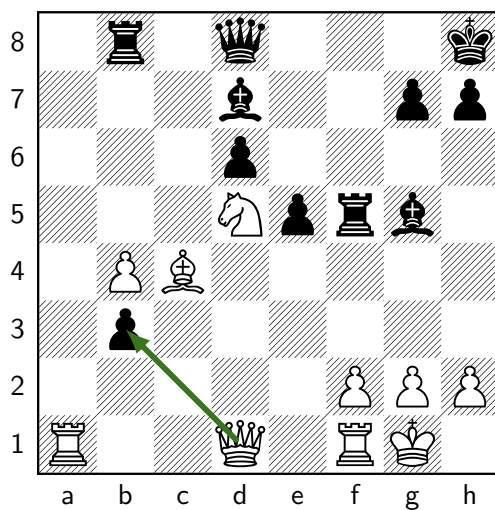


(d)

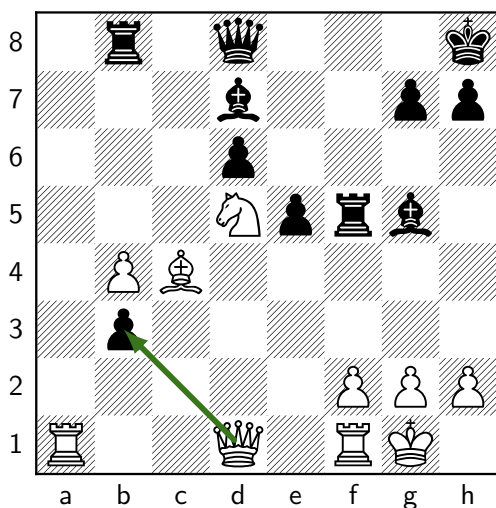
Figure 6: **Move 12, Black: exd4.** (a) Our position; Melanie plays **exd4** (**e5d4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1342` (ply 25) [structural distance $d=90$, QAP obj=128]: the master played **cx d4**; under μ our Pe_4 plays the role of that game’s Pc_4 ; cross-type roles: our ph_5 as their Ra_8 , our Nb_8 as their Ph_7 ; transferred move **e5d4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1343` (ply 25) [structural distance $d=90$, QAP obj=128]: the master played **cx d4**; under μ our Pe_4 plays the role of that game’s Pc_4 ; cross-type roles: our ph_5 as their Ra_8 , our Nb_8 as their Ph_7 ; transferred move **e5d4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1780` (ply 25) [structural distance $d=90$, QAP obj=128]: the master played **cx d4**; under μ our Pe_4 plays the role of that game’s Pc_4 ; cross-type roles: our ph_5 as their Ra_8 , our Nb_8 as their Ph_7 ; transferred move **e5d4** (**endorses this move**). Synthesis: 13 retrieved analogs transferred to **exd4** (summed QAP objective 1365, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



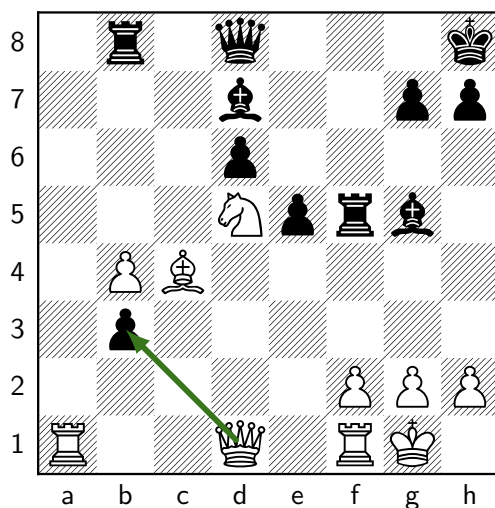
(a)



(b)

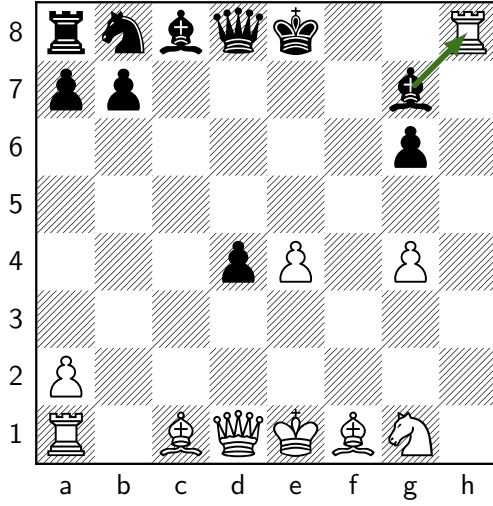


(c)

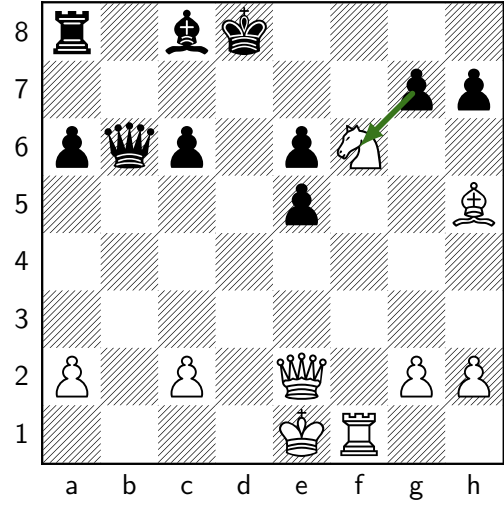


(d)

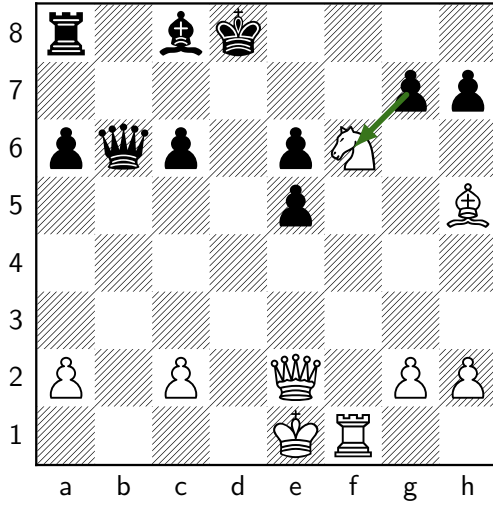
Figure 7: **Move 13, Black: hxg6**. (a) Our position; Melanie plays **hxg6** (**h7g6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#92` (ply 40) [structural distance $d=103$, QAP obj=63]: the master played **Qxb3**; under μ our *Ph2* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Qd8* as their *Rf1*, our *Ra8* as their *Ph2*; transferred move **h7g6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2525` (ply 40) [structural distance $d=103$, QAP obj=63]: the master played **Qxb3**; under μ our *Ph2* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Qd8* as their *Rf1*, our *Ra8* as their *Ph2*; transferred move **h7g6** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2725` (ply 40) [structural distance $d=103$, QAP obj=63]: the master played **Qxb3**; under μ our *Ph2* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Qd8* as their *Rf1*, our *Ra8* as their *Ph2*; transferred move **h7g6** (**endorses this move**). Synthesis: 9 retrieved analogs transferred to **hxg6** (summed QAP objective 409, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



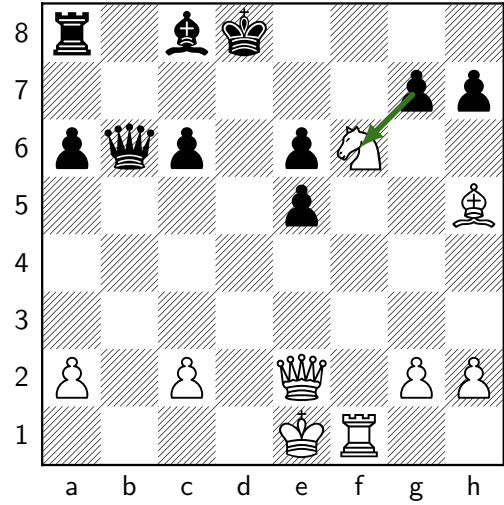
(a)



(b)

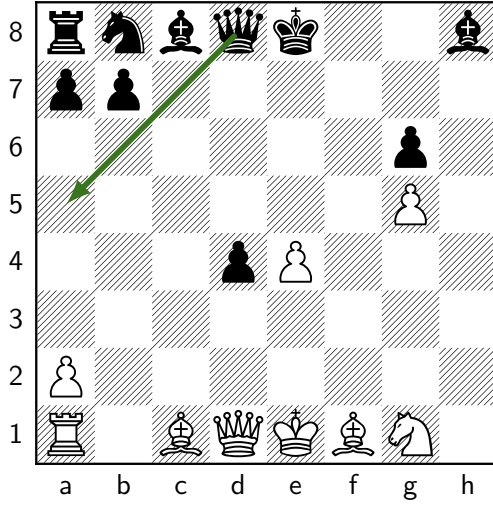


(c)

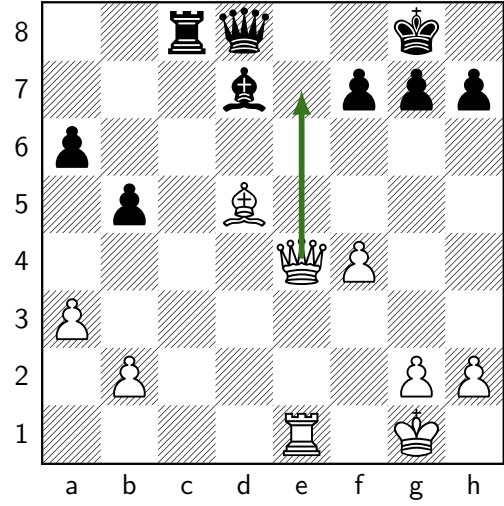


(d)

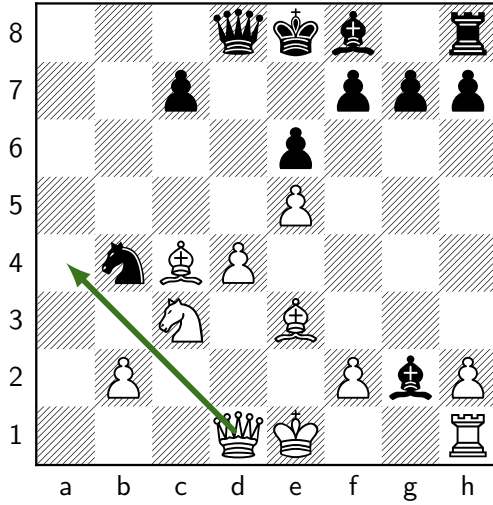
Figure 8: **Move 14, Black: Bxh8**. (a) Our position; Melanie plays Bxh8 (g7h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2913` (ply 43) [structural distance $d=90$, QAP obj=8]: the master played gxf6; under μ our Bg2 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our pa2 as their Ra8, our Ra8 as their Ph7; transferred move g7h8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1024` (ply 43) [structural distance $d=90$, QAP obj=8]: the master played gxf6; under μ our Bg2 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our pa2 as their Ra8, our Ra8 as their Ph7; transferred move g7h8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1` (ply 43) [structural distance $d=90$, QAP obj=8]: the master played gxf6; under μ our Bg2 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our pa2 as their Ra8, our Ra8 as their Ph7; transferred move g7h8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bxh8 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



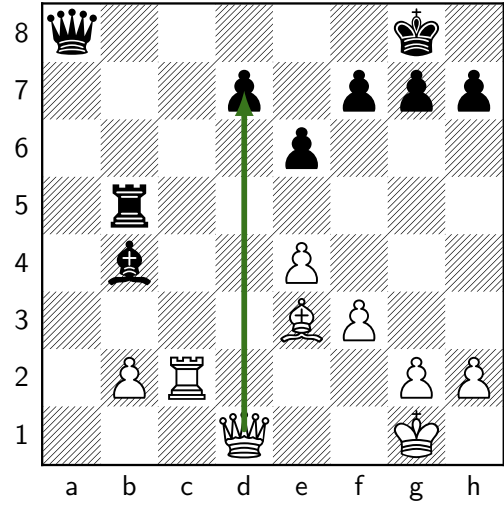
(a)



(b)

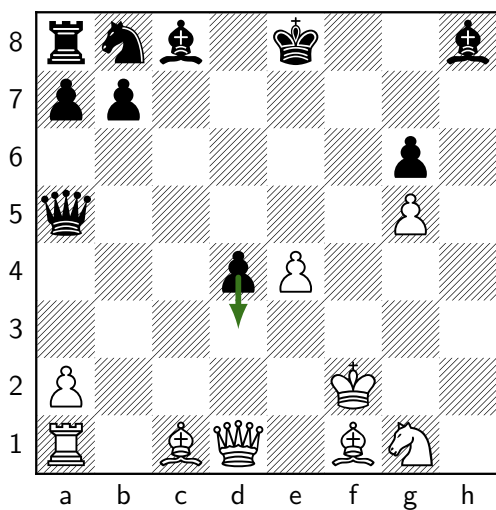


(c)

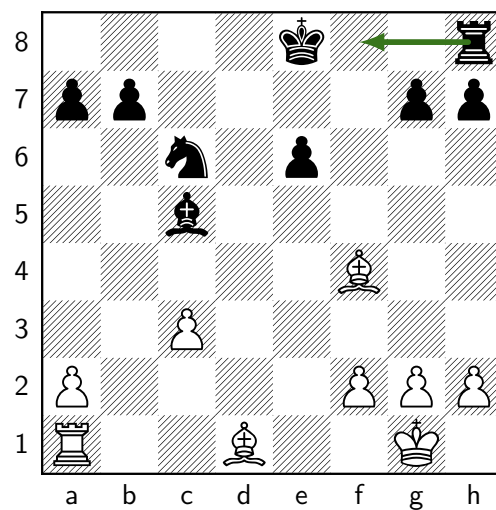


(d)

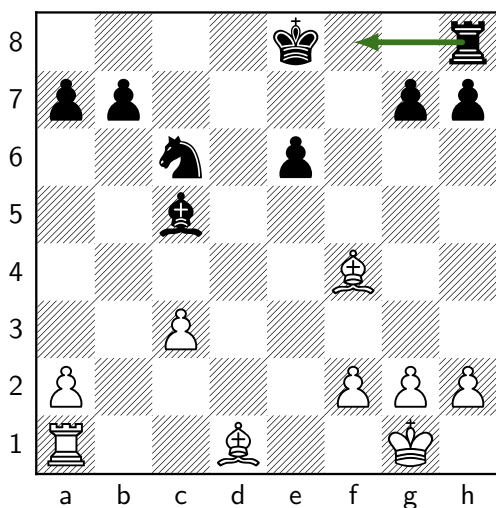
Figure 9: **Move 15, Black: Qa5+.** (a) Our position; Melanie plays Qa5+ (d8a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0001#3624` (ply 44) [structural distance $d=80$, QAP obj=38]: the master played Qe7; under μ our Qd1 plays the role of that game’s Qe4; cross-type roles: our Nb8 as their Re1, our ng1 as their Ph2; transferred move d8a5 (**endorses this move**). (c) Retrieved from `ccl1_top_p0001#805` (ply 28) [structural distance $d=89$, QAP obj=32]: the master played Qa4+; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pg6 as their Rh1, our ra1 as their Pb2; transferred move d8a5 (**endorses this move**). (d) Retrieved from `ccl1_top_p0000#5` (ply 42) [structural distance $d=83$, QAP obj=28]: the master played Qxd7; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Bc8 as their Rc2, our Nb8 as their Pg2; transferred move d8a5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Qa5+ (summed QAP objective 112, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



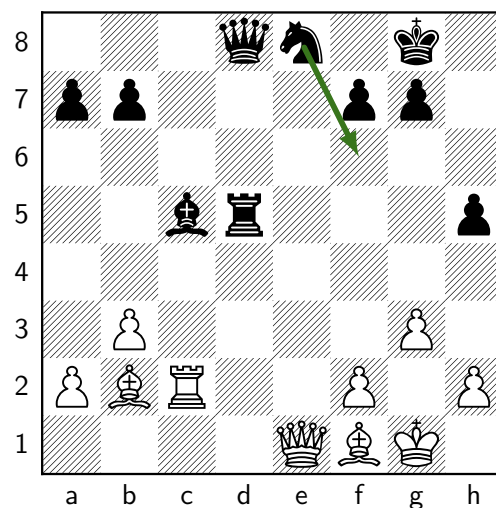
(a)



(b)

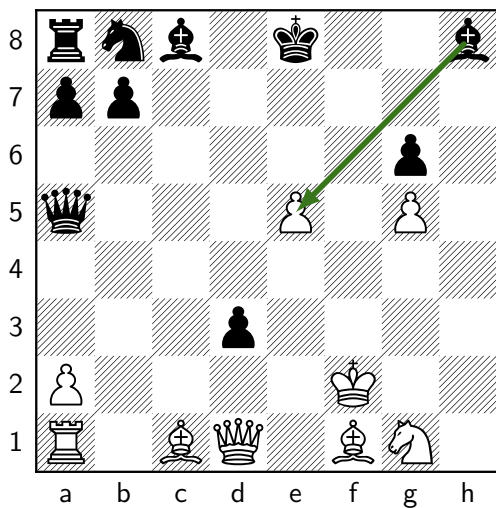


(c)

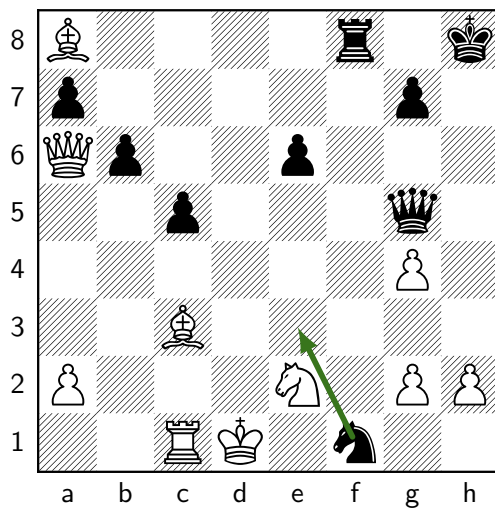


(d)

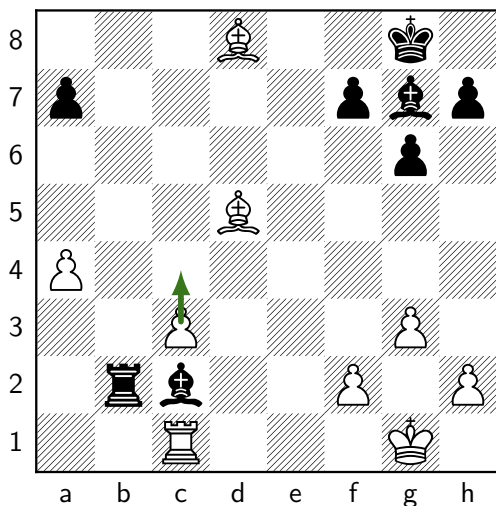
Figure 10: **Move 16, Black: d3**. (a) Our position; Melanie plays d3 (d4d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1064` (ply 31) [structural distance $d=101$, QAP obj=42]: the master played Rf8; under μ our Pd5 plays the role of that game’s Rh1 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pg7, our Bc8 as their Ph7; transferred move d4d3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1065` (ply 31) [structural distance $d=101$, QAP obj=42]: the master played Rf8; under μ our Pd5 plays the role of that game’s Rh1 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pg7, our Bc8 as their Ph7; transferred move d4d3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1729` (ply 45) [structural distance $d=93$, QAP obj=38]: the master played Nf6; under μ our Pd5 plays the role of that game’s Ne1 (a **cross-type** correspondence); cross-type roles: our ra1 as their Qd8, our Nb8 as their Pb7; transferred move d4d3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to d3 (summed QAP objective 122, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



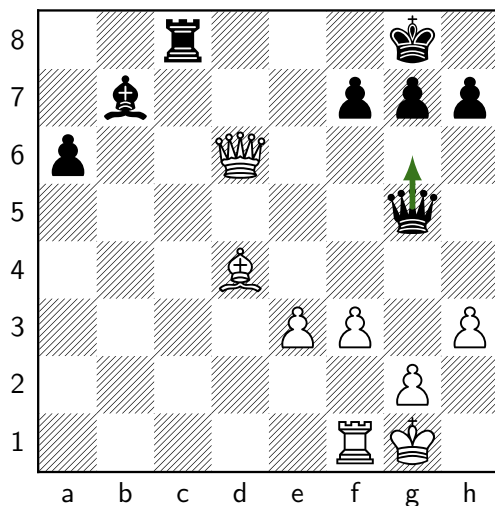
(a)



(b)

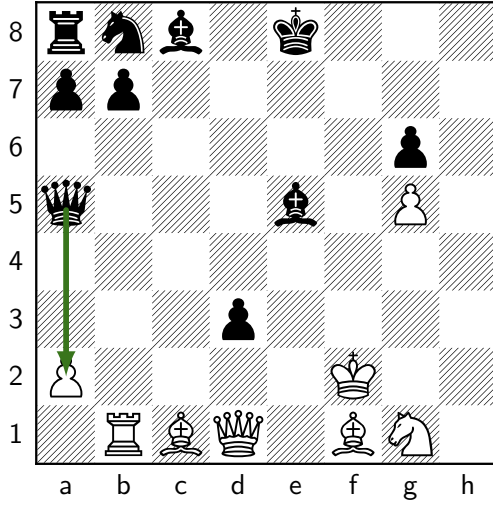


(c)

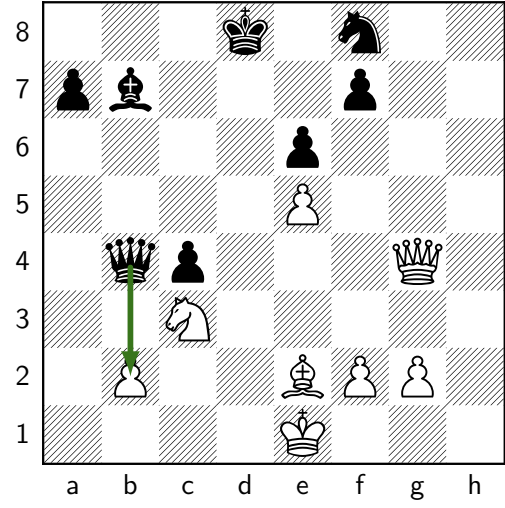


(d)

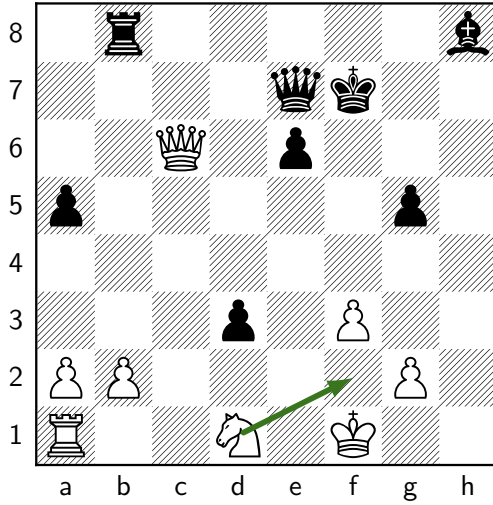
Figure 11: **Move 17, Black: Bxe5.** (a) Our position; Melanie plays **Bxe5** (**h8e5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2451` (ply 49) [structural distance $d=90$, QAP obj=38]: the master played **Ne3+**; under μ our *Bh1* plays the role of that game’s *Nf8* (a **cross-type** correspondence); cross-type roles: our *bc1* as their *Rf8*, our *bf1* as their *qa6*; transferred move **h8e5** (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3059` (ply 44) [structural distance $d=99$, QAP obj=16]: the master played **c4**; under μ our *Bh1* plays the role of that game’s *Pc3* (a **cross-type** correspondence); cross-type roles: our *Nb8* as their *Rc1*, our *qd1* as their *rb2*; transferred move **h8e5** (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2023` (ply 49) [structural distance $d=103$, QAP obj=78]: the master played **Qg6**; under μ our *Qa4* plays the role of that game’s *Qg4*; cross-type roles: our *Pa7* as their *Rc8*, our *Ra8* as their *Ph7*; transferred move **a5b6**. Synthesis: 2 retrieved analogs transferred to **Bxe5** (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



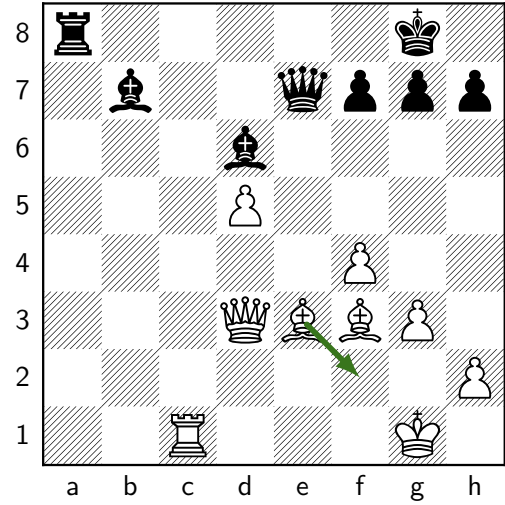
(a)



(b)

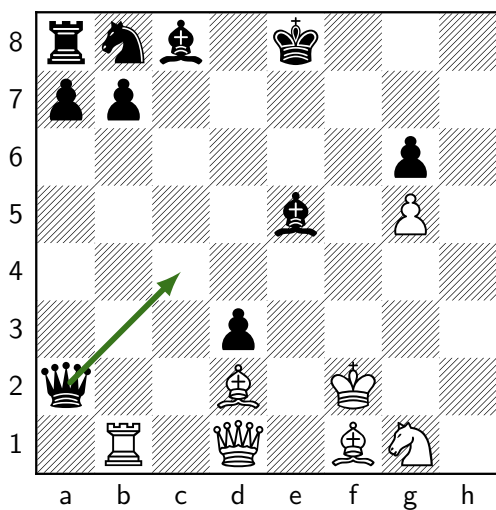


(c)

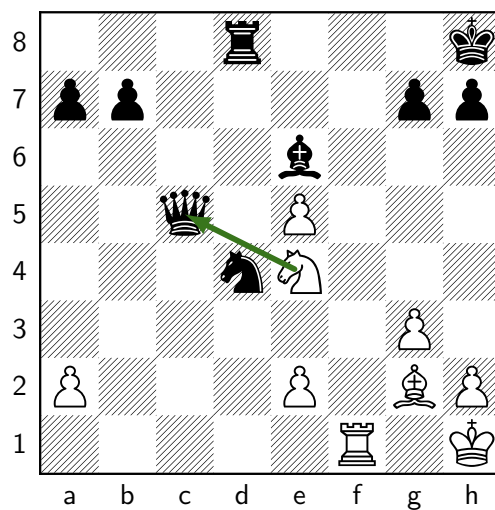


(d)

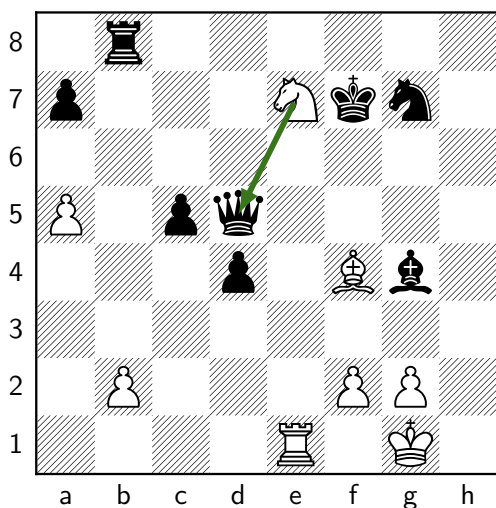
Figure 12: **Move 18, Black: Qxa2+.** (a) Our position; Melanie plays Qxa2+ (a5a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1682` (ply 51) [structural distance $d=83$, QAP obj=10]: the master played Qxb2; under μ our Qa4 plays the role of that game’s Qb5; cross-type roles: our Be5 as their Nf8, our Ra8 as their Pa7; transferred move a5a2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2039` (ply 50) [structural distance $d=91$, QAP obj=49]: the master played Nf2; under μ our Bc1 plays the role of that game’s Nd1 (a **cross-type** correspondence); cross-type roles: our qd1 as their pd3, our bc1 as their Pf3; transferred move c8d7. (d) Retrieved from `ccr1_top_p0000#1691` (ply 50) [structural distance $d=97$, QAP obj=41]: the master played Bf2; under μ our Pd6 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our Be5 as their Rc1, our rb1 as their bb7; transferred move d3d2. Synthesis: 1 retrieved analog transferred to Qxa2+ (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



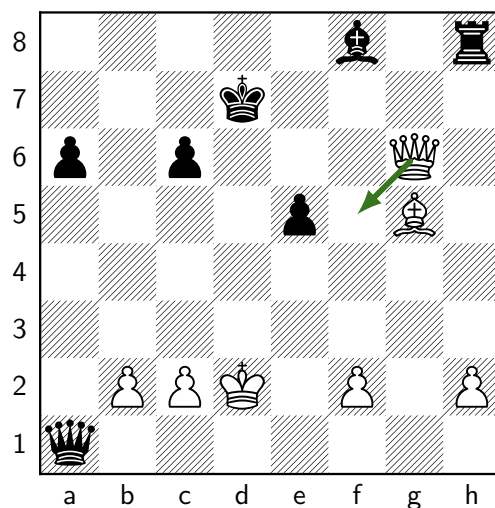
(a)



(b)

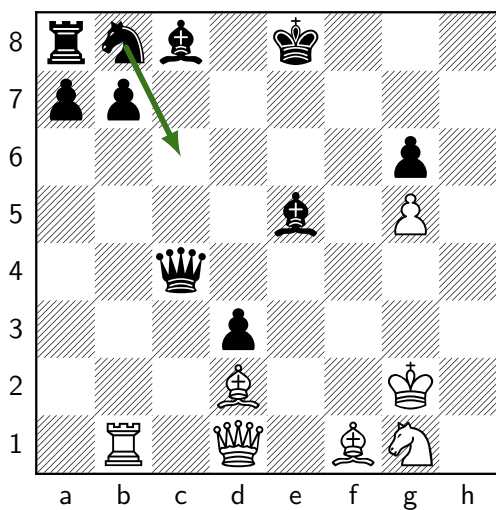


(c)

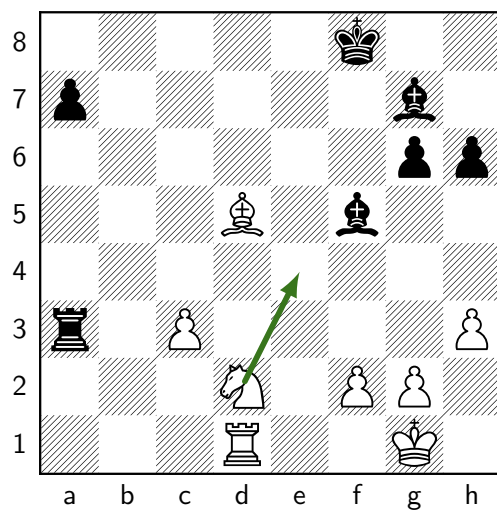


(d)

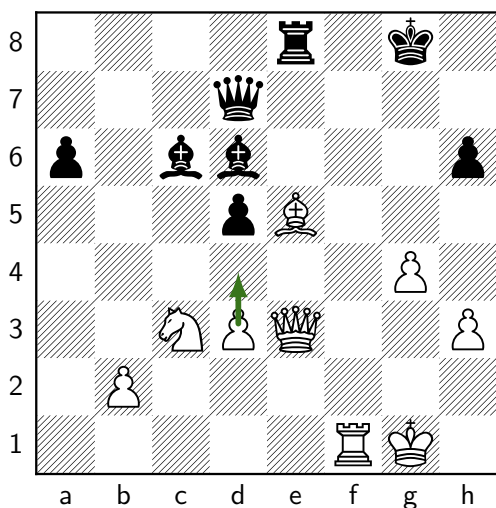
Figure 13: **Move 19, Black: Qc4.** (a) Our position; Melanie plays Qc4 (a2c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1815` (ply 52) [structural distance $d=104$, QAP obj=19]: the master played Nxc5; under μ our Qa7 plays the role of that game’s Ne4 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Rf1, our ng1 as their Bg2; transferred move a2c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3389` (ply 52) [structural distance $d=102$, QAP obj=19]: the master played Nxd5; under μ our Qa7 plays the role of that game’s Ne7 (a **cross-type** correspondence); cross-type roles: our Pd3 as their Re1, our Bc8 as their Pf2; transferred move a2c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2762` (ply 52) [structural distance $d=104$, QAP obj=2]: the master played Qf5+; under μ our Qa7 plays the role of that game’s Qg6; cross-type roles: our pg5 as their qa1, our Ra8 as their Pf2; transferred move a2c4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qc4 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



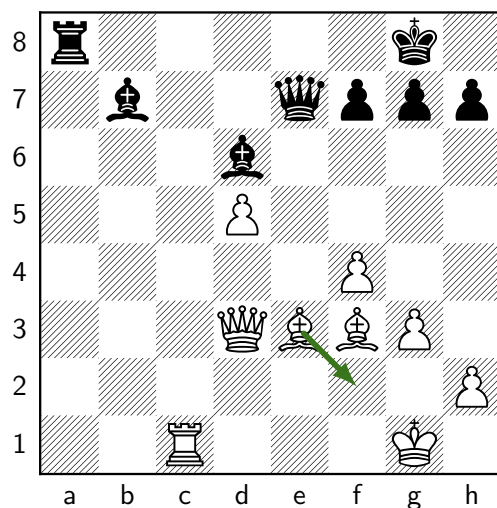
(a)



(b)

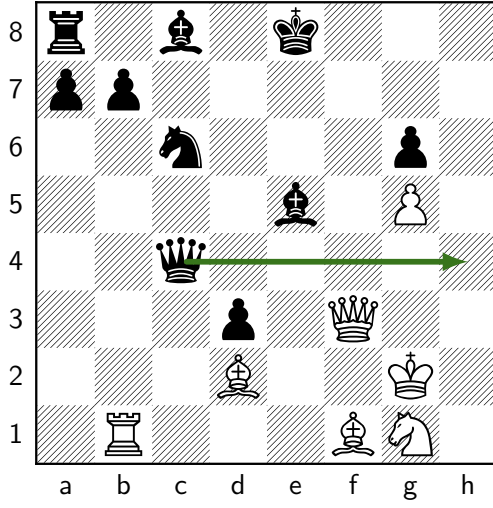


(c)

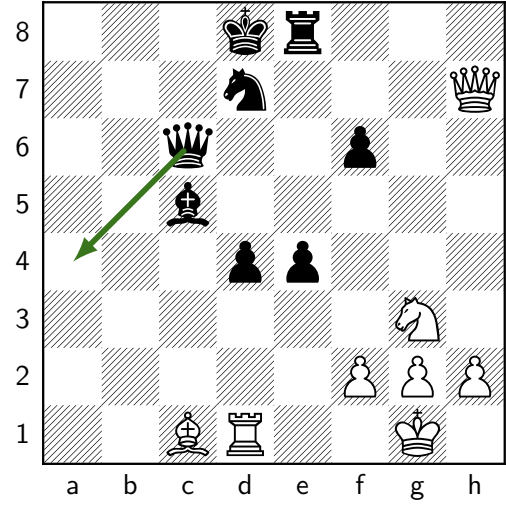


(d)

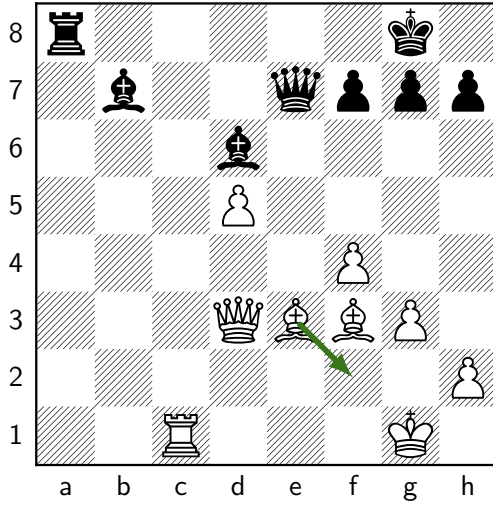
Figure 14: **Move 20, Black: Nc6**. (a) Our position; Melanie plays Nc6 (b8c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2836` (ply 50) [structural distance $d=109$, QAP obj=36]: the master played Ne4; under μ our Nb1 plays the role of that game’s Nd2; cross-type roles: our Qc4 as their Pg2, our bf1 as their ra3; transferred move b8c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3740` (ply 52) [structural distance $d=102$, QAP obj=20]: the master played d4; under μ our Nb1 plays the role of that game’s Pd3 (a **cross-type** correspondence); cross-type roles: our rb1 as their Nc3, our bd2 as their Pg4; transferred move b8c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1691` (ply 50) [structural distance $d=87$, QAP obj=43]: the master played Bf2; under μ our Pb2 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Rc1, our Pa7 as their Bf3; transferred move b7b6. Synthesis: 2 retrieved analogs transferred to Nc6 (summed QAP objective 56, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



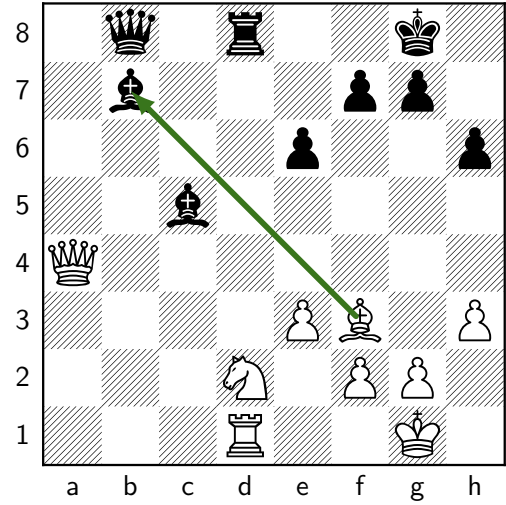
(a)



(b)

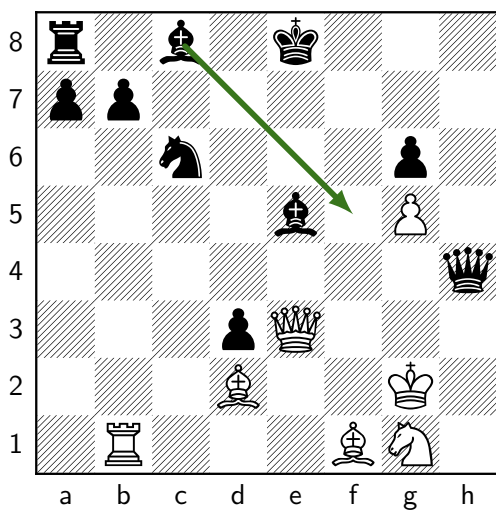


(c)

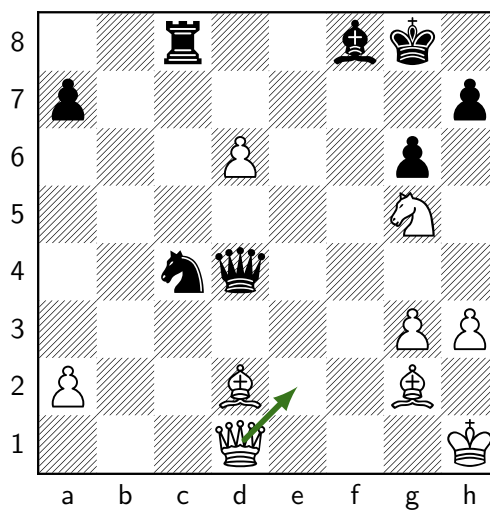


(d)

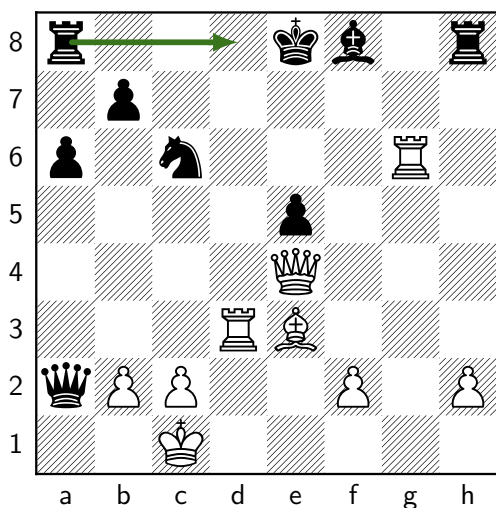
Figure 15: **Move 21, Black: Qh4**. (a) Our position; Melanie plays Qh4 (c4h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2351` (ply 57) [structural distance $d=108$, QAP obj=56]: the master played Qa4; under μ our Qc5 plays the role of that game’s Qc3; cross-type roles: our Pb7 as their Re8, our bf1 as their ng3; transferred move c4a4. (c) Retrieved from `ccr1_top_p0000#1691` (ply 50) [structural distance $d=104$, QAP obj=54]: the master played Bf2; under μ our Nc3 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our Pb7 as their Rc1, our Ra8 as their Ph2; transferred move c6d8. (d) Retrieved from `ccr1_top_p0001#3698` (ply 52) [structural distance $d=113$, QAP obj=53]: the master played Bxb7; under μ our Nc3 plays the role of that game’s Bf3 (a **cross-type** correspondence); cross-type roles: our Qc4 as their Rd1, our Pd3 as their Nd2; transferred move c6a5. Synthesis: no retrieved analog transferred to Qh4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



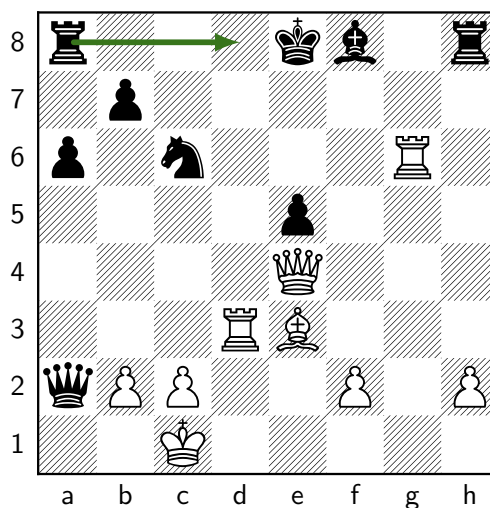
(a)



(b)

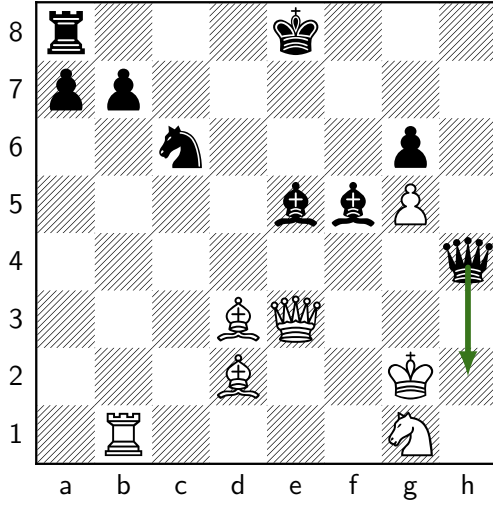


(c)

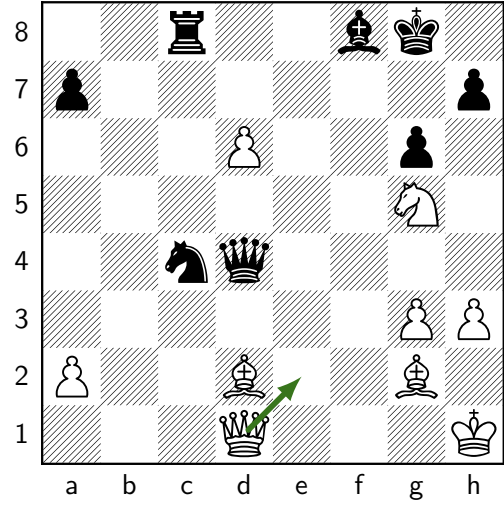


(d)

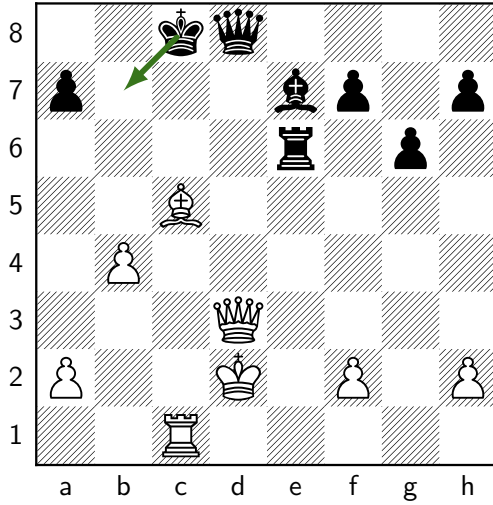
Figure 16: **Move 22, Black: Bf5**. (a) Our position; Melanie plays Bf5 (c8f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#412` (ply 58) [structural distance $d=92$, QAP obj=89]: the master played Qe2; under μ our Nc3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pa2, our Pg6 as their Bg2; transferred move c6a5. (c) Retrieved from `ccr1_top_p0000#2522` (ply 41) [structural distance $d=102$, QAP obj=63]: the master played Rd8; under μ our Pb2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our rb1 as their Bf8, our Pd3 as their Nc6; transferred move b7b6. (d) Retrieved from `ccr1_top_p0000#3252` (ply 41) [structural distance $d=102$, QAP obj=63]: the master played Rd8; under μ our Pb2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our rb1 as their Bf8, our Pd3 as their Nc6; transferred move b7b6. Synthesis: no retrieved analog transferred to Bf5 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



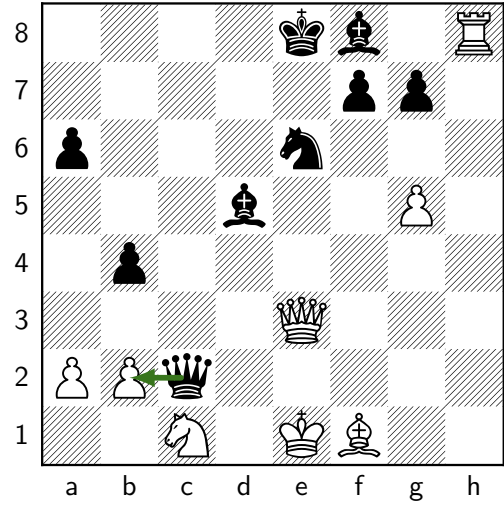
(a)



(b)

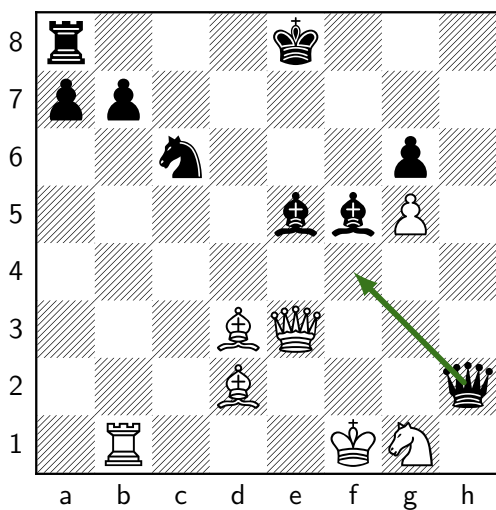


(c)

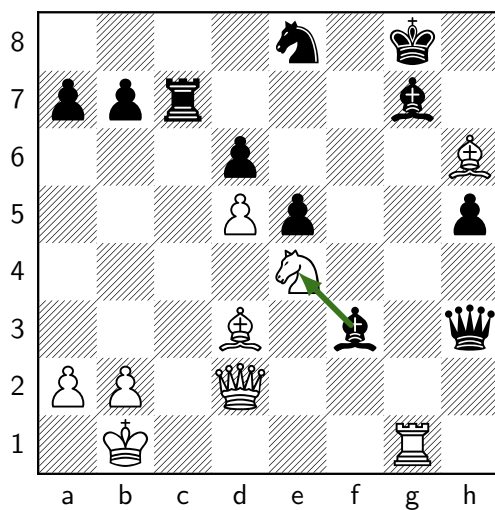


(d)

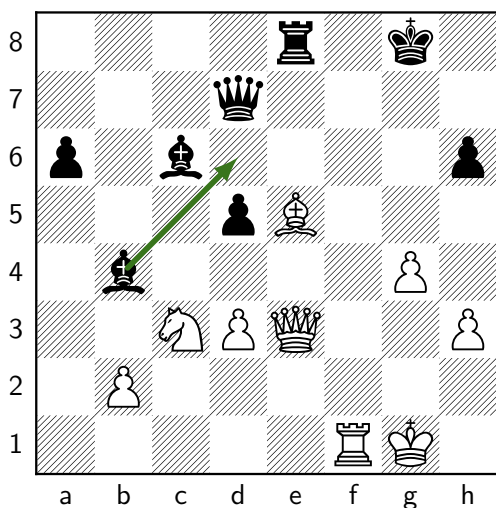
Figure 17: **Move 23, Black: Qh2+.** (a) Our position; Melanie plays Qh2+ (h4h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#412` (ply 58) [structural distance $d=102$, QAP obj=74]: the master played Qe2; under μ our Nc3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Bg2, our Bf5 as their Pg3; transferred move c6a5. (c) Retrieved from `ccr1_top_p0003#386` (ply 53) [structural distance $d=115$, QAP obj=73]: the master played Kb7; under μ our Ke1 plays the role of that game’s Kc1; cross-type roles: our Bf5 as their Qd8, our Bf5 as their Pa7; transferred move e8f7. (d) Retrieved from `ccr1_top_p0003#1629` (ply 61) [structural distance $d=113$, QAP obj=65]: the master played Qxb2; under μ our Qh5 plays the role of that game’s Qc7; cross-type roles: our Bf5 as their rh8, our rb1 as their Pg7; transferred move h4a4. Synthesis: no retrieved analog transferred to Qh2+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



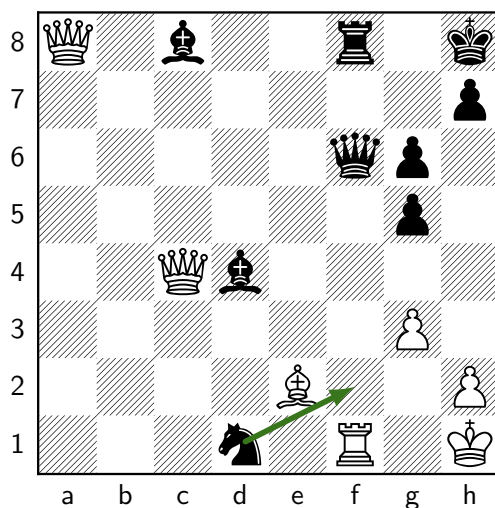
(a)



(b)

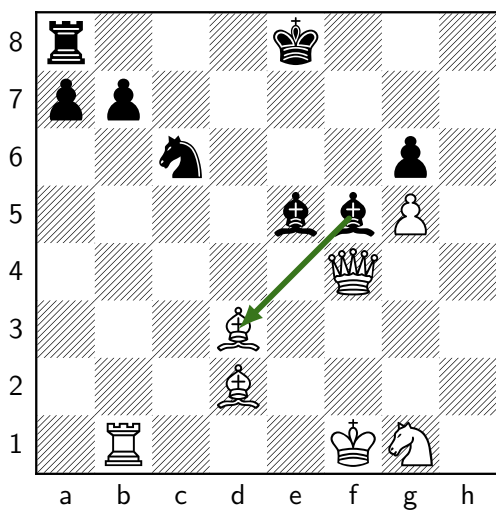


(c)

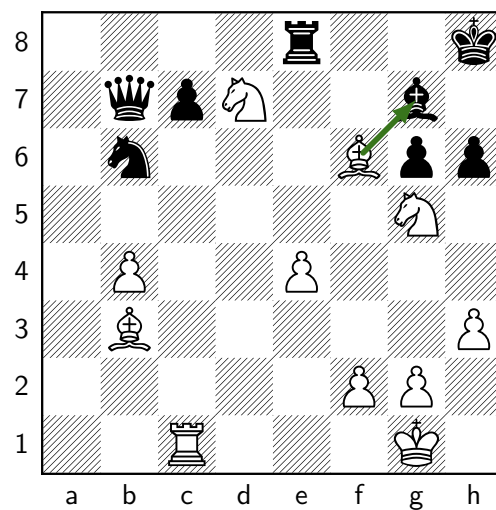


(d)

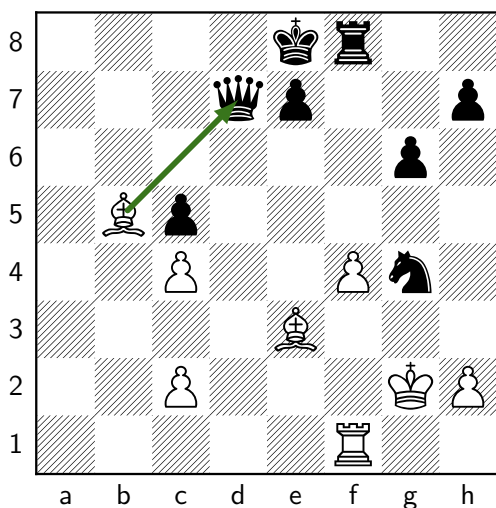
Figure 18: **Move 24, Black: Qf4+.** (a) Our position; Melanie plays Qf4+ (h2f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1258` (ply 51) [structural distance $d=112$, QAP obj=41]: the master played Bxe4; under μ our Qh7 plays the role of that game’s Bf6 (a **cross-type** correspondence); cross-type roles: our Pa7 as their Ne8, our Be5 as their Pb7; transferred move h2f4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3740` (ply 51) [structural distance $d=105$, QAP obj=41]: the master played Bd6; under μ our Qh7 plays the role of that game’s Bb5 (a **cross-type** correspondence); cross-type roles: our Bf5 as their Re8, our Pb7 as their Bc6; transferred move h2f4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2980` (ply 63) [structural distance $d=117$, QAP obj=82]: the master played Nf2+; under μ our Bf4 plays the role of that game’s Nd8 (a **cross-type** correspondence); cross-type roles: our Nc6 as their Rf8, our Ra8 as their Ph7; transferred move f5h3. Synthesis: 2 retrieved analogs transferred to Qf4+ (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



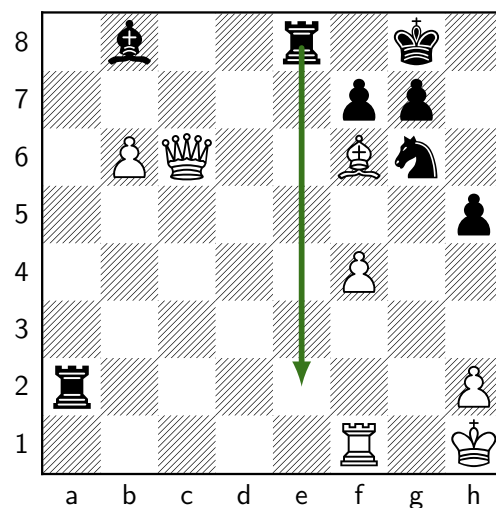
(a)



(b)

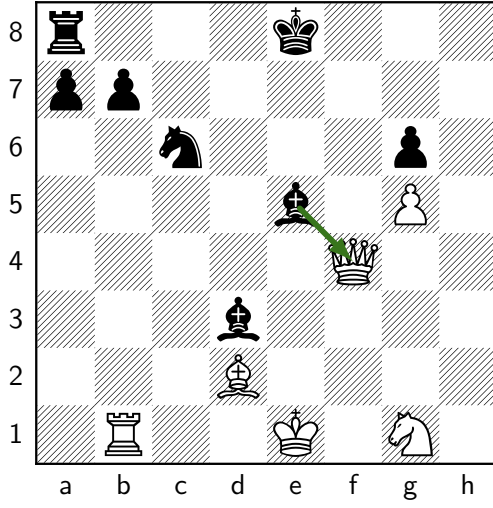


(c)

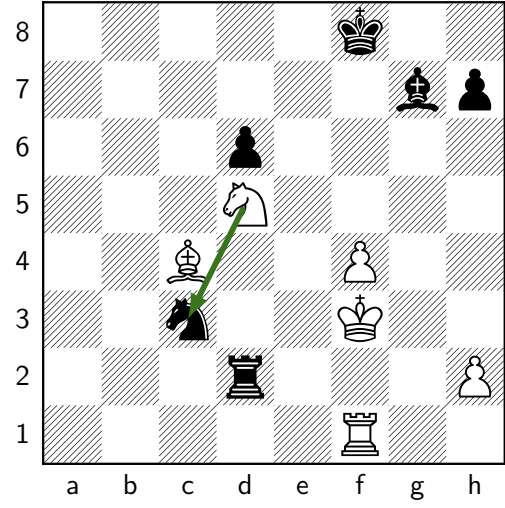


(d)

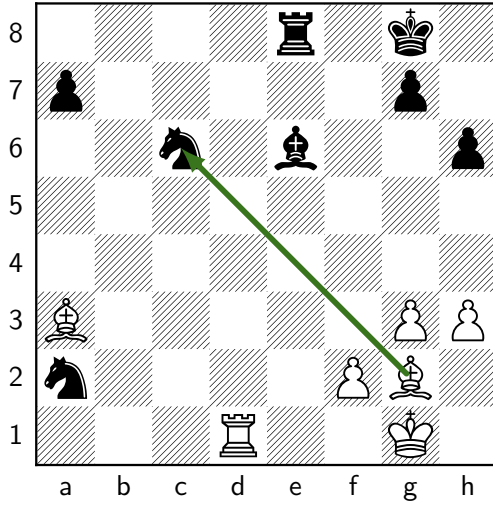
Figure 19: **Move 25, Black: Bxd3+.** (a) Our position; Melanie plays Bxd3+ (f5d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#1247 (ply 64) [structural distance $d=106$, QAP obj=22]: the master played Bxg7+; under μ our Bf4 plays the role of that game’s Bf6; cross-type roles: our ng1 as their Bb3, our Nc6 as their Pb4; transferred move f5d3 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#2831 (ply 42) [structural distance $d=107$, QAP obj=16]: the master played Bxd7+; under μ our Bf4 plays the role of that game’s Bb5; cross-type roles: our Be5 as their Rf1, our Nc6 as their Ph2; transferred move f5d3 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#2398 (ply 61) [structural distance $d=109$, QAP obj=13]: the master played Ree2; under μ our Bf4 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Pb7 as their Bb8, our Be5 as their Pg7; transferred move f5d3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bxd3+ (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



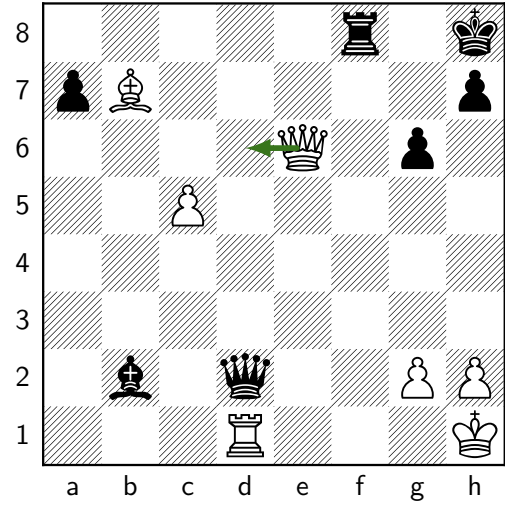
(a)



(b)

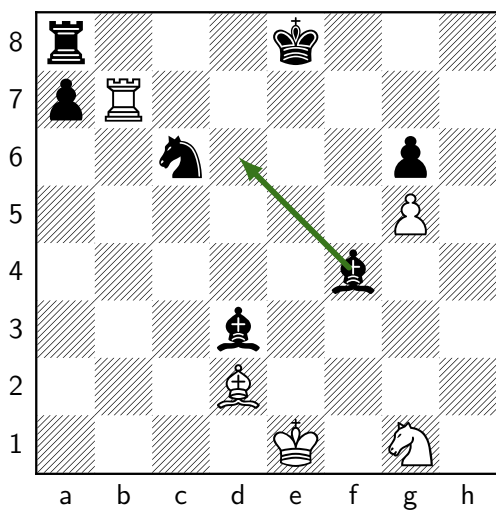


(c)

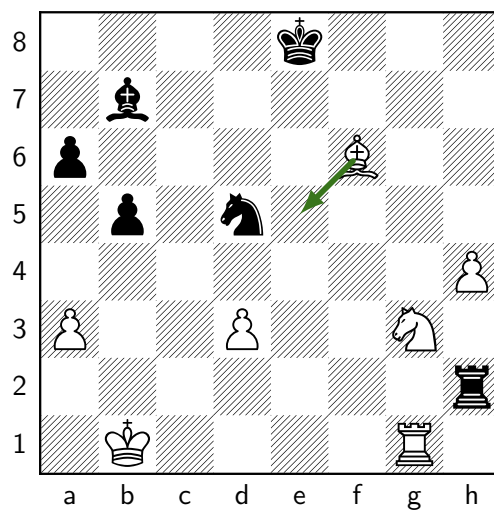


(d)

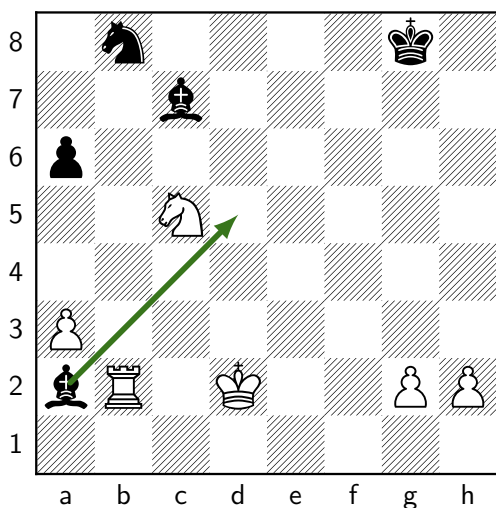
Figure 20: **Move 26, Black: Bxf4**. (a) Our position; Melanie plays Bxf4 (e5f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3065 (ply 66) [structural distance $d=74$, QAP obj=8]: the master played Nxc3; under μ our Be4 plays the role of that game’s Nd5 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Rf1, our qf4 as their nc3; transferred move e5f4 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#1010 (ply 60) [structural distance $d=79$, QAP obj=3]: the master played Bxc6; under μ our Be4 plays the role of that game’s Bg2; cross-type roles: our Nc6 as their Pf2, our qf4 as their nc6; transferred move e5f4 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#814 (ply 58) [structural distance $d=75$, QAP obj=29]: the master played Qd6; under μ our Bd6 plays the role of that game’s Qe6 (a **cross-type** correspondence); cross-type roles: our Be5 as their Rd1, our pg5 as their bb2; transferred move d3f5. Synthesis: 2 retrieved analogs transferred to Bxf4 (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



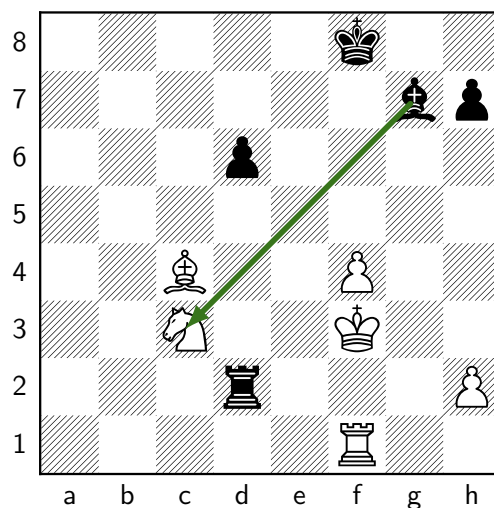
(a)



(b)

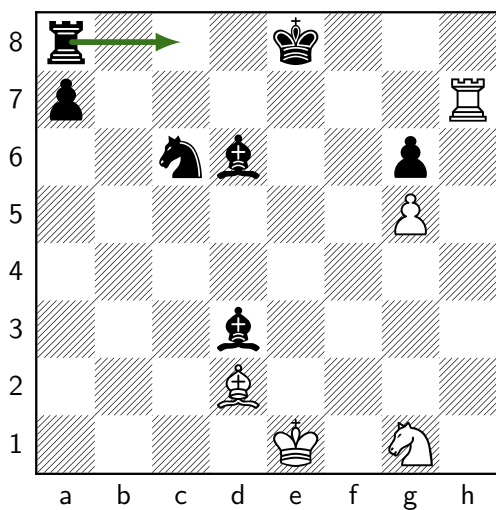


(c)

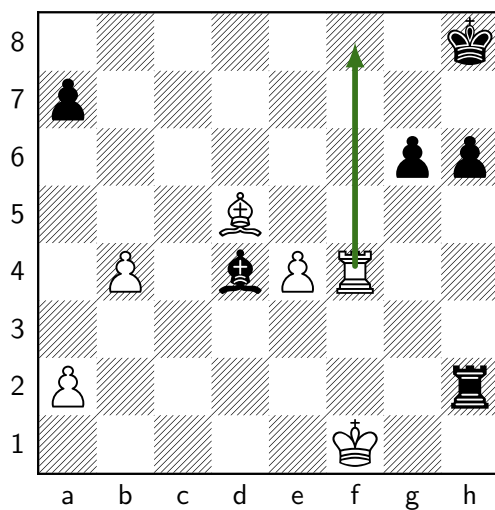


(d)

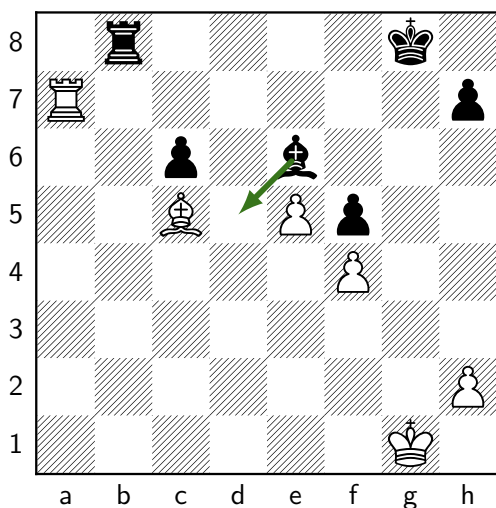
Figure 21: **Move 27, Black: Bd6 (f4d6)**. (a) Our position; Melanie plays Bd6 (f4d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2224` (ply 64) [structural distance $d=47$, QAP obj=27]: the master played Be5; under μ our Bf5 plays the role of that game’s Bf6; cross-type roles: our Nc6 as their Pa3, our Bd3 as their Ng3; transferred move f4d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#385` (ply 67) [structural distance $d=45$, QAP obj=14]: the master played Bd5; under μ our Bf5 plays the role of that game’s Ba7; cross-type roles: our Pa7 as their Nb8, our Nc6 as their Pa6; transferred move f4d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3065` (ply 67) [structural distance $d=50$, QAP obj=12]: the master played Bxc3; under μ our Bf5 plays the role of that game’s Bg2; cross-type roles: our Ra8 as their Ph7, our rb7 as their bc4; transferred move f4d6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bd6 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



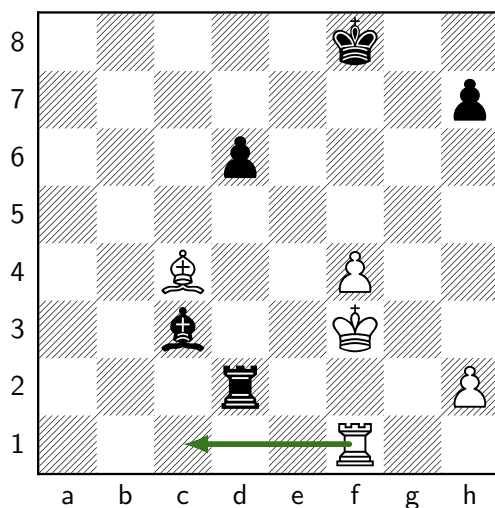
(a)



(b)

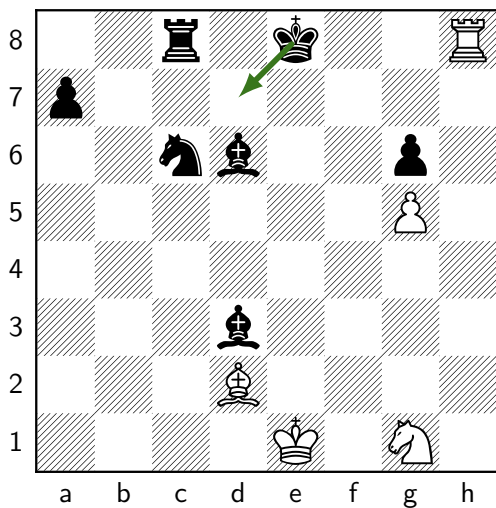


(c)

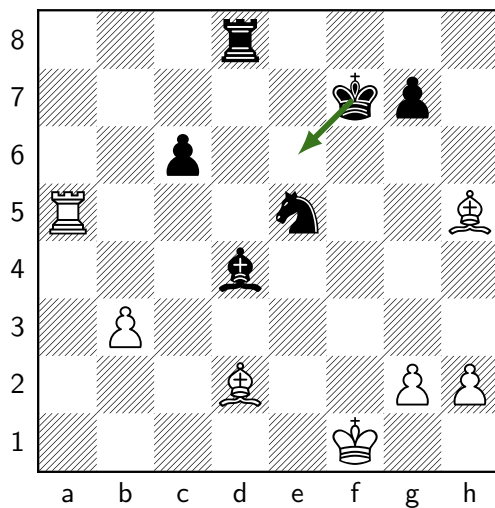


(d)

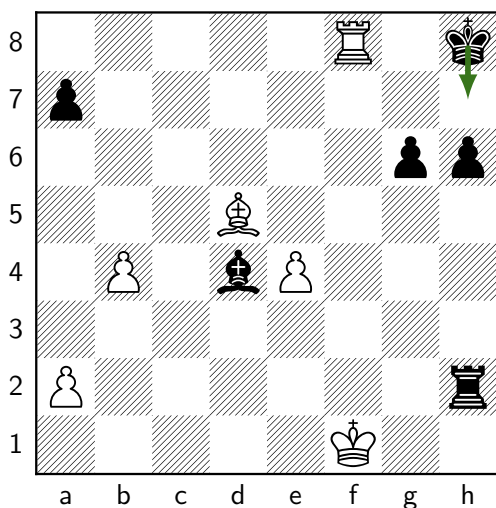
Figure 22: **Move 28, Black: Rc8.** (a) Our position; Melanie plays Rc8 (a8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2756` (ply 60) [structural distance $d=25$, QAP obj=23]: the master played Rf8+; under μ our Ra1 plays the role of that game’s Rf4; cross-type roles: our Bd6 as their Pb4, our Nc6 as their Bd5; transferred move a8c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2647` (ply 71) [structural distance $d=31$, QAP obj=21]: the master played Bd5; under μ our Ra1 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our Bd6 as their Rb8, our Nc6 as their Pc6; transferred move a8c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3065` (ply 68) [structural distance $d=38$, QAP obj=14]: the master played Rc1; under μ our Ra1 plays the role of that game’s Rf1; cross-type roles: our Nc6 as their ph7; transferred move a8c8 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Rc8 (summed QAP objective 100, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



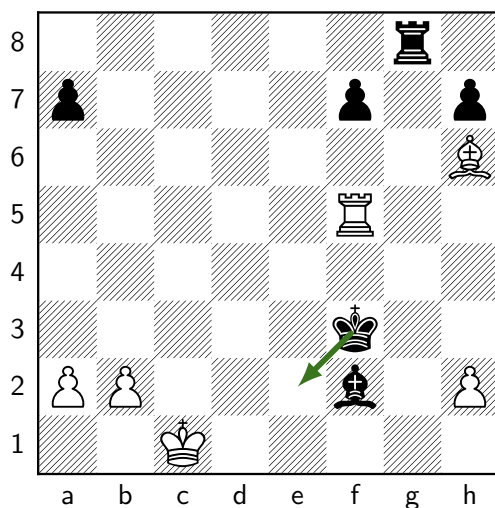
(a)



(b)

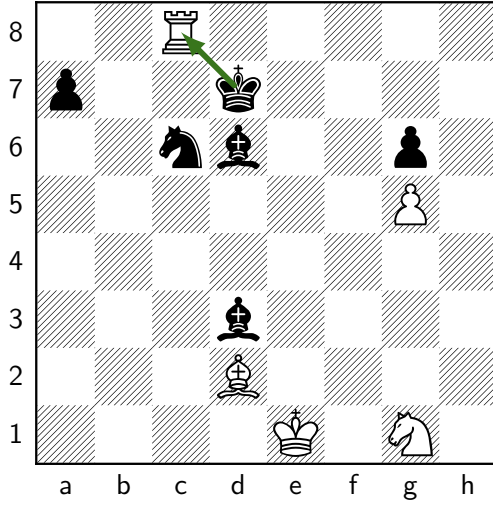


(c)

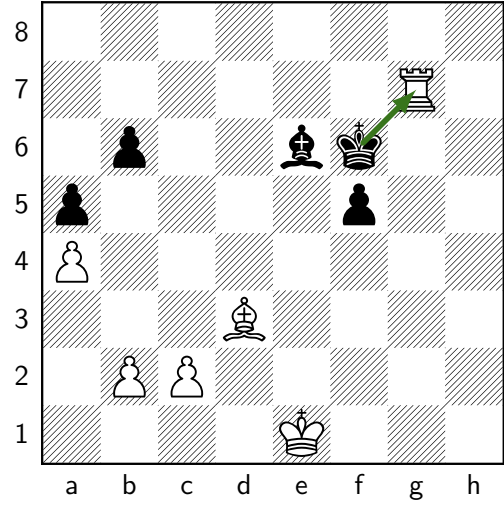


(d)

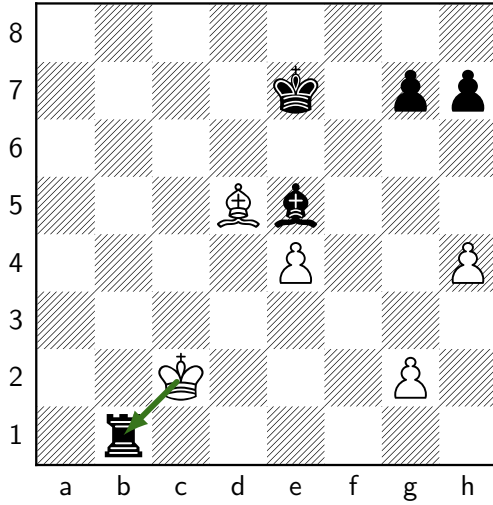
Figure 23: **Move 29, Black: Kd7.** (a) Our position; Melanie plays Kd7 (e8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2499` (ply 71) [structural distance $d=31$, QAP obj=48]: the master played Ke6; under μ our Ke1 plays the role of that game’s Kf2; cross-type roles: our Pa7 as their Rd8, our Nc6 as their Pc6; transferred move e8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2756` (ply 61) [structural distance $d=29$, QAP obj=47]: the master played Kh7; under μ our Ke1 plays the role of that game’s Kh1; cross-type roles: our Nc6 as their bd5, our Bd6 as their pb4; transferred move e8d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1966` (ply 49) [structural distance $d=39$, QAP obj=46]: the master played Ke2; under μ our Ke1 plays the role of that game’s Kf6; cross-type roles: our Bd3 as their Rg8, our Nc6 as their Pa7; transferred move e8d7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kd7 (summed QAP objective 180, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



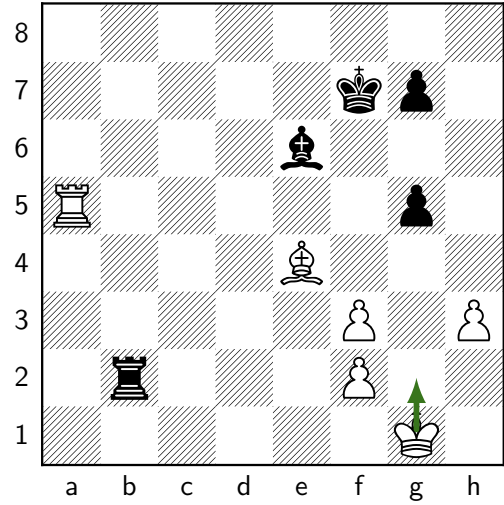
(a)



(b)

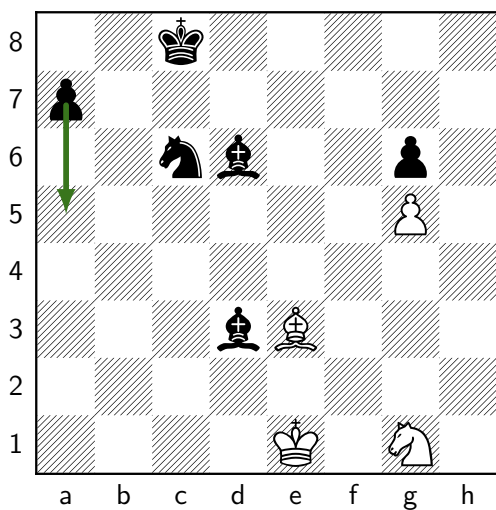


(c)

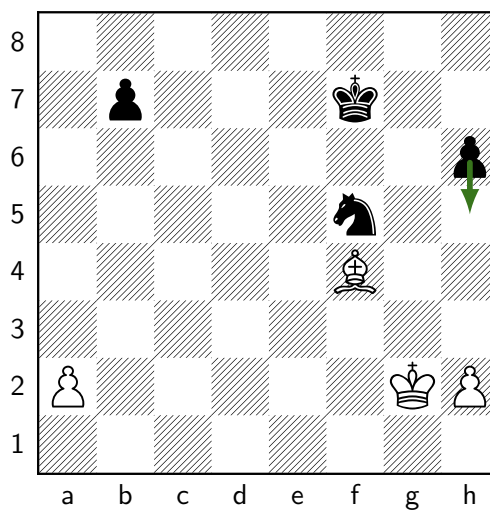


(d)

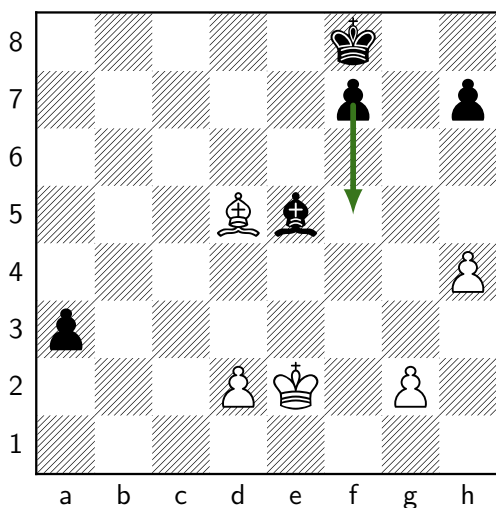
Figure 24: **Move 30, Black: Kxc8.** (a) Our position; Melanie plays Kxc8 (d7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2474` (ply 75) [structural distance $d=34$, QAP obj=23]: the master played Kxg7; under μ our *Kd2* plays the role of that game’s *Kf3*; cross-type roles: our *Bd3* as their *Pa5*, our *Nc6* as their *Pf5*; transferred move d7c8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1854` (ply 62) [structural distance $d=33$, QAP obj=20]: the master played Kxb1; under μ our *Kd2* plays the role of that game’s *Kc2*; cross-type roles: our *Nc6* as their *Pg2*, our *Pg6* as their *be5*; transferred move d7c8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2468` (ply 60) [structural distance $d=38$, QAP obj=4]: the master played Kg2; under μ our *Kd2* plays the role of that game’s *Kg1*; cross-type roles: our *Bd6* as their *Pf2*, our *Nc6* as their *Be4*; transferred move d7c8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kxc8 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



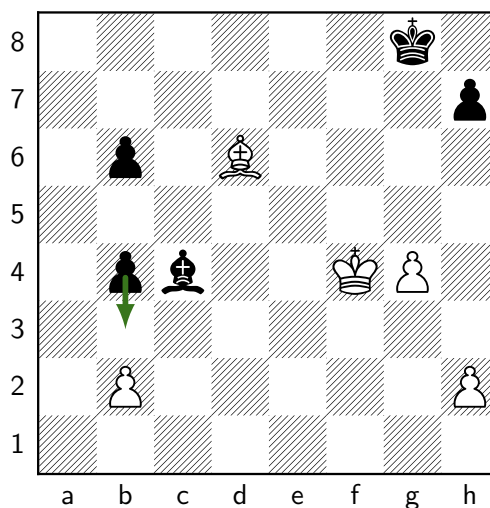
(a)



(b)

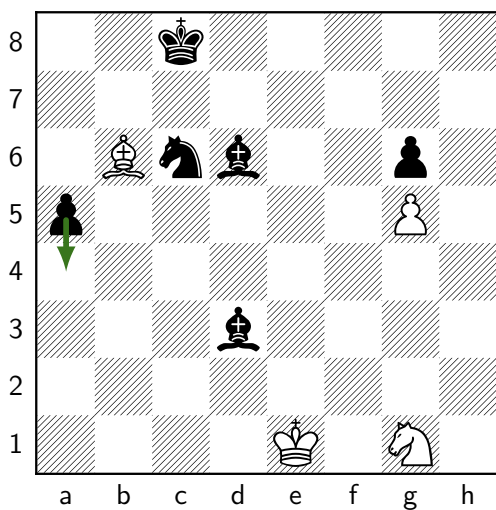


(c)

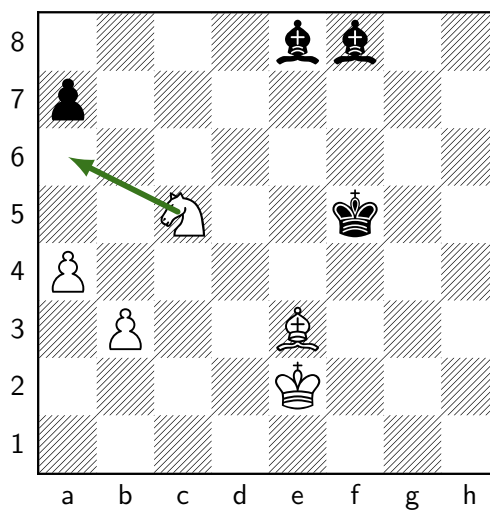


(d)

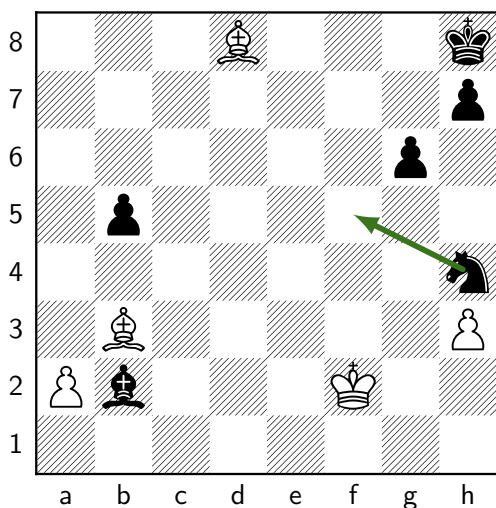
Figure 25: **Move 31, Black: a5**. (a) Our position; Melanie plays a5 (a7a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3630` (ply 73) [structural distance $d=26$, QAP obj=13]: the master played h5; under μ our *Pa2* plays the role of that game’s *Ph3*; cross-type roles: our *Bd6* as their *pa2*, our *ng1* as their *ph2*; transferred move a7a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#713` (ply 77) [structural distance $d=28$, QAP obj=12]: the master played f5; under μ our *Pa2* plays the role of that game’s *Pf2*; cross-type roles: our *Nc6* as their *Ph7*, our *Bd6* as their *ph4*; transferred move a7a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2141` (ply 69) [structural distance $d=27$, QAP obj=10]: the master played b3; under μ our *Pa2* plays the role of that game’s *Pb5*; cross-type roles: our *Nc6* as their *Pb6*, our *Bd6* as their *pg4*; transferred move a7a5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to a5 (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



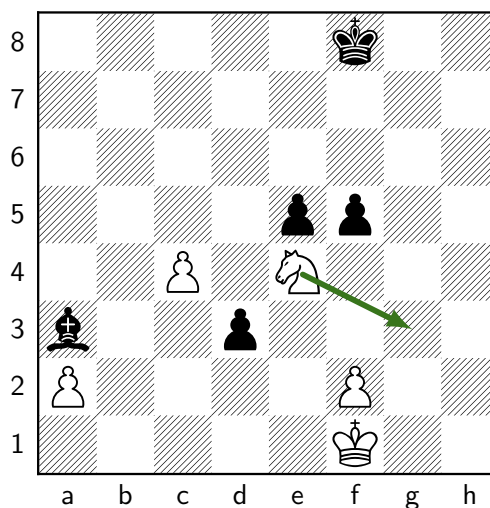
(a)



(b)

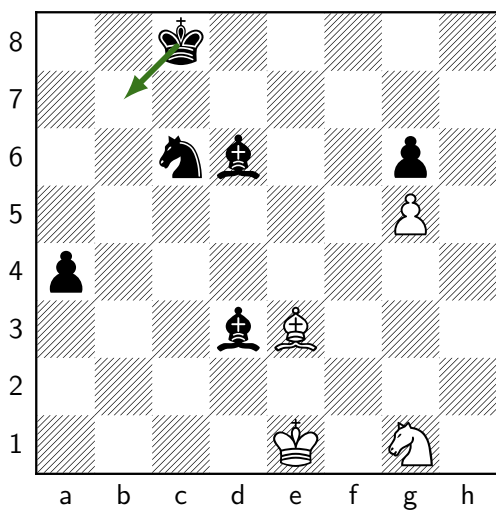


(c)

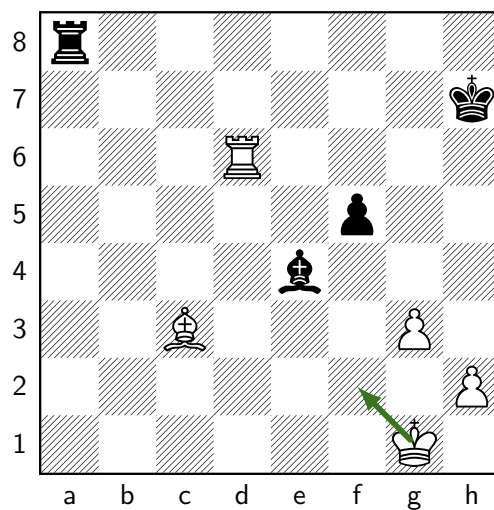


(d)

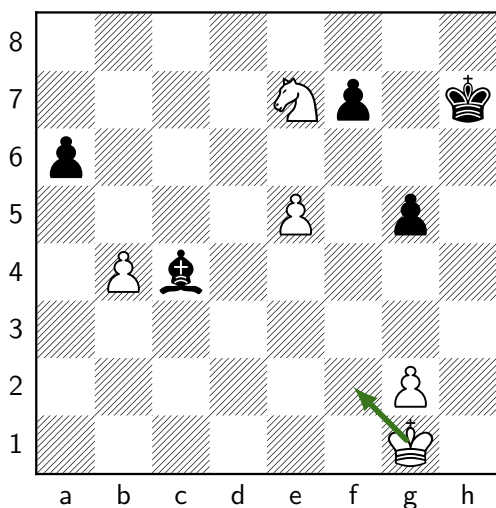
Figure 26: **Move 32, Black: a4**. (a) Our position; Melanie plays a4 (a5a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2916` (ply 72) [structural distance $d=23$, QAP obj=20]: the master played Na6; under μ our Pa4 plays the role of that game’s Nc5 (a **cross-type** correspondence); cross-type roles: our Nc6 as their Be3, our Bd3 as their Pa4; transferred move a5a4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1068` (ply 59) [structural distance $d=24$, QAP obj=14]: the master played Nf5; under μ our Pa4 plays the role of that game’s Nh5 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Ph7, our Nc6 as their Pg6; transferred move a5a4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2015` (ply 78) [structural distance $d=26$, QAP obj=7]: the master played Ng3; under μ our Pa4 plays the role of that game’s Ne4 (a **cross-type** correspondence); cross-type roles: our Bd6 as their pd3, our ng1 as their Pc4; transferred move a5a4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to a4 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



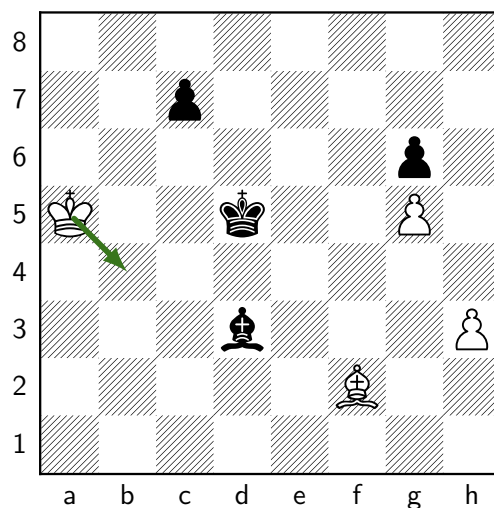
(a)



(b)

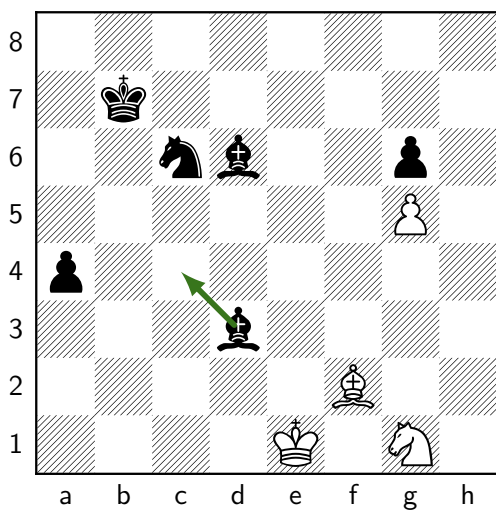


(c)

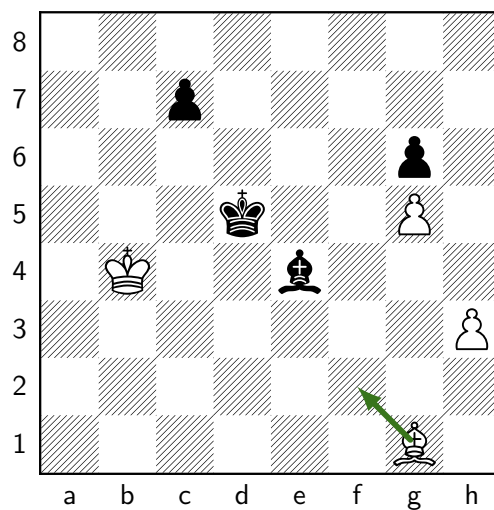


(d)

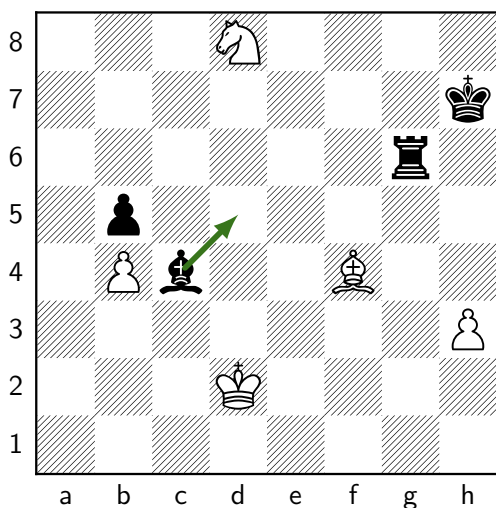
Figure 27: **Move 33, Black: Kb7**. (a) Our position; Melanie plays Kb7 (c8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1785` (ply 72) [structural distance $d=21$, QAP obj=3]: the master played Kf2; under μ our Kc1 plays the role of that game’s Kg1; cross-type roles: our Bd6 as their Ph2, our Nc6 as their Rd6; transferred move c8b7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3714` (ply 74) [structural distance $d=22$, QAP obj=3]: the master played Kf2; under μ our Kc1 plays the role of that game’s Kg1; cross-type roles: our Nc6 as their Pg2, our Bd6 as their Pe5; transferred move c8b7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#979` (ply 74) [structural distance $d=18$, QAP obj=3]: the master played Kb4; under μ our Kc1 plays the role of that game’s Ka5; cross-type roles: our Nc6 as their Ph3, our ng1 as their pc7; transferred move c8b7 (**endorses this move**). Synthesis: 19 retrieved analogs transferred to Kb7 (summed QAP objective 35, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



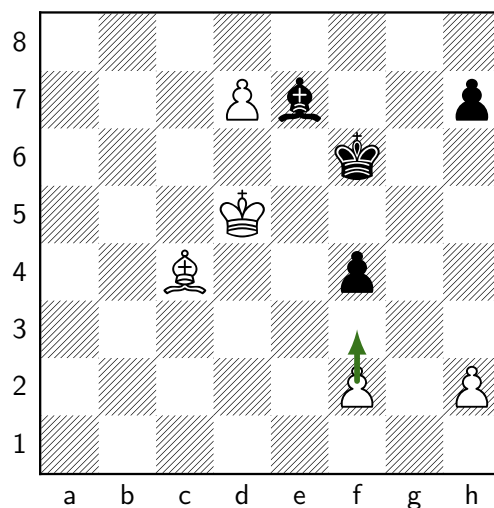
(a)



(b)

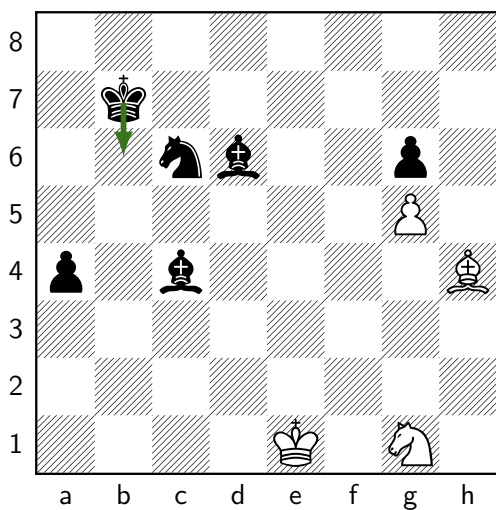


(c)

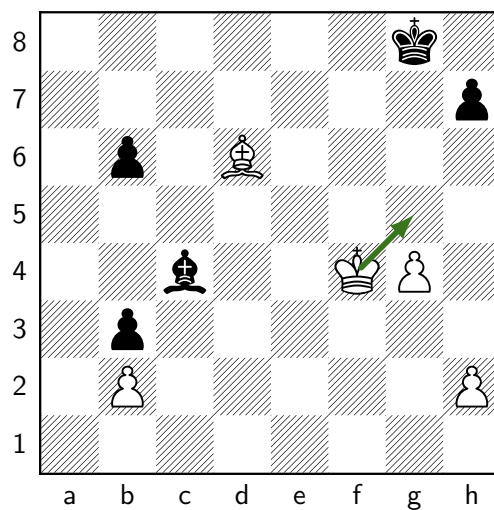


(d)

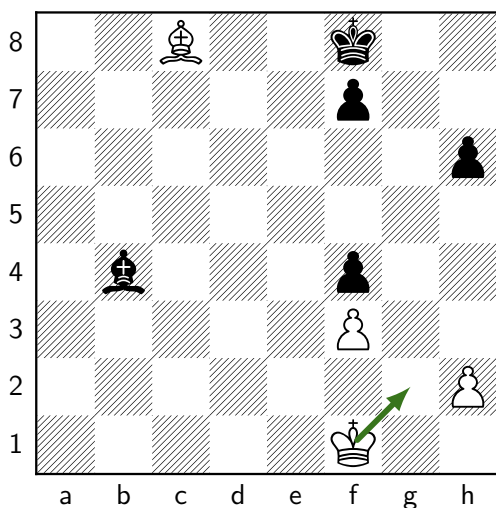
Figure 28: **Move 34, Black: Bc4.** (a) Our position; Melanie plays Bc4 (d3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#979` (ply 80) [structural distance $d=22$, QAP obj=4]: the master played Bf2; under μ our Bd6 plays the role of that game’s Bg1; cross-type roles: our Nc6 as their Ph3, our ng1 as their pc7; transferred move d3c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2219` (ply 79) [structural distance $d=24$, QAP obj=3]: the master played Bd5; under μ our Bd6 plays the role of that game’s Bc5; cross-type roles: our Bd6 as their Rg6, our Nc6 as their ph3; transferred move d3c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2795` (ply 76) [structural distance $d=23$, QAP obj=3]: the master played f3; under μ our Bd6 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our ng1 as their Bc4, our Nc6 as their Pd7; transferred move d3c4 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bc4 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



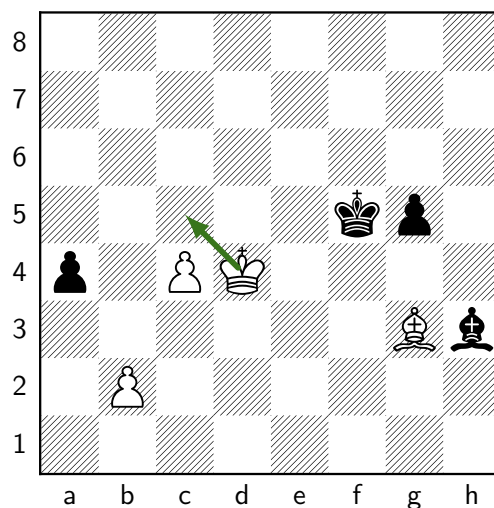
(a)



(b)

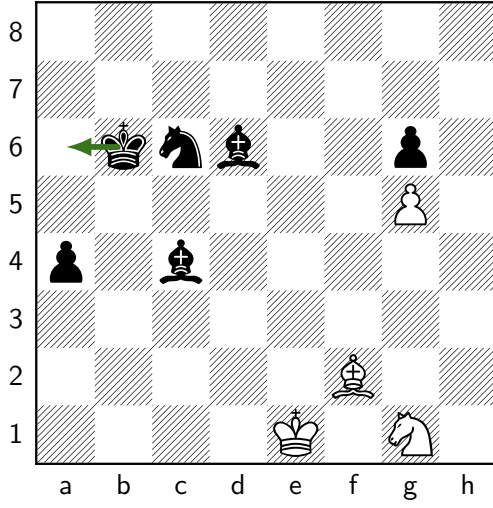


(c)

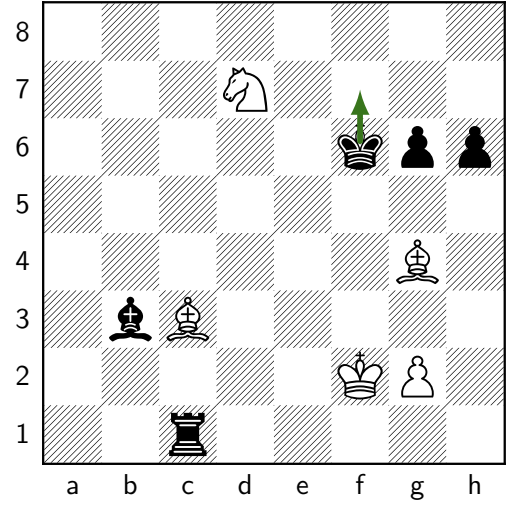


(d)

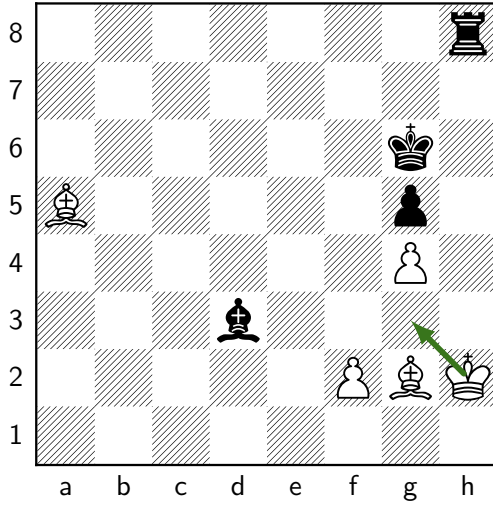
Figure 29: **Move 35, Black: Kb6**. (a) Our position; Melanie plays Kb6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2141` (ply 70) [structural distance $d=21$, QAP obj=3]: the master played Kg5; under μ our Kb2 plays the role of that game’s Kf4; cross-type roles: our Nc6 as their Ph2, our Bd6 as their Pg4; transferred move b7b6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#882` (ply 82) [structural distance $d=18$, QAP obj=2]: the master played Kg2; under μ our Kb2 plays the role of that game’s Kf1; cross-type roles: our Bd6 as their Pf3, our Nc6 as their ph6; transferred move b7b6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2374` (ply 82) [structural distance $d=18$, QAP obj=2]: the master played Kc5; under μ our Kb2 plays the role of that game’s Kd4; cross-type roles: our Nc6 as their Pc4, our ng1 as their pg5; transferred move b7b6 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kb6 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



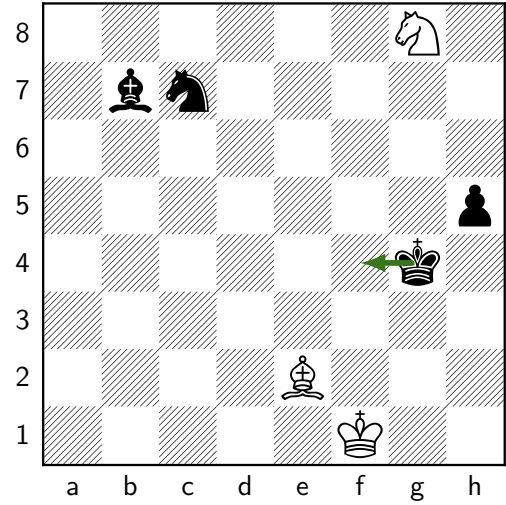
(a)



(b)

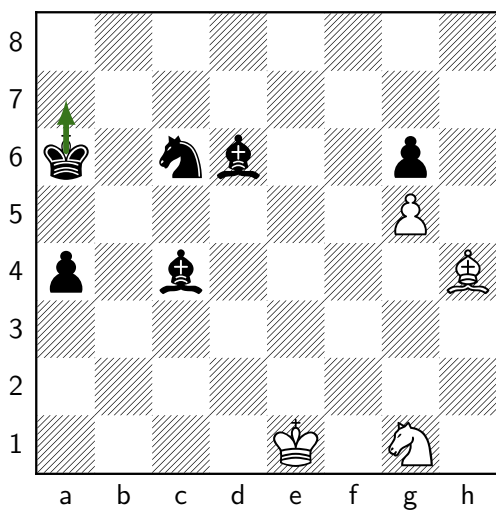


(c)

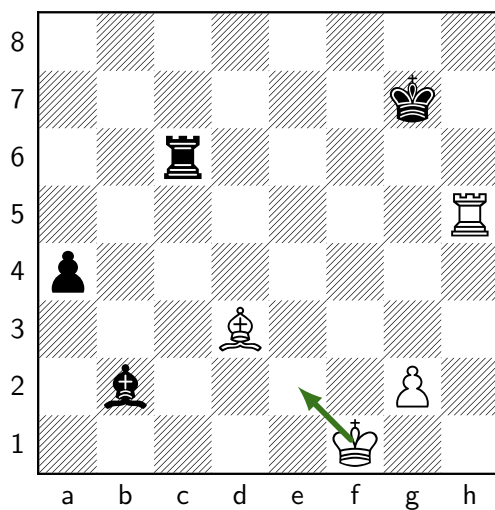


(d)

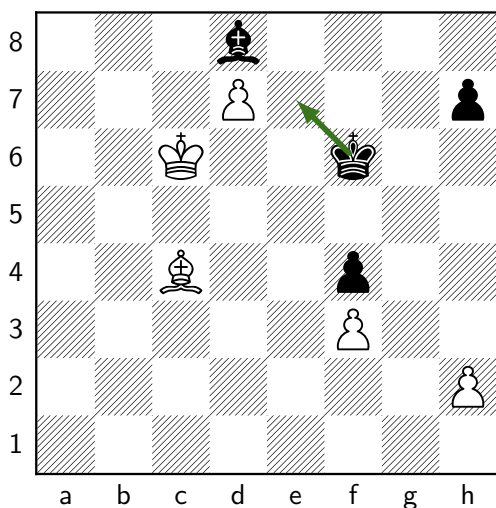
Figure 30: **Move 36, Black: Ka6.** (a) Our position; Melanie plays Ka6 (b6a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1664` (ply 83) [structural distance $d=29$, QAP obj=64]: the master played Kf7; under μ our Kb3 plays the role of that game’s Kf3; cross-type roles: our bf2 as their nd7, our Nc6 as their Pg6; transferred move b6a6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#344` (ply 84) [structural distance $d=24$, QAP obj=45]: the master played Kg3; under μ our Kb3 plays the role of that game’s Kh2; cross-type roles: our Nc6 as their Bg2, our pg5 as their bd3; transferred move b6b5. (d) Retrieved from `ccrl_top_p0001#3914` (ply 83) [structural distance $d=24$, QAP obj=38]: the master played Kf4; under μ our Kb3 plays the role of that game’s Kg5; cross-type roles: our Pg6 as their ng8, our Nc6 as their Ph5; transferred move b6b7. Synthesis: 1 retrieved analog transferred to Ka6 (summed QAP objective 64, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



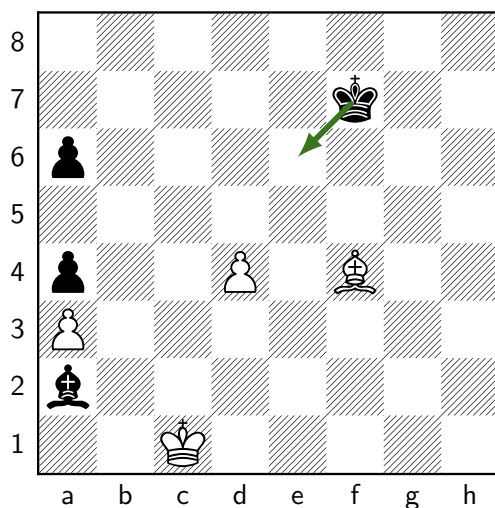
(a)



(b)

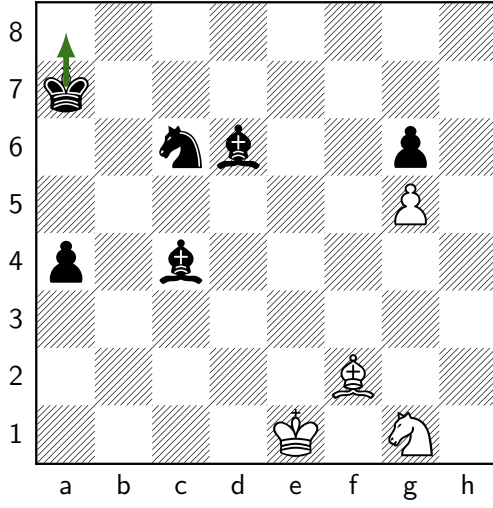


(c)

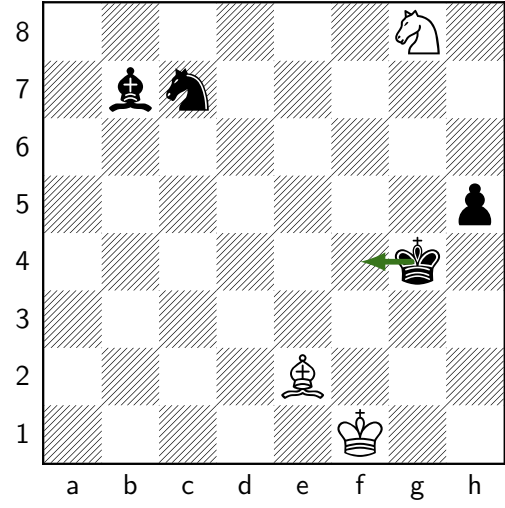


(d)

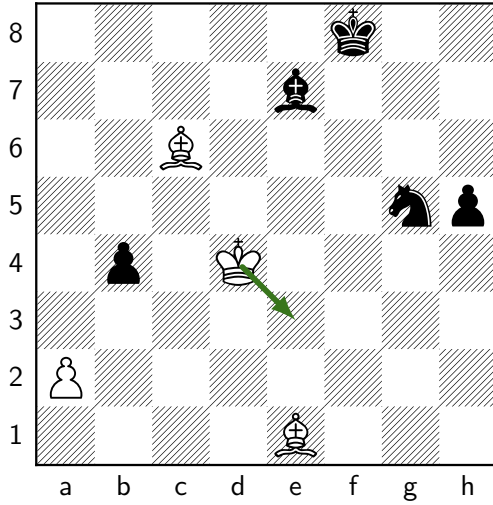
Figure 31: **Move 37, Black: Ka7**. (a) Our position; Melanie plays Ka7 (a6a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2720` (ply 88) [structural distance $d=21$, QAP obj=4]: the master played Ke2; under μ our Ka3 plays the role of that game’s Kf1; cross-type roles: our Nc6 as their Pg2, our Bd6 as their Rh5; transferred move a6a7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2795` (ply 79) [structural distance $d=22$, QAP obj=4]: the master played Ke7; under μ our Ka3 plays the role of that game’s Kf3; cross-type roles: our ng1 as their pd7, our Nc6 as their Ph7; transferred move a6a7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1688` (ply 77) [structural distance $d=14$, QAP obj=3]: the master played Ke6; under μ our Ka3 plays the role of that game’s Kf2; cross-type roles: our Nc6 as their Pa6, our ng1 as their pd4; transferred move a6a7 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Ka7 (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



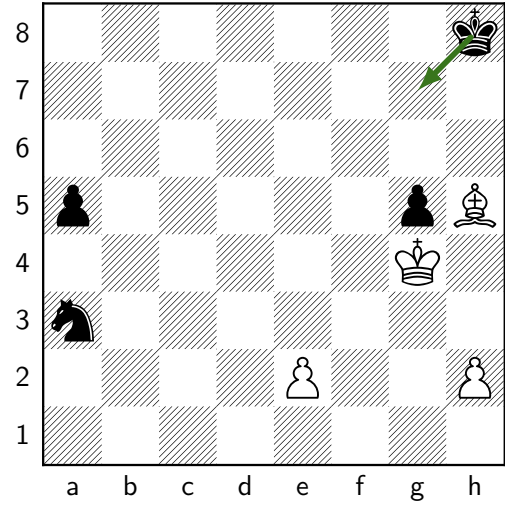
(a)



(b)

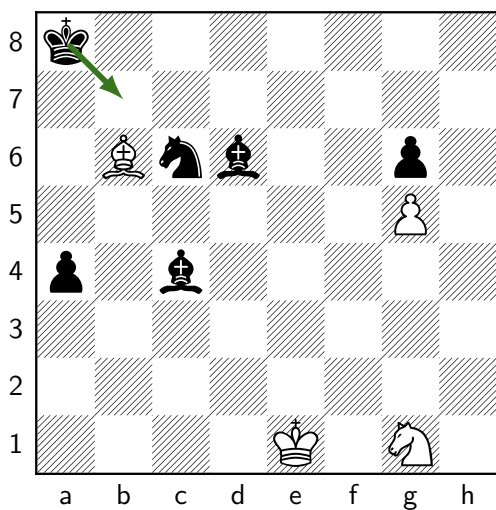


(c)

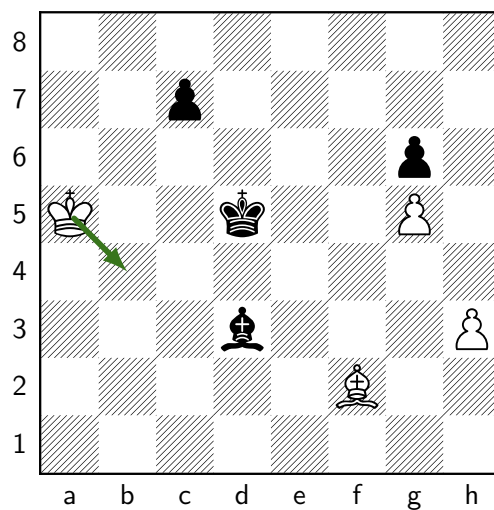


(d)

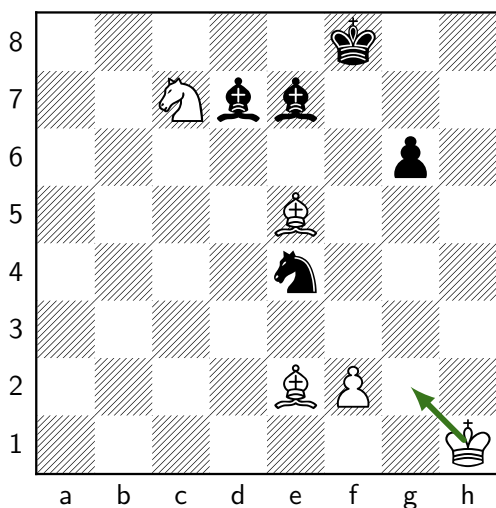
Figure 32: **Move 38, Black: Ka8**. (a) Our position; Melanie plays Ka8 (a7a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3914` (ply 83) [structural distance $d=26$, QAP obj=31]: the master played Kf4; under μ our Ka2 plays the role of that game’s Kg5; cross-type roles: our Pg6 as their ng8, our Nc6 as their Ph5; transferred move a7a8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1068` (ply 84) [structural distance $d=19$, QAP obj=4]: the master played Ke3; under μ our Ka2 plays the role of that game’s Kd4; cross-type roles: our Nc6 as their ph5; transferred move a7a8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2070` (ply 77) [structural distance $d=26$, QAP obj=2]: the master played Kg7; under μ our Ka2 plays the role of that game’s Kh1; cross-type roles: our Bc4 as their Pg5, our ng1 as their pe2; transferred move a7a8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ka8 (summed QAP objective 37, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



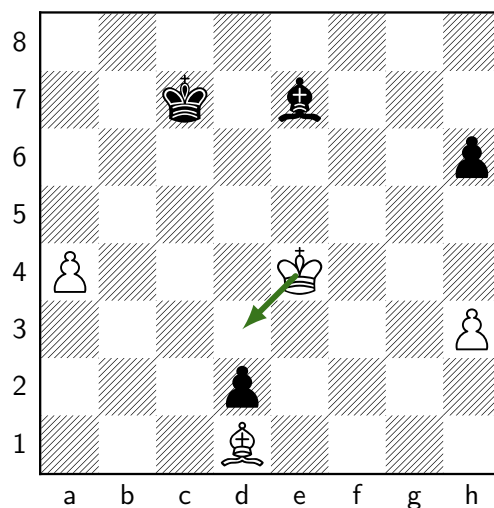
(a)



(b)

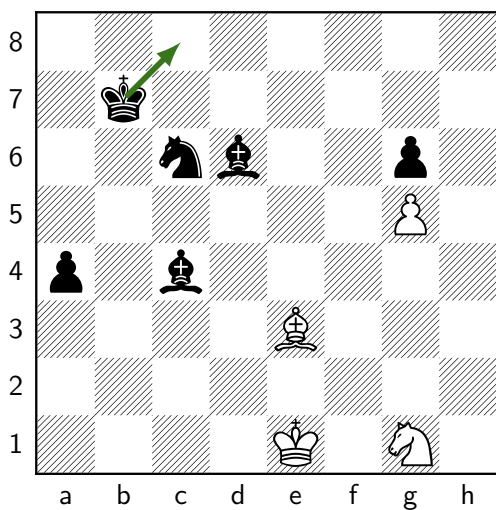


(c)

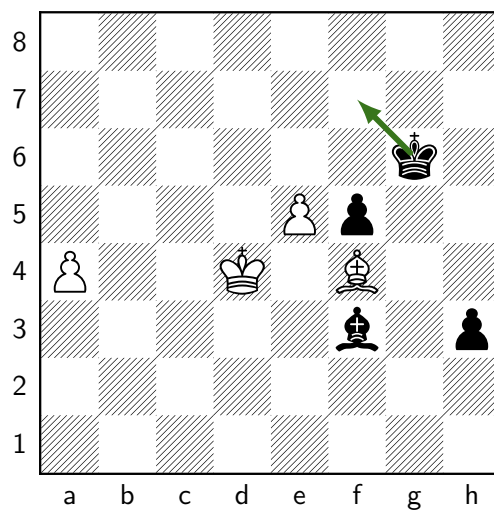


(d)

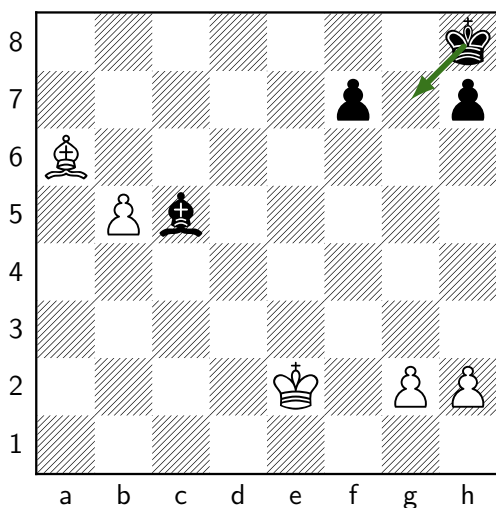
Figure 33: **Move 39, Black: Kb7.** (a) Our position; Melanie plays Kb7 (a8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#979` (ply 74) [structural distance $d=14$, QAP obj=2]: the master played Kb4; under μ our *Ka1* plays the role of that game’s *Ka5*; cross-type roles: our *pg5* as their *Bf2*, our *Bd6* as their *Ph3*; transferred move a8b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2925` (ply 90) [structural distance $d=18$, QAP obj=2]: the master played Kg2; under μ our *Ka1* plays the role of that game’s *Kh1*; cross-type roles: our *pg5* as their *ne4*, our *ng1* as their *be7*; transferred move a8b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#502` (ply 88) [structural distance $d=19$, QAP obj=2]: the master played Kd3; under μ our *Ka1* plays the role of that game’s *Ke4*; cross-type roles: our *Bd6* as their *pd2*, our *ng1* as their *Pa4*; transferred move a8b7 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Kb7 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



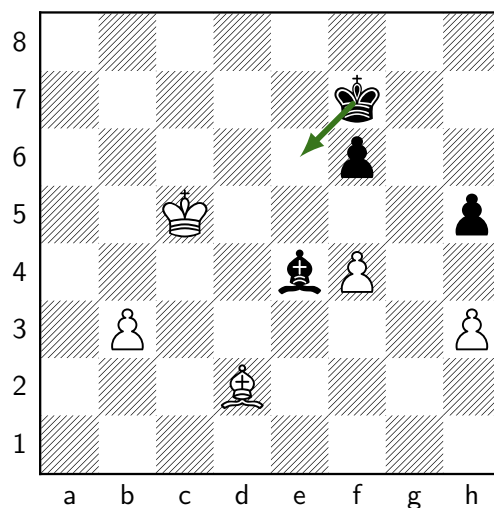
(a)



(b)

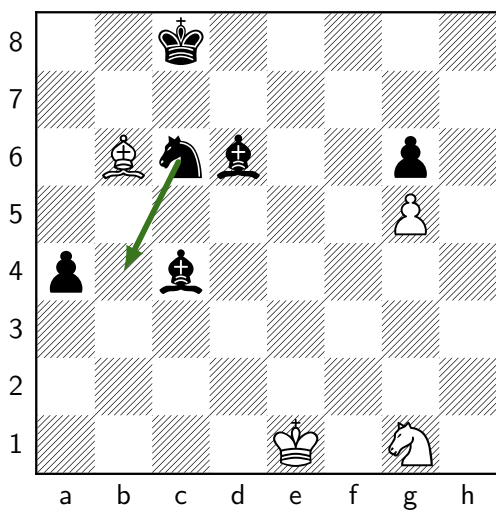


(c)

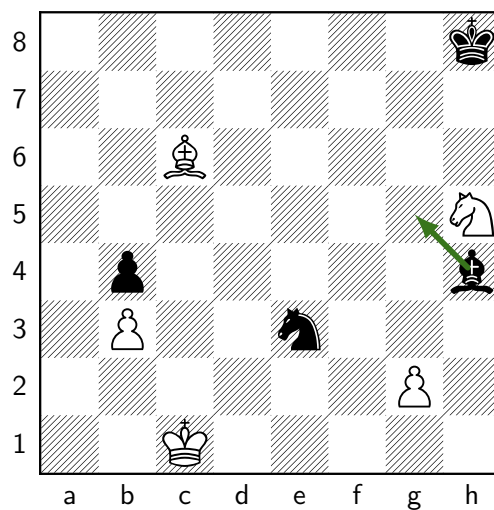


(d)

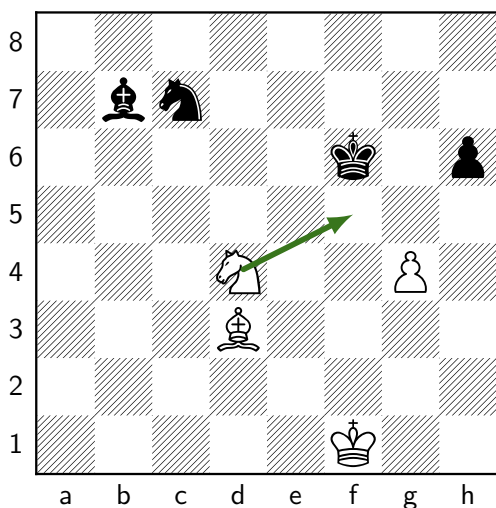
Figure 34: **Move 40, Black: Kc8.** (a) Our position; Melanie plays Kc8 (b7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#376` (ply 91) [structural distance $d=22$, QAP obj=3]: the master played Kf7; under μ our Kb2 plays the role of that game’s Kg3; cross-type roles: our ng1 as their pe5, our Nc6 as their Pf5; transferred move b7c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1635` (ply 81) [structural distance $d=20$, QAP obj=2]: the master played Kg7; under μ our Kb2 plays the role of that game’s Kh1; cross-type roles: our Nc6 as their Ph7, our ng1 as their pb5; transferred move b7c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#801` (ply 89) [structural distance $d=21$, QAP obj=2]: the master played Ke6; under μ our Kb2 plays the role of that game’s Kf2; cross-type roles: our Nc6 as their Pf6, our be3 as their pf4; transferred move b7c8 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kc8 (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



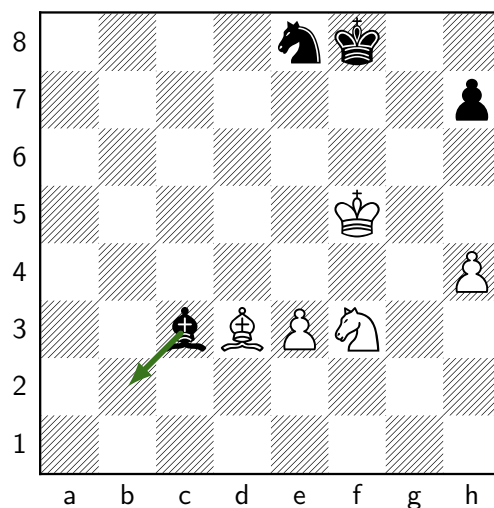
(a)



(b)

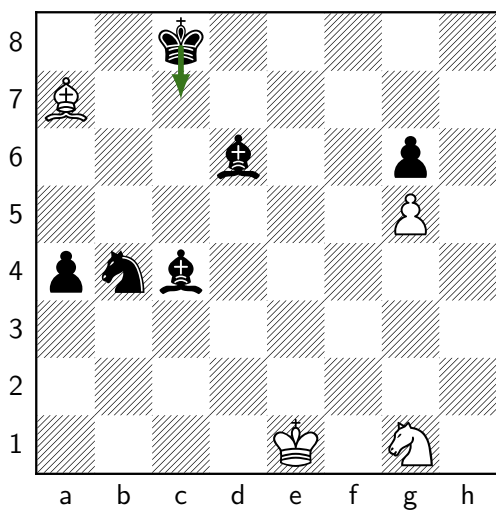


(c)

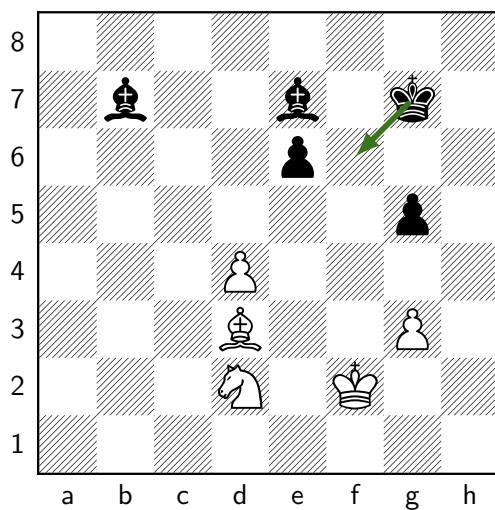


(d)

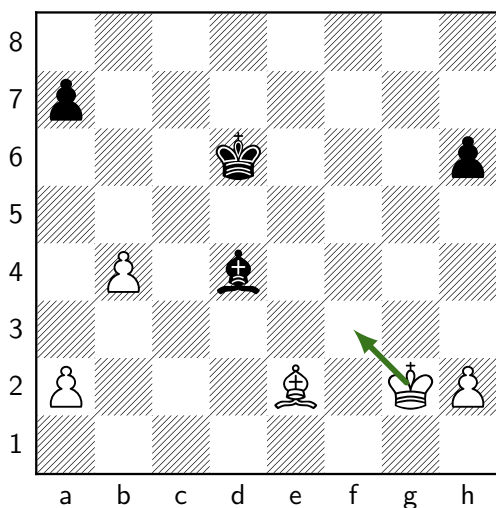
Figure 35: **Move 41, Black: Nb4**. (a) Our position; Melanie plays Nb4 (c6b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1169` (ply 91) [structural distance $d=13$, QAP obj=3]: the master played Bg5; under μ our Nc3 plays the role of that game’s Bh5 (a **cross-type** correspondence); cross-type roles: our ng1 as their bc6, our bb6 as their nh5; transferred move c6b4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3914` (ply 76) [structural distance $d=16$, QAP obj=2]: the master played Nf5; under μ our Nc3 plays the role of that game’s Nd4; transferred move c6b4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3047` (ply 87) [structural distance $d=19$, QAP obj=1]: the master played Bb2; under μ our Nc3 plays the role of that game’s Bc6 (a **cross-type** correspondence); cross-type roles: our Bc4 as their Ne8, our Bd6 as their Ph7; transferred move c6b4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nb4 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



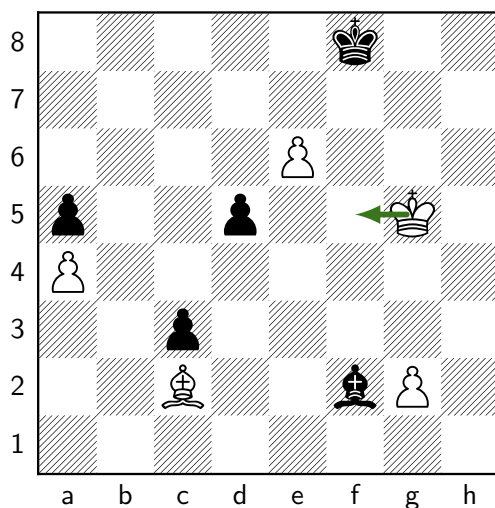
(a)



(b)

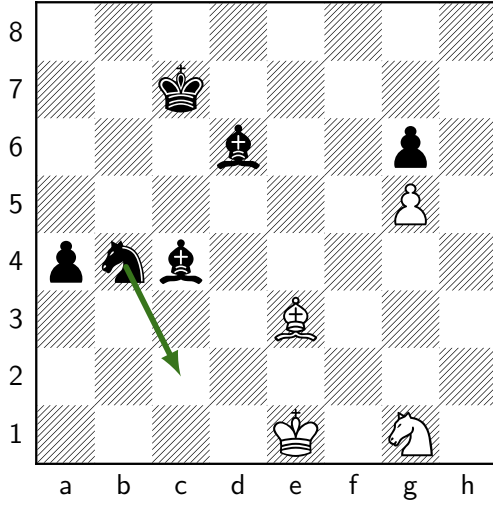


(c)

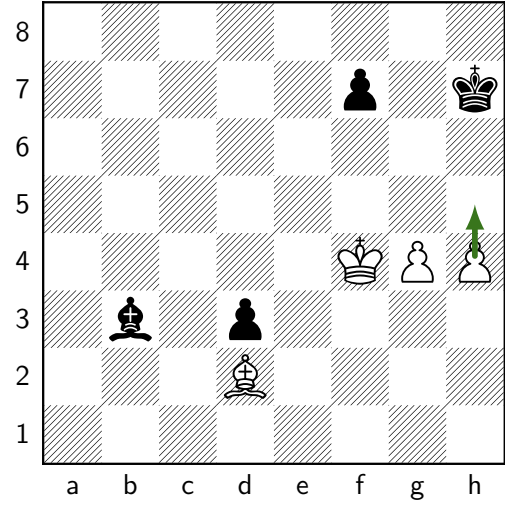


(d)

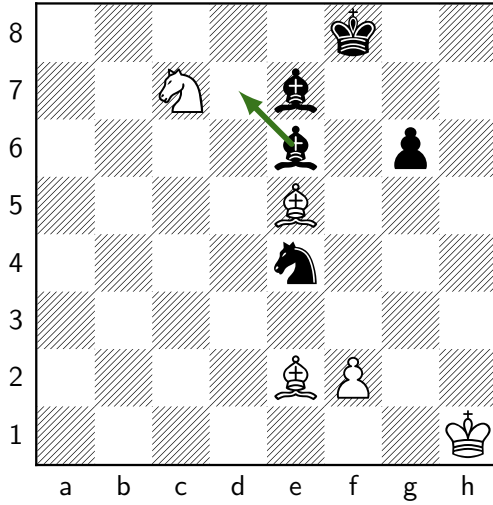
Figure 36: **Move 42, Black: Kc7.** (a) Our position; Melanie plays Kc7 (c8c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#962` (ply 99) [structural distance $d=19$, QAP obj=2]: the master played Kf6; under μ our *Kc1* plays the role of that game’s *Kg2*; cross-type roles: our *Nb4* as their *Pg5*; transferred move c8c7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#871` (ply 94) [structural distance $d=19$, QAP obj=1]: the master played Kf3; under μ our *Kc1* plays the role of that game’s *Kg2*; cross-type roles: our *Bd6* as their *Pa2*, our *pg5* as their *Be2*; transferred move c8c7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#572` (ply 96) [structural distance $d=20$, QAP obj=1]: the master played Kf5; under μ our *Kc1* plays the role of that game’s *Kg5*; cross-type roles: our *pg5* as their *bf2*, our *Nb4* as their *Pa4*; transferred move c8c7 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kc7 (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



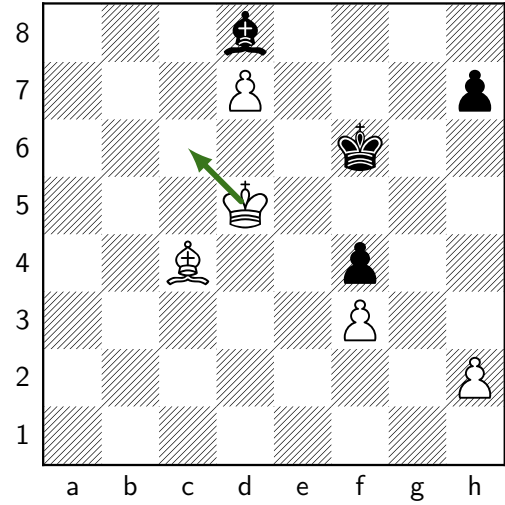
(a)



(b)

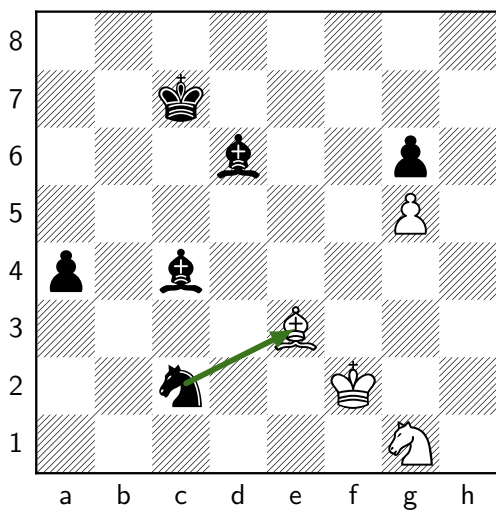


(c)

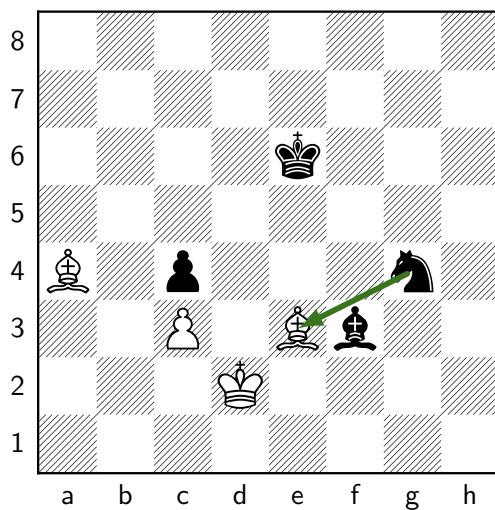


(d)

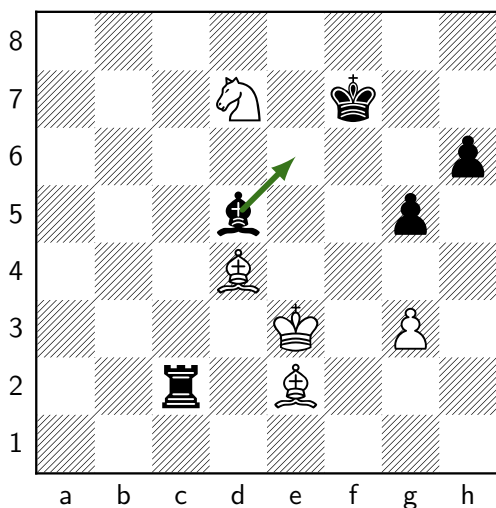
Figure 37: **Move 43, Black: Nc2+.** (a) Our position; Melanie plays Nc2+ (b4c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3706` (ply 94) [structural distance $d=22$, QAP obj=1]: the master played h5; under μ our Nb5 plays the role of that game’s Ph4 (a **cross-type** correspondence); cross-type roles: our Pg6 as their bb3, our be3 as their pd3; transferred move b4c2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2925` (ply 89) [structural distance $d=20$, QAP obj=4]: the master played Bd7; under μ our Bc5 plays the role of that game’s Be3; cross-type roles: our Pg6 as their Ne4, our Nb4 as their be2; transferred move c4g8. (d) Retrieved from `ccrl_top_p0002#2795` (ply 78) [structural distance $d=20$, QAP obj=3]: the master played Kc6; under μ our Kc2 plays the role of that game’s Kd5; cross-type roles: our be3 as their Pf3, our ng1 as their Pd7; transferred move c7c6. Synthesis: 1 retrieved analog transferred to Nc2+ (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



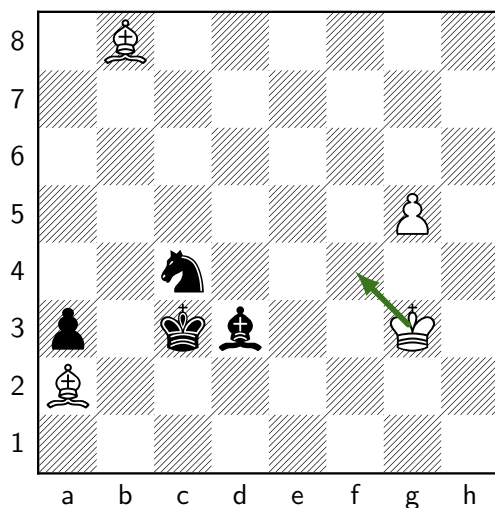
(a)



(b)

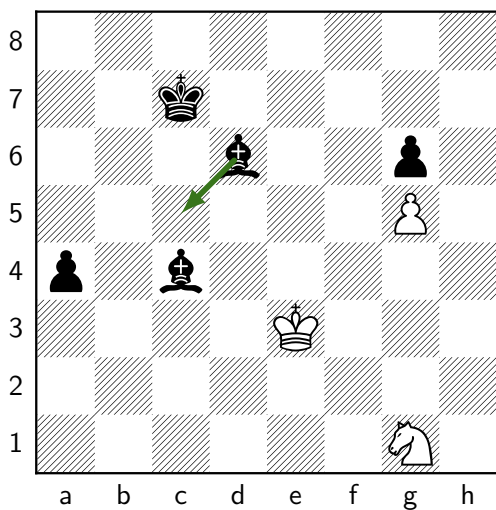


(c)

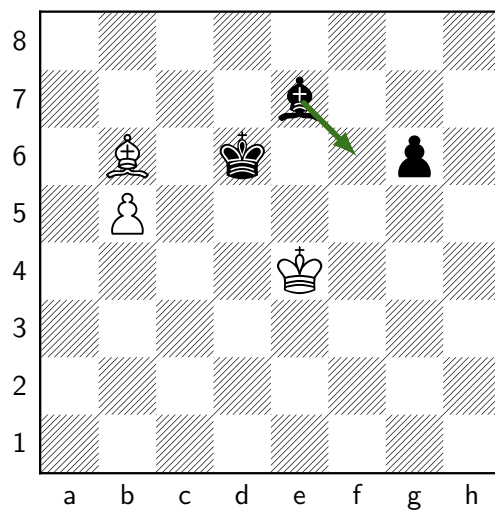


(d)

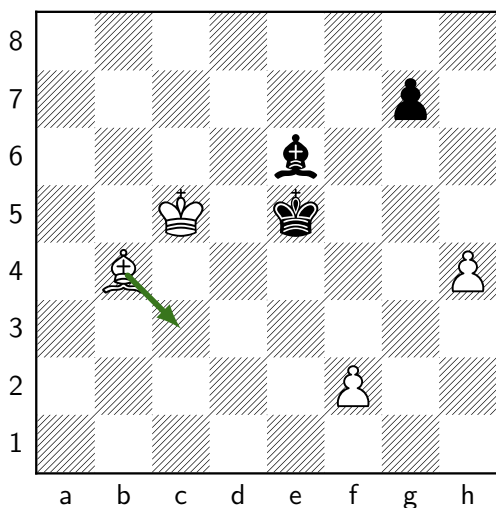
Figure 38: **Move 44, Black: Nxe3.** (a) Our position; Melanie plays Nxe3 (c2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1657` (ply 91) [structural distance $d=21$, QAP obj=25]: the master played Nxe3; under μ our Nc7 plays the role of that game’s Ng5; cross-type roles: our ng1 as their ba4; transferred move c2e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1664` (ply 91) [structural distance $d=27$, QAP obj=21]: the master played Be6; under μ our Bd3 plays the role of that game’s Bd4; cross-type roles: our Pa4 as their nd7, our Bc4 as their Pg5; transferred move d6e5. (d) Retrieved from `ccr1_top_p0002#376` (ply 94) [structural distance $d=24$, QAP obj=21]: the master played Kf4; under μ our Kc2 plays the role of that game’s Kg3; cross-type roles: our Nc2 as their Ba2, our Bd6 as their pa3; transferred move c7c6. Synthesis: 1 retrieved analog transferred to Nxe3 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



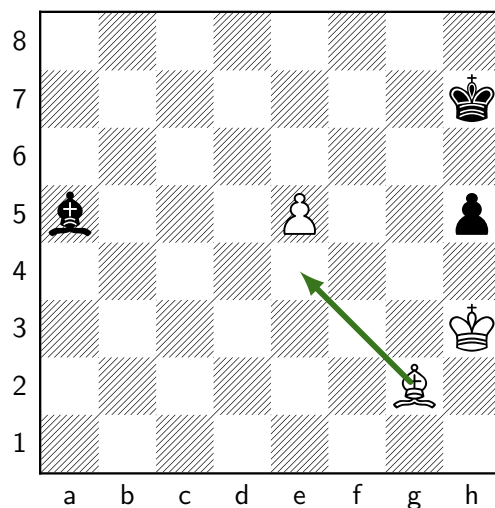
(a)



(b)

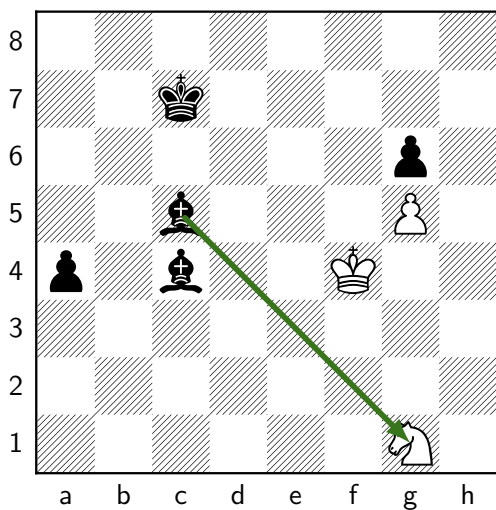


(c)

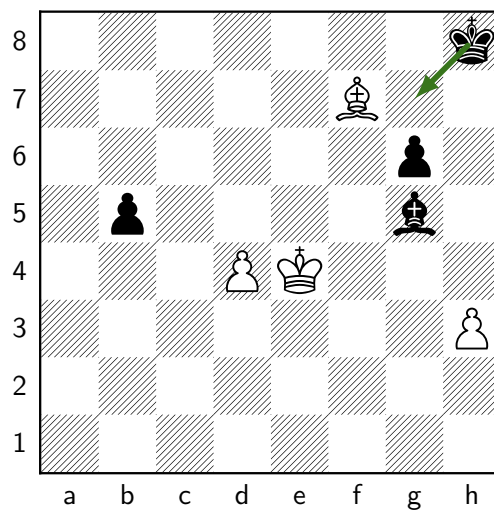


(d)

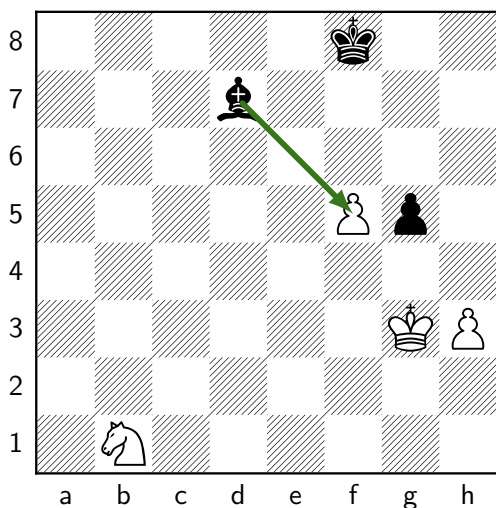
Figure 39: **Move 45, Black: Bc5+.** (a) Our position; Melanie plays Bc5+ (d6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1647` (ply 89) [structural distance $d=13$, QAP obj=3]: the master played Bf6; under μ our Bd3 plays the role of that game’s Be2; cross-type roles: our pg5 as their bb6, our Bc4 as their Pg6; transferred move d6c5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2648` (ply 88) [structural distance $d=10$, QAP obj=2]: the master played Bc3+; under μ our Bd3 plays the role of that game’s Bb4; transferred move d6c5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3496` (ply 102) [structural distance $d=11$, QAP obj=2]: the master played Be4+; under μ our Bd3 plays the role of that game’s Bg2; cross-type roles: our Pg6 as their ba5; transferred move d6c5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bc5+ (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



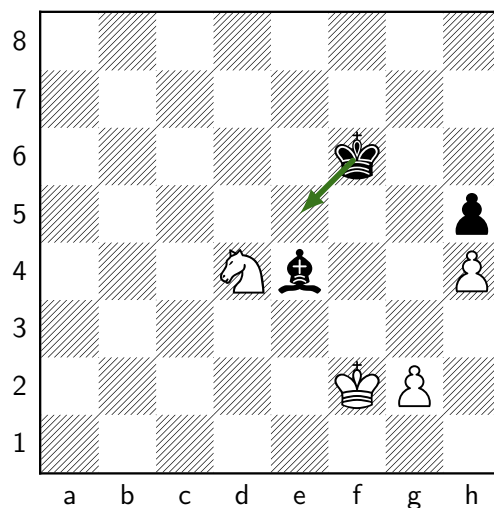
(a)



(b)

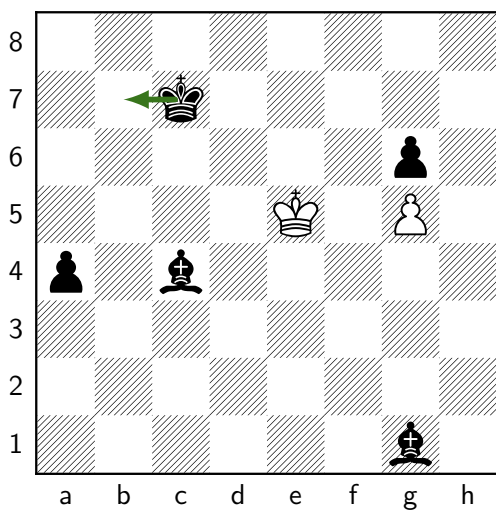


(c)

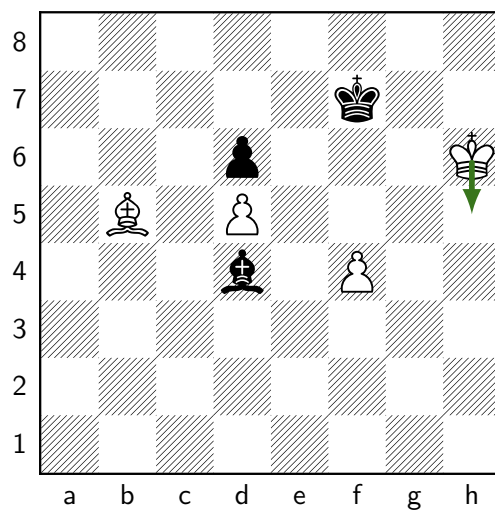


(d)

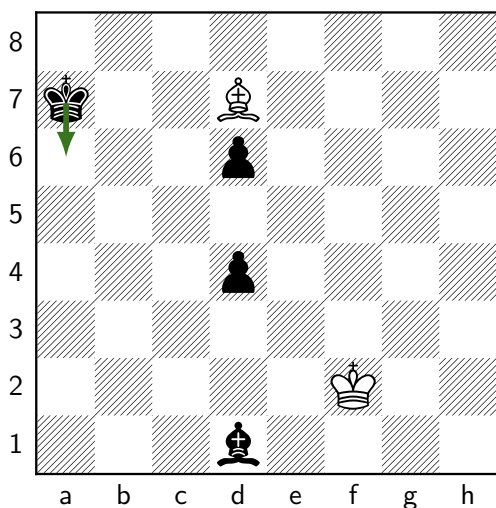
Figure 40: **Move 46, Black: Bxg1.** (a) Our position; Melanie plays Bxg1 (c5g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2079` (ply 89) [structural distance $d=11$, QAP obj=15]: the master played Kg7; under μ our Kc2 plays the role of that game’s Kh1; cross-type roles: our ng1 as their Pg6; transferred move c7d7. (c) Retrieved from `ccr1_top_p0002#928` (ply 91) [structural distance $d=7$, QAP obj=15]: the master played Bxf5; under μ our Bc4 plays the role of that game’s Bd2; cross-type roles: our ng1 as their pf5, our Pa4 as their nb1; transferred move c5d4. (d) Retrieved from `ccr1_top_p0003#1758` (ply 107) [structural distance $d=11$, QAP obj=13]: the master played Ke5; under μ our Kc2 plays the role of that game’s Kf3; cross-type roles: our Pa4 as their nd4, our ng1 as their pg2; transferred move c7c6. Synthesis: no retrieved analog transferred to Bxg1 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



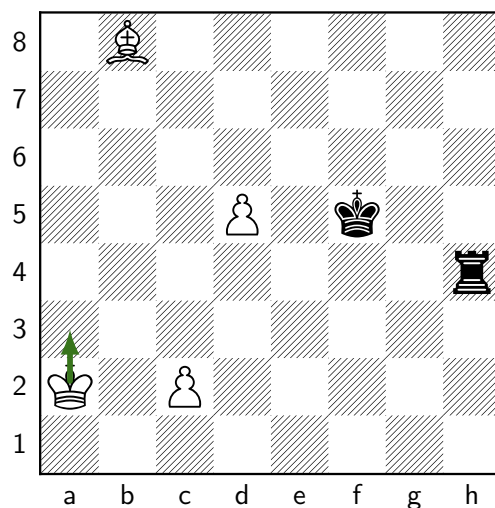
(a)



(b)

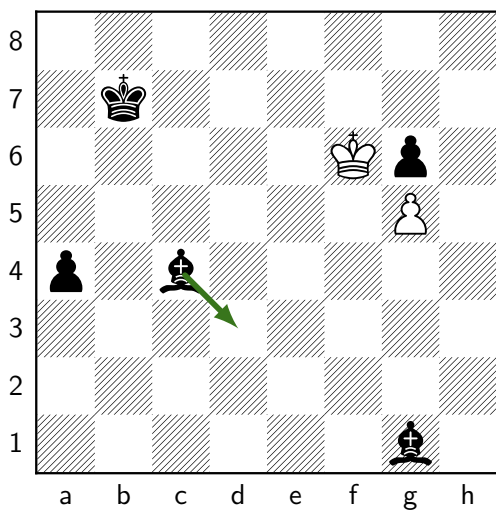


(c)

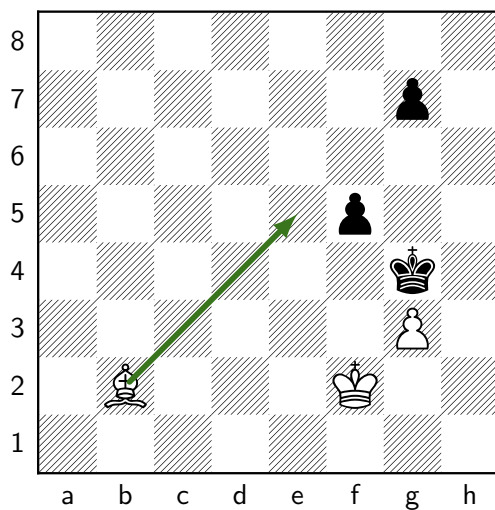


(d)

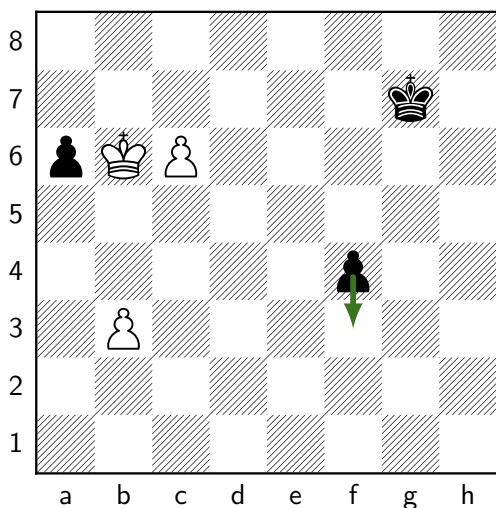
Figure 41: **Move 47, Black: Kb7.** (a) Our position; Melanie plays Kb7 (**c7b7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#337` (ply 108) [structural distance $d=4$, QAP obj=2]: the master played Kh5; under μ our *Kc2* plays the role of that game’s *Kh6*; cross-type roles: our *Pa4* as their *bd4*, our *Bc4* as their *Pf4*; transferred move **c7b7 (endorses this move)**. (c) Retrieved from `ccrl_top_p0003#2609` (ply 101) [structural distance $d=5$, QAP obj=2]: the master played Ka6; under μ our *Kc2* plays the role of that game’s *Ka2*; cross-type roles: our *pg5* as their *bd7*, our *Bc4* as their *Pd4*; transferred move **c7b7 (endorses this move)**. (d) Retrieved from `ccrl_top_p0000#3159` (ply 92) [structural distance $d=5$, QAP obj=2]: the master played Ka3; under μ our *Kc2* plays the role of that game’s *Ka2*; cross-type roles: our *pg5* as their *rh4*, our *Bc4* as their *Pd5*; transferred move **c7b7 (endorses this move)**. Synthesis: 12 retrieved analogs transferred to Kb7 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



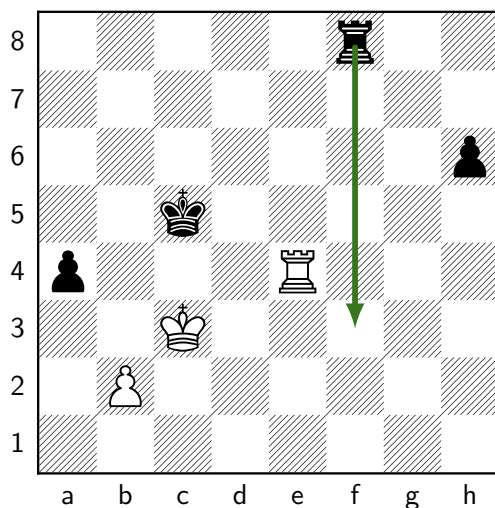
(a)



(b)

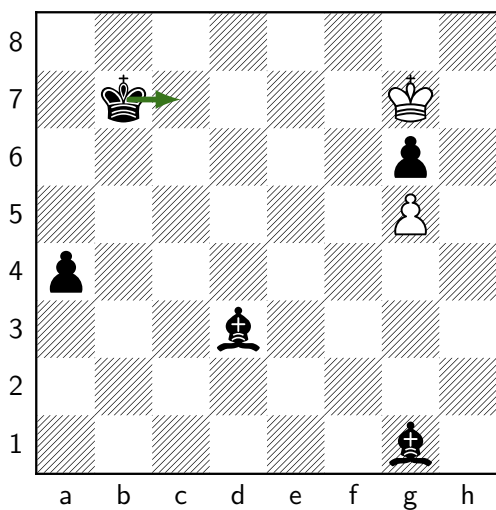


(c)

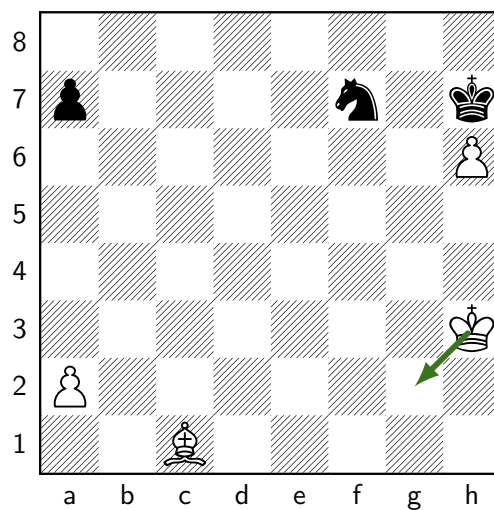


(d)

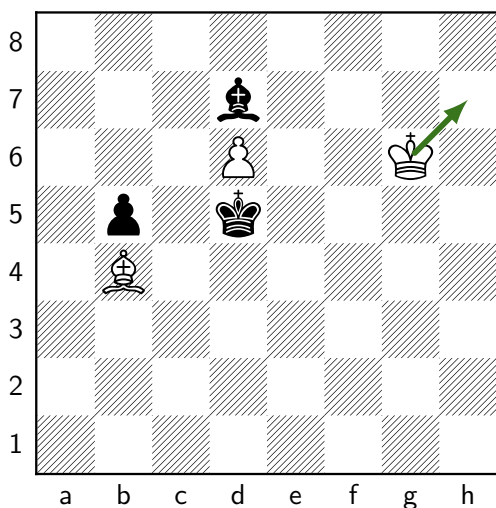
Figure 42: **Move 48, Black: Bd3.** (a) Our position; Melanie plays Bd3 (c4d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3572` (ply 106) [structural distance $d=10$, QAP obj=22]: the master played Be5; under μ our Bc5 plays the role of that game’s Bb2; transferred move c4d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1582` (ply 85) [structural distance $d=9$, QAP obj=20]: the master played f3; under μ our Bc5 plays the role of that game’s Pf5 (a **cross-type** correspondence); transferred move c4d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1913` (ply 97) [structural distance $d=10$, QAP obj=3]: the master played Rf3+; under μ our Bc5 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our kf6 as their re4; transferred move c4d3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bd3 (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



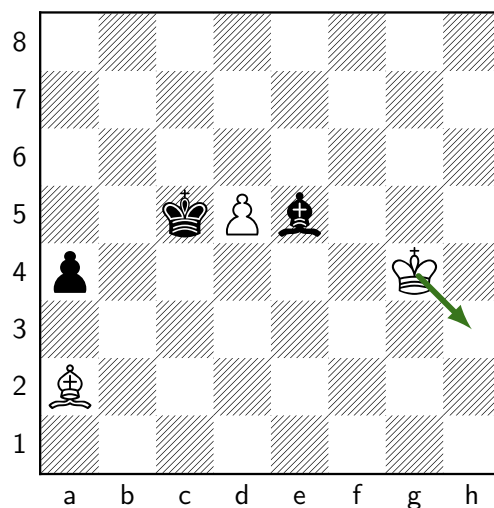
(a)



(b)

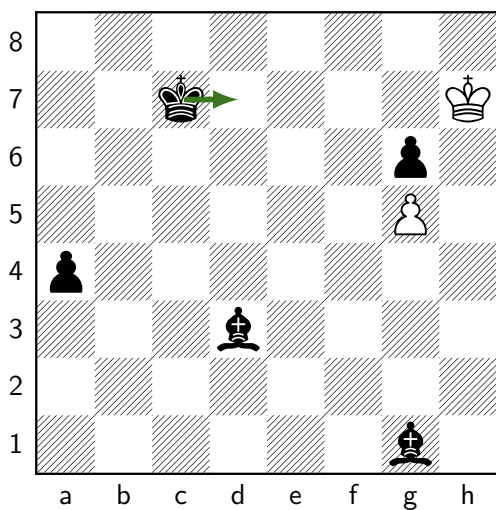


(c)

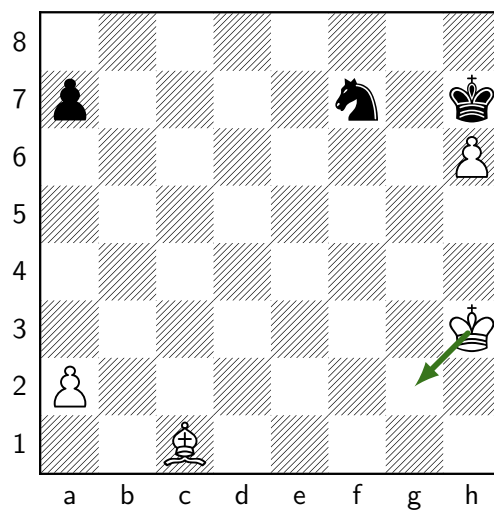


(d)

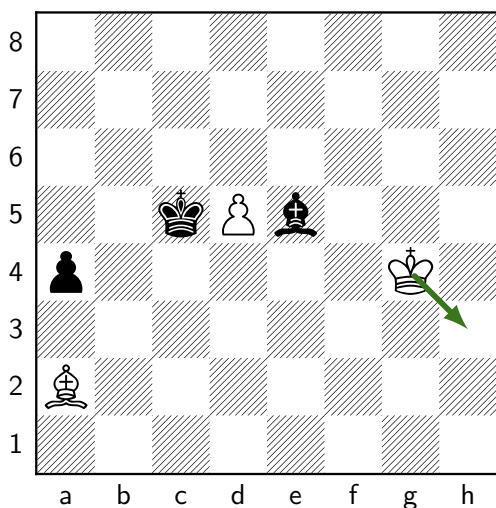
Figure 43: **Move 49, Black: Kc7.** (a) Our position; Melanie plays Kc7 (b7c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3222` (ply 86) [structural distance $d=8$, QAP obj=25]: the master played Kg2; under μ our Kb2 plays the role of that game’s Kh3; cross-type roles: our kg7 as their nf7; transferred move b7c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3134` (ply 102) [structural distance $d=9$, QAP obj=22]: the master played Kh7; under μ our Kb2 plays the role of that game’s Kg6; cross-type roles: our pg5 as their bd7; transferred move b7c7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2669` (ply 108) [structural distance $d=6$, QAP obj=22]: the master played Kh3; under μ our Kb2 plays the role of that game’s Kg4; cross-type roles: our Pa4 as their be5; transferred move b7c7 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kc7 (summed QAP objective 122, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



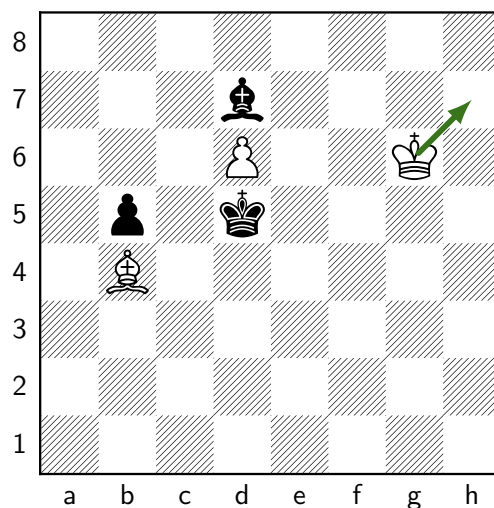
(a)



(b)

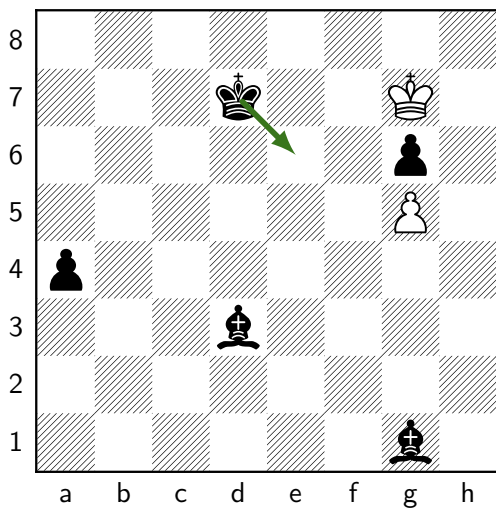


(c)

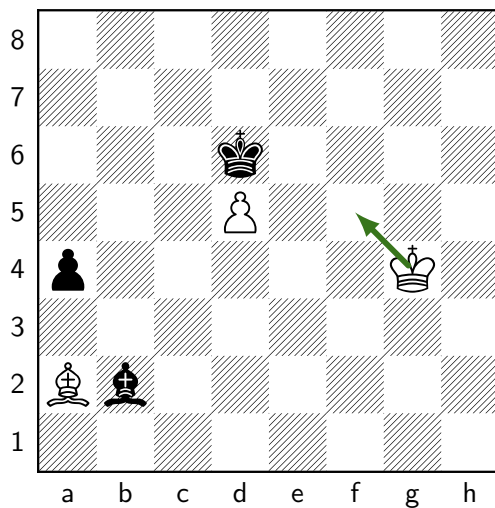


(d)

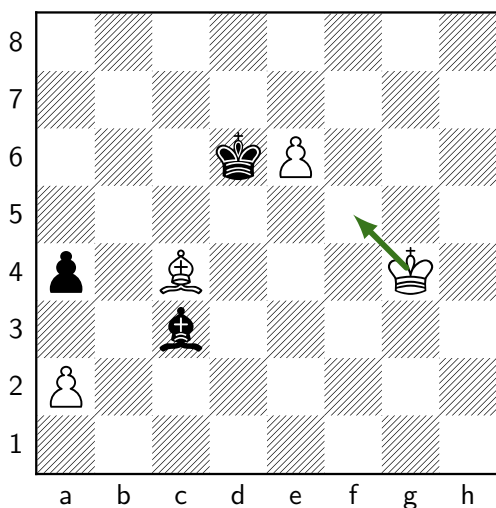
Figure 44: **Move 50, Black: Kd7.** (a) Our position; Melanie plays Kd7 (c7d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3222` (ply 86) [structural distance $d=7$, QAP obj=25]: the master played Kg2; under μ our Kc2 plays the role of that game’s Kh3; cross-type roles: our kh7 as their nf7; transferred move c7d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2669` (ply 108) [structural distance $d=6$, QAP obj=22]: the master played Kh3; under μ our Kc2 plays the role of that game’s Kg4; cross-type roles: our Pa4 as their be5; transferred move c7d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3134` (ply 102) [structural distance $d=9$, QAP obj=22]: the master played Kh7; under μ our Kc2 plays the role of that game’s Kg6; cross-type roles: our pg5 as their bd7; transferred move c7d7 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kd7 (summed QAP objective 142, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



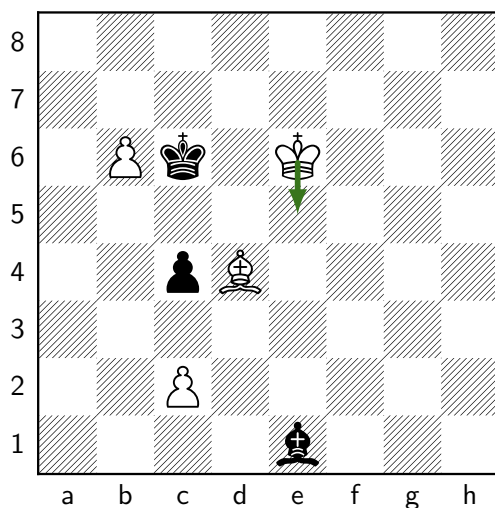
(a)



(b)

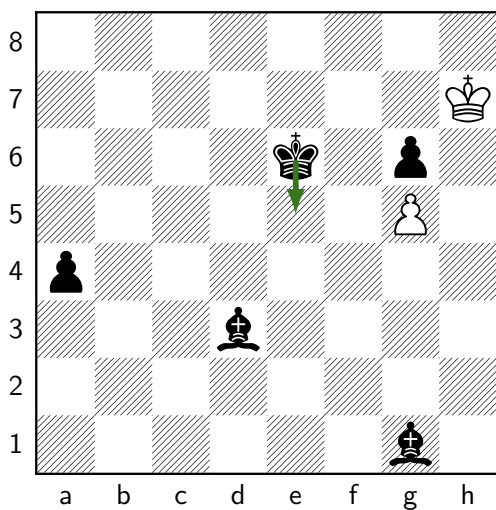


(c)

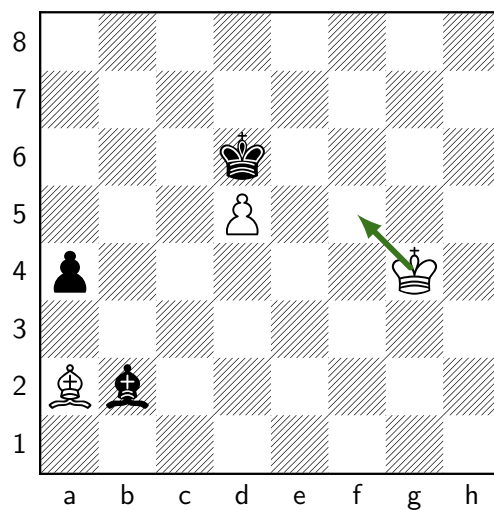


(d)

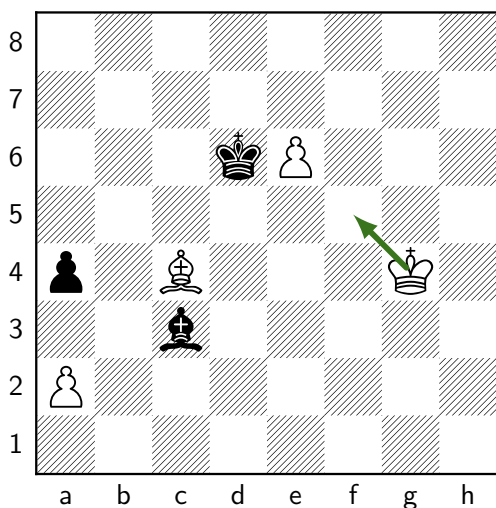
Figure 45: **Move 51, Black: Ke6**. (a) Our position; Melanie plays Ke6 (d7e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2669` (ply 104) [structural distance $d=6$, QAP obj=22]: the master played Kf5; under μ our *Kd2* plays the role of that game’s *Kg4*; cross-type roles: our *Pa4* as their *bb2*; transferred move d7e6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#336` (ply 116) [structural distance $d=8$, QAP obj=20]: the master played Kf5; under μ our *Kd2* plays the role of that game’s *Kg4*; cross-type roles: our *Bg1* as their *Pa2*, our *Pa4* as their *bc3*; transferred move d7e6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3621` (ply 100) [structural distance $d=6$, QAP obj=20]: the master played Ke5; under μ our *Kd2* plays the role of that game’s *Ke6*; cross-type roles: our *Pa4* as their *be1*, our *Bg1* as their *Pc2*; transferred move d7e6 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Ke6 (summed QAP objective 109, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



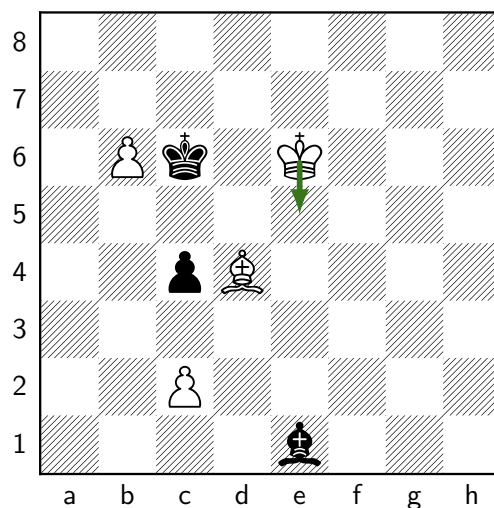
(a)



(b)

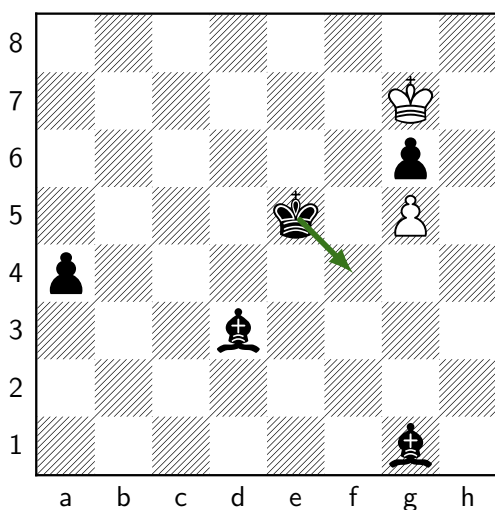


(c)

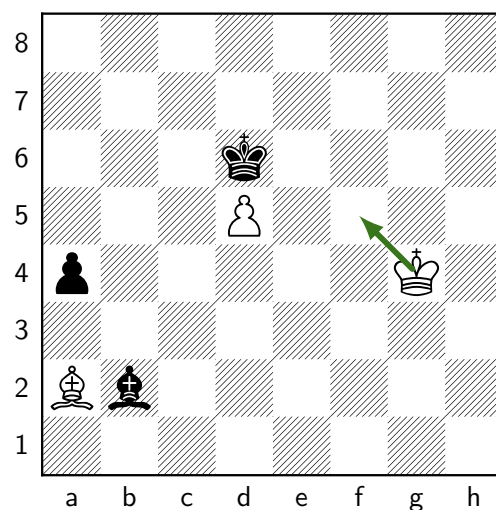


(d)

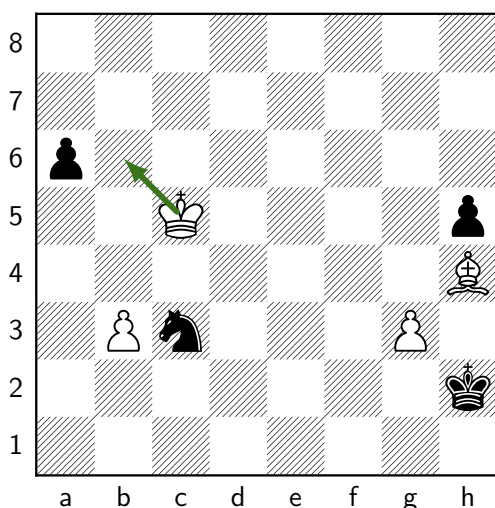
Figure 46: **Move 52, Black: Ke5 (e6e5).** (a) Our position; Melanie plays Ke5 (e6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2669` (ply 104) [structural distance $d=6$, QAP obj=22]: the master played Kf5; under μ our *Ke3* plays the role of that game’s *Kg4*; cross-type roles: our *Pa4* as their *bb2*; transferred move **e6e5 (endorses this move)**. (c) Retrieved from `ccr1_top_p0003#336` (ply 116) [structural distance $d=7$, QAP obj=21]: the master played Kf5; under μ our *Ke3* plays the role of that game’s *Kg4*; cross-type roles: our *Bg1* as their *Pa2*, our *Pa4* as their *bc3*; transferred move **e6e5 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#3621` (ply 100) [structural distance $d=6$, QAP obj=21]: the master played Ke5; under μ our *Ke3* plays the role of that game’s *Ke6*; cross-type roles: our *Pa4* as their *be1*, our *Bg1* as their *Pc2*; transferred move **e6e5 (endorses this move)**. Synthesis: 9 retrieved analogs transferred to Ke5 (summed QAP objective 99, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



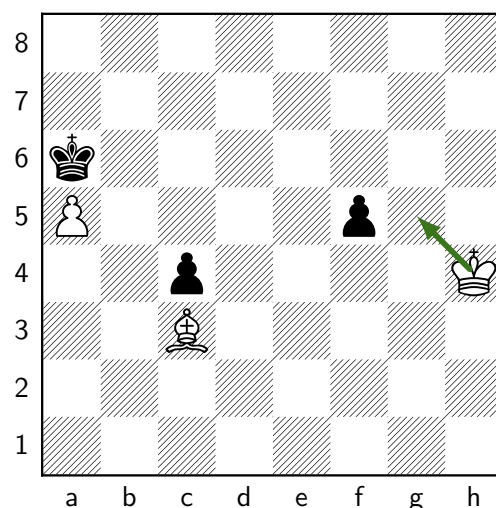
(a)



(b)

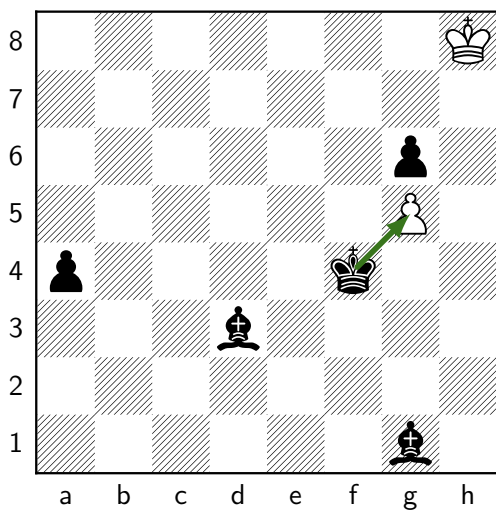


(c)

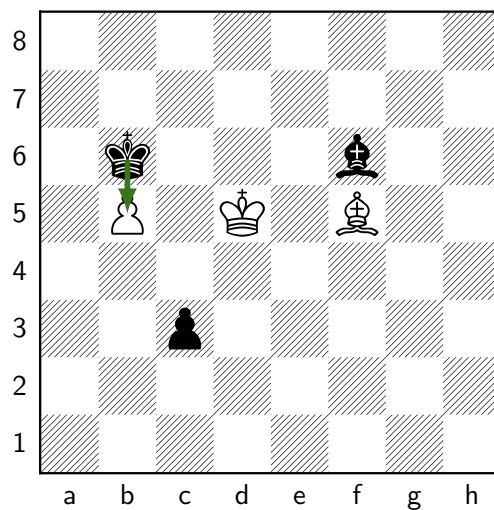


(d)

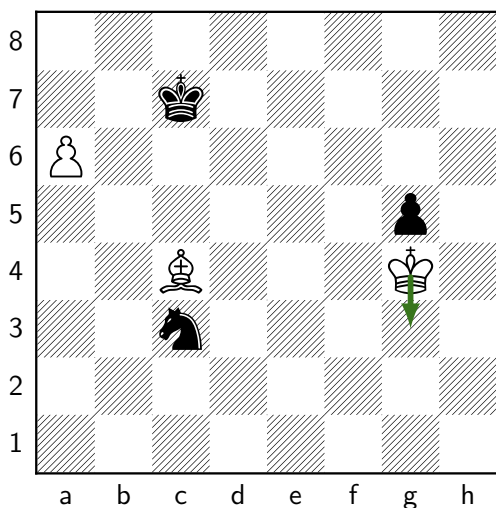
Figure 47: **Move 53, Black: Kf4**. (a) Our position; Melanie plays Kf4 (e5f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2669` (ply 104) [structural distance $d=6$, QAP obj=22]: the master played Kf5; under μ our Ke_4 plays the role of that game's Kg_4 ; cross-type roles: our Pa_4 as their bb_2 ; transferred move e5f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#954` (ply 102) [structural distance $d=8$, QAP obj=21]: the master played Kb6; under μ our Ke_4 plays the role of that game's Kc_5 ; cross-type roles: our Bg_1 as their nc_3 ; transferred move e5f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1285` (ply 120) [structural distance $d=5$, QAP obj=21]: the master played Kg5; under μ our Ke_4 plays the role of that game's Kh_4 ; transferred move e5f4 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kf4 (summed QAP objective 120, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



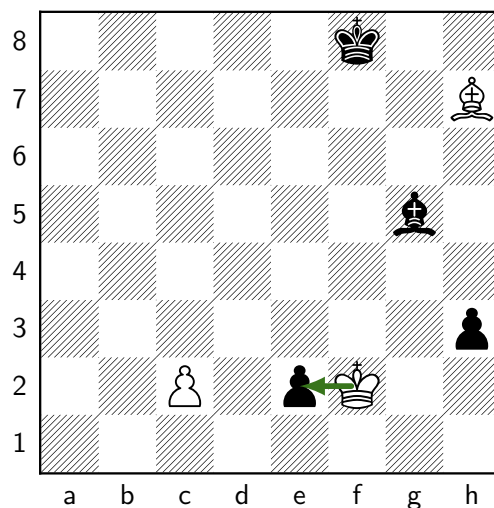
(a)



(b)

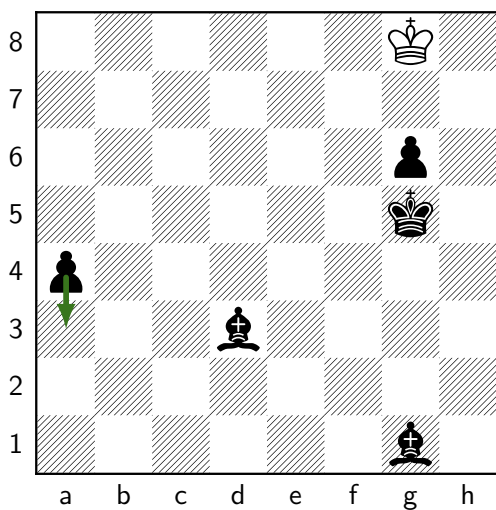


(c)

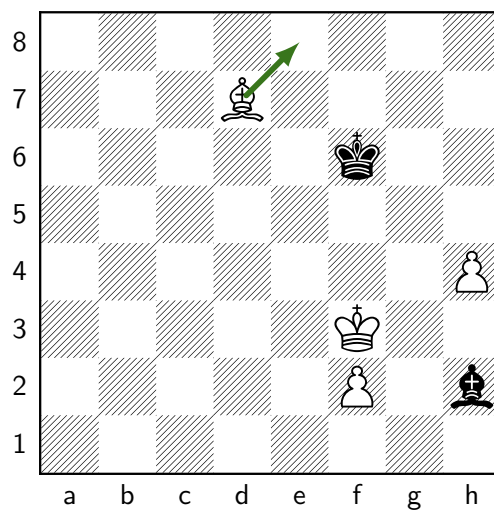


(d)

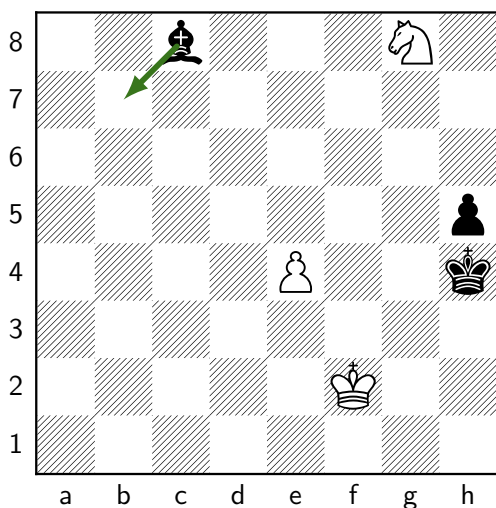
Figure 48: **Move 54, Black: Kxg5.** (a) Our position; Melanie plays Kxg5 (f4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2615` (ply 115) [structural distance $d=7$, QAP obj=20]: the master played Kxb5; under μ our Kf5 plays the role of that game’s Kb3; cross-type roles: our Pa4 as their bf5; transferred move f4g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3094` (ply 96) [structural distance $d=6$, QAP obj=19]: the master played Kg3; under μ our Kf5 plays the role of that game’s Kg4; cross-type roles: our Pa4 as their nc3; transferred move f4g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3920` (ply 98) [structural distance $d=6$, QAP obj=19]: the master played Kxe2; under μ our Kf5 plays the role of that game’s Kf2; transferred move f4g5 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kxg5 (summed QAP objective 110, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



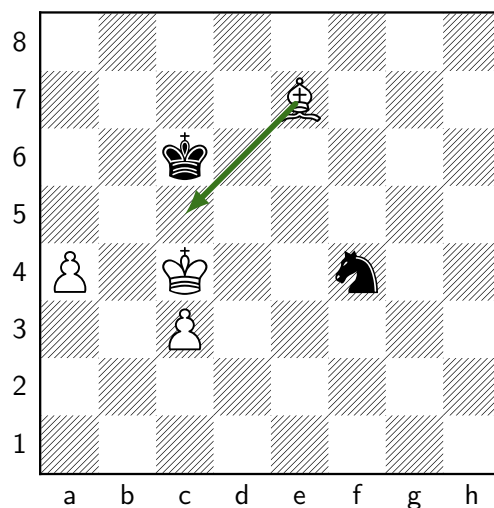
(a)



(b)



(c)



(d)

Figure 49: **Move 55, Black: a3**. (a) Our position; Melanie plays a3 (a4a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#109` (ply 100) [structural distance $d=8$, QAP obj=2]: the master played Be8; under μ our Pa5 plays the role of that game’s Bd7 (a **cross-type** correspondence); cross-type roles: our Bg1 as their Ph4; transferred move a4a3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2633` (ply 109) [structural distance $d=8$, QAP obj=2]: the master played Bb7; under μ our Pa5 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Bd3 as their ng8, our Bg1 as their pe4; transferred move a4a3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1766` (ply 122) [structural distance $d=9$, QAP obj=2]: the master played Bc5; under μ our Pa5 plays the role of that game’s Be7 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Pa4, our Bg1 as their nf4; transferred move a4a3 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to a3 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

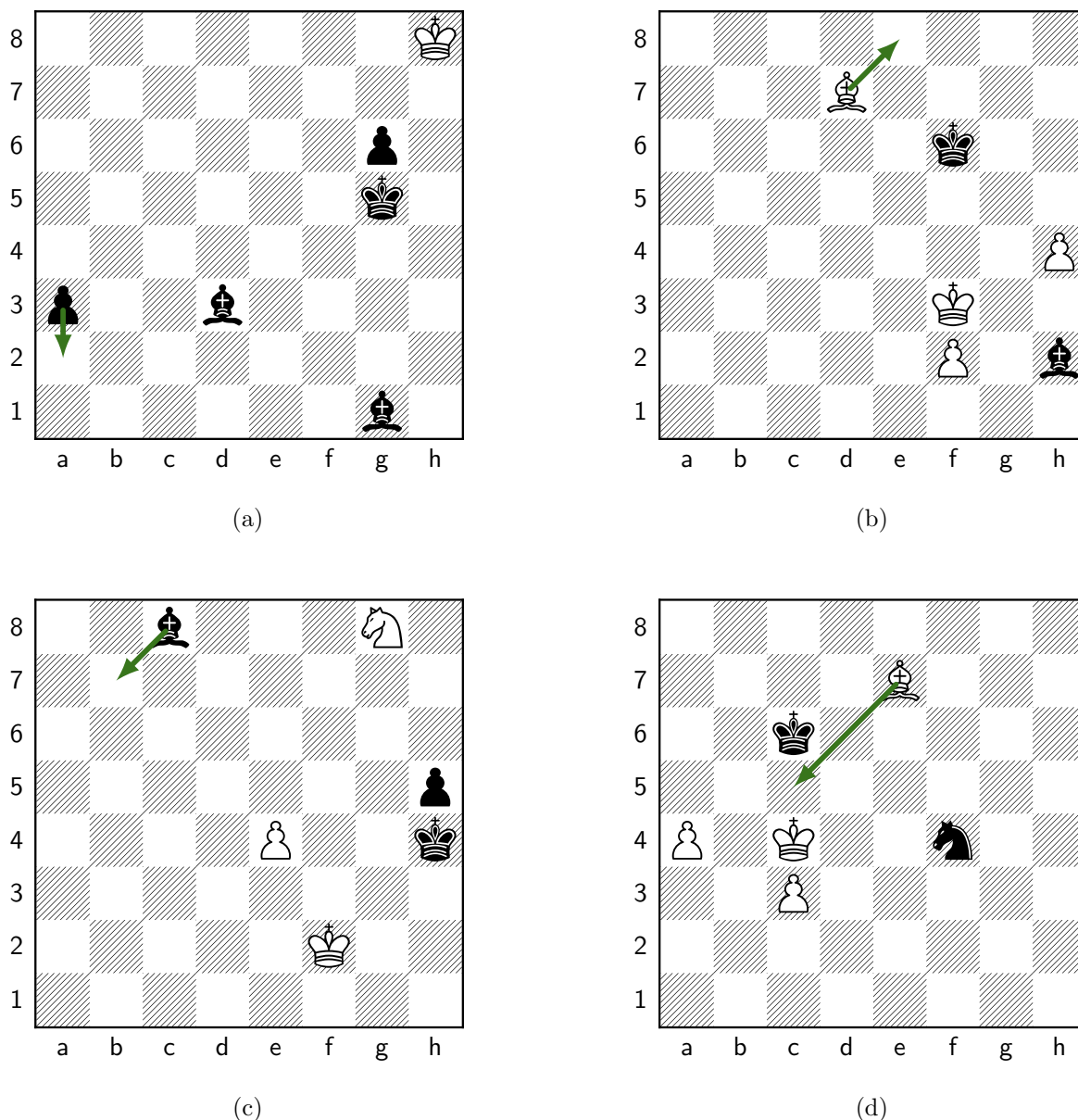
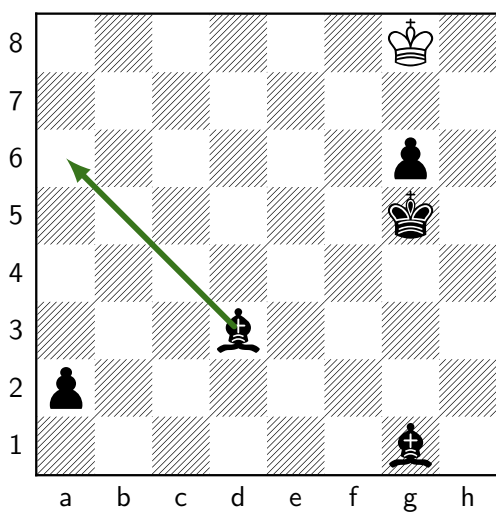
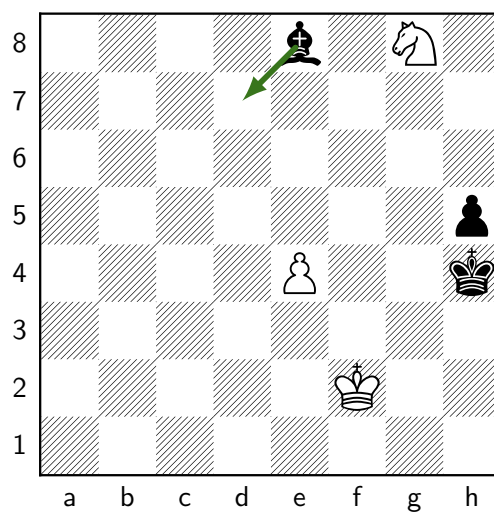


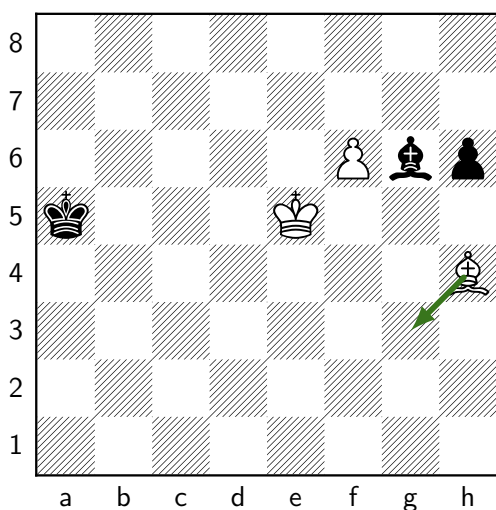
Figure 50: **Move 56, Black: a2**. (a) Our position; Melanie plays a2 (a3a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#109` (ply 100) [structural distance $d=8$, QAP obj=2]: the master played Bc8; under μ our Pa6 plays the role of that game’s Bc7 (a **cross-type** correspondence); cross-type roles: our Bg1 as their Ph4; transferred move a3a2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2633` (ply 109) [structural distance $d=8$, QAP obj=2]: the master played Bb7; under μ our Pa6 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Bd3 as their ng8, our Bg1 as their pe4; transferred move a3a2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1766` (ply 122) [structural distance $d=9$, QAP obj=2]: the master played Bc5; under μ our Pa6 plays the role of that game’s Be7 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Pa4, our Bg1 as their nf4; transferred move a3a2 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to a2 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



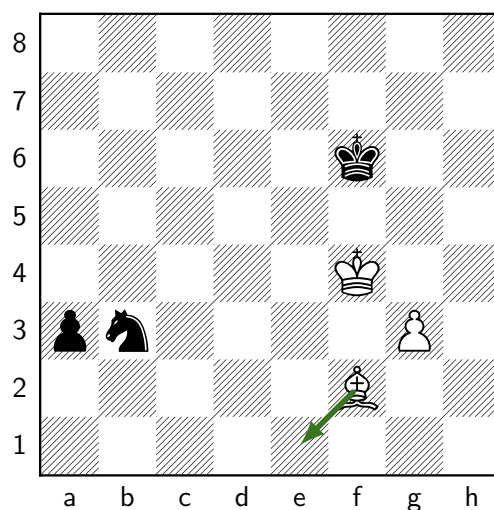
(a)



(b)

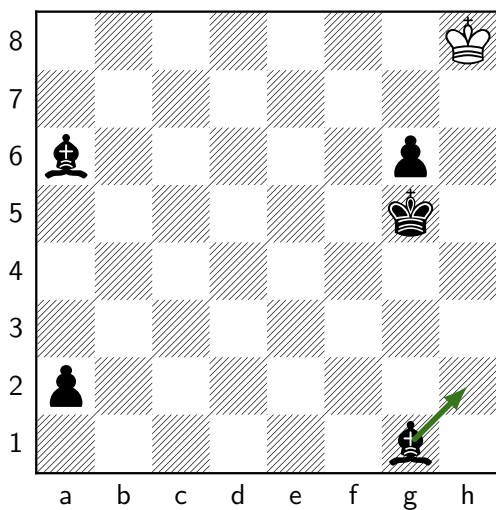


(c)

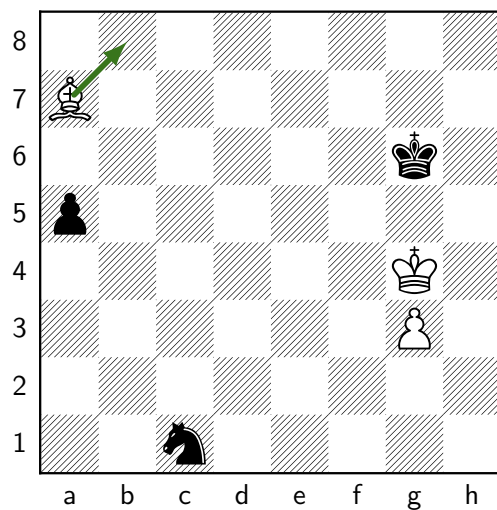


(d)

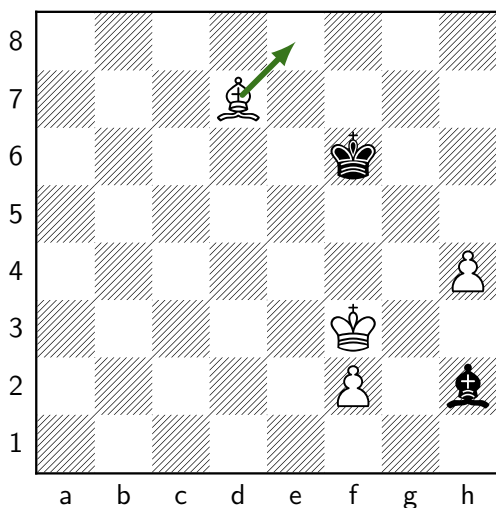
Figure 51: **Move 57, Black: Ba6.** (a) Our position; Melanie plays Ba6 (d3a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2633` (ply 117) [structural distance $d=4$, QAP obj=3]: the master played Bd7; under μ our Bd6 plays the role of that game’s Be1; cross-type roles: our Pa2 as their ng8, our Bg1 as their pe4; transferred move d3a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3972` (ply 112) [structural distance $d=7$, QAP obj=1]: the master played Bg3; under μ our Bd6 plays the role of that game’s Bh4; transferred move d3a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2369` (ply 110) [structural distance $d=9$, QAP obj=1]: the master played Be1; under μ our Bd6 plays the role of that game’s Bf2; cross-type roles: our Bg1 as their nb3; transferred move d3a6 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ba6 (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



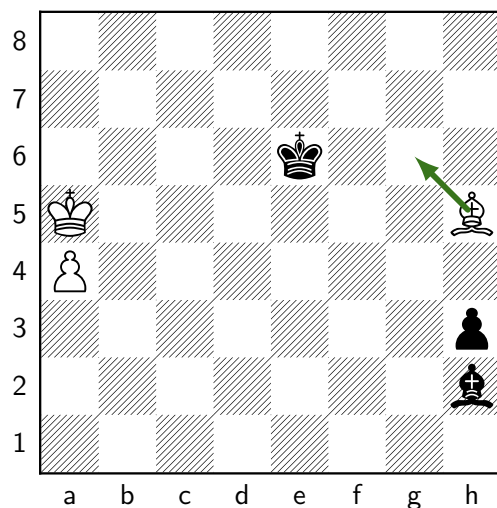
(a)



(b)

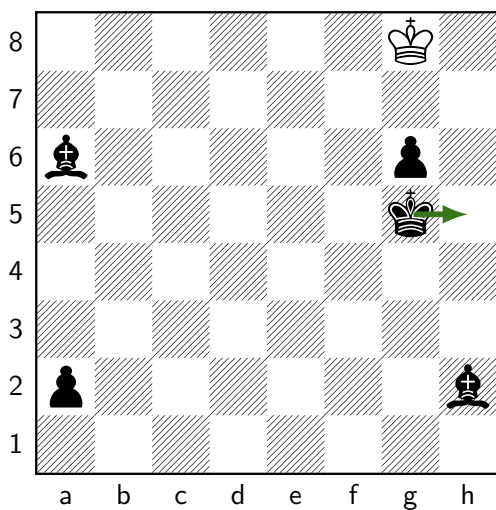


(c)

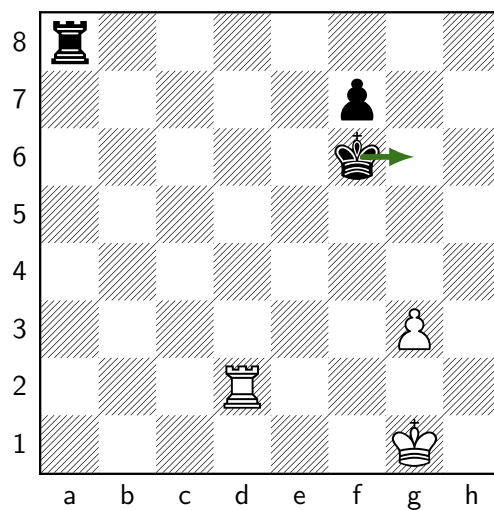


(d)

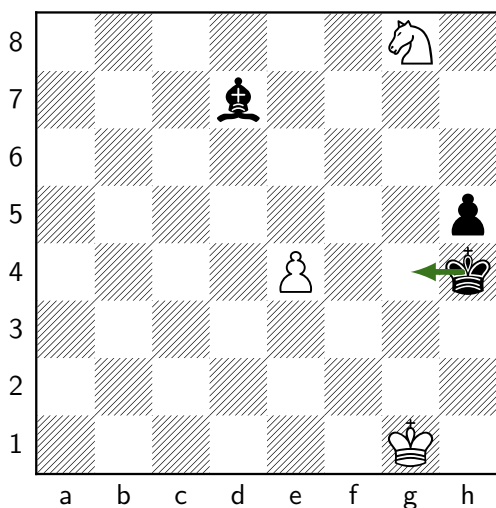
Figure 52: **Move 58, Black: Bh2**. (a) Our position; Melanie plays Bh2 (g1h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2369` (ply 102) [structural distance $d=5$, QAP obj=2]: the master played Bb8; under μ our Bg8 plays the role of that game’s Ba7; cross-type roles: our Ba6 as their nc1; transferred move g1h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#109` (ply 100) [structural distance $d=4$, QAP obj=2]: the master played Be8; under μ our Bg8 plays the role of that game’s Bd7; transferred move g1h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2626` (ply 106) [structural distance $d=7$, QAP obj=2]: the master played Bg6; under μ our Bg8 plays the role of that game’s Bh5; transferred move g1h2 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Bh2 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



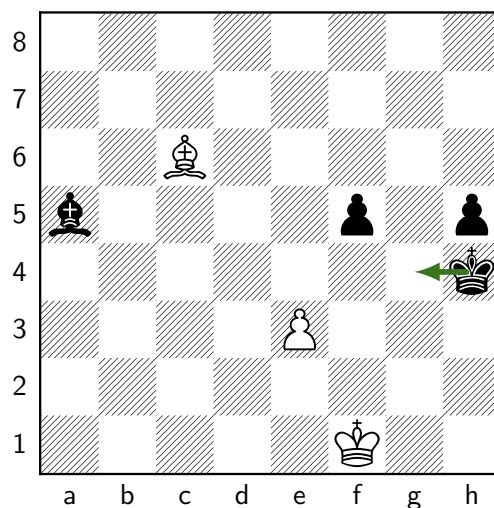
(a)



(b)

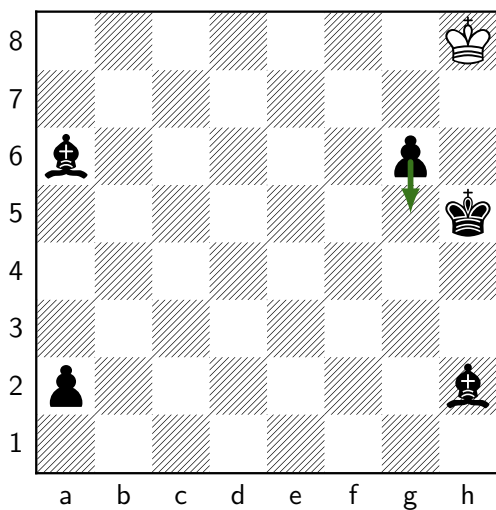


(c)

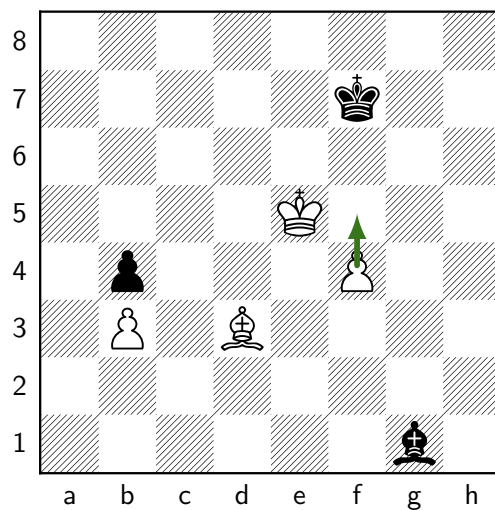


(d)

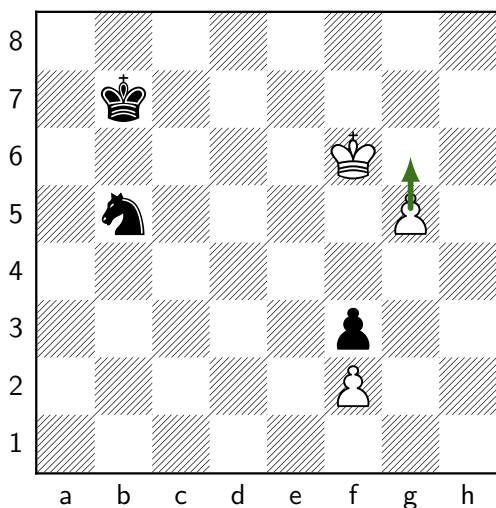
Figure 53: **Move 59, Black: Kh5.** (a) Our position; Melanie plays Kh5 (g5h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2403` (ply 123) [structural distance $d=8$, QAP obj=2]: the master played Kg6; under μ our Kg4 plays the role of that game’s Kf3; cross-type roles: our Ba6 as their Ra8, our Bh2 as their pg3; transferred move g5h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2633` (ply 119) [structural distance $d=4$, QAP obj=2]: the master played Kg4; under μ our Kg4 plays the role of that game’s Kh5; cross-type roles: our Ba6 as their ng8, our Pa2 as their Bd7; transferred move g5h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#311` (ply 105) [structural distance $d=5$, QAP obj=2]: the master played Kg4; under μ our Kg4 plays the role of that game’s Kh5; cross-type roles: our Pa2 as their Ba5, our Bh2 as their Pf5; transferred move g5h5 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kh5 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



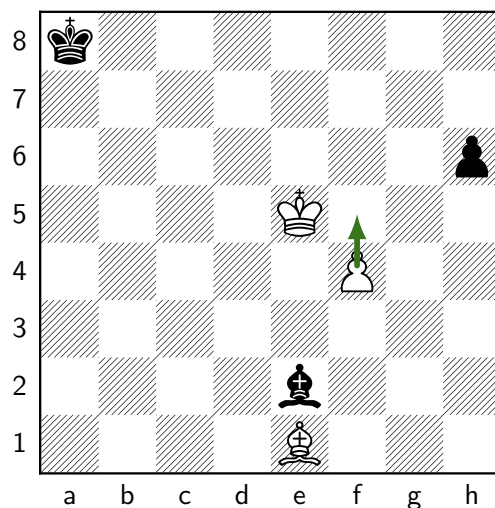
(a)



(b)

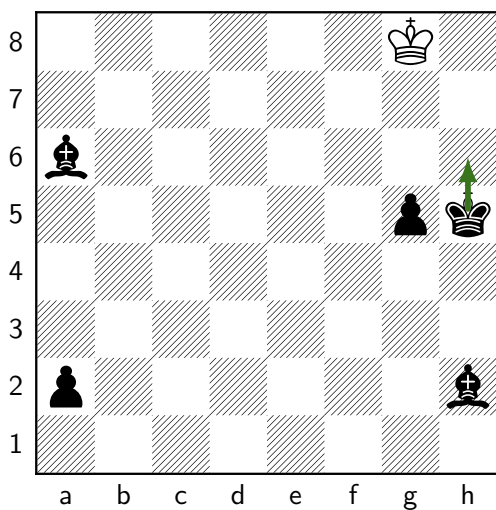


(c)

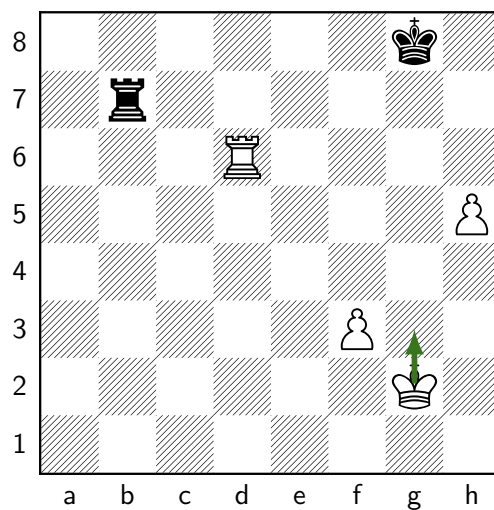


(d)

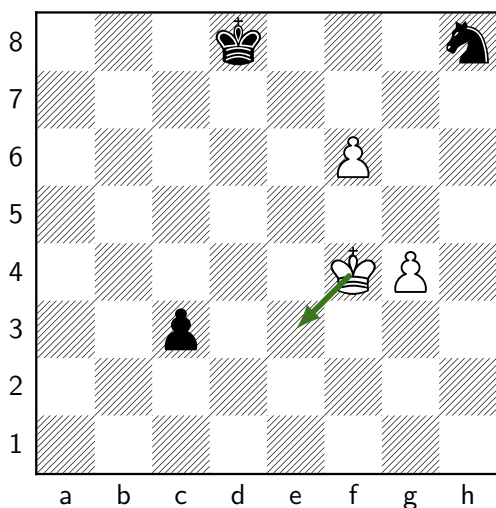
Figure 54: **Move 60, Black: g5.** (a) Our position; Melanie plays g5 (g6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2168` (ply 114) [structural distance $d=5$, QAP obj=2]: the master played f5; under μ our $Pg3$ plays the role of that game’s $Pf4$; transferred move g6g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3908` (ply 120) [structural distance $d=6$, QAP obj=2]: the master played g6; under μ our $Pg3$ plays the role of that game’s $Pg5$; cross-type roles: our $Ba6$ as their $Pf2$, our $Bh2$ as their $nb5$; transferred move g6g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3972` (ply 102) [structural distance $d=5$, QAP obj=2]: the master played f5; under μ our $Pg3$ plays the role of that game’s $Pf4$; cross-type roles: our $Pa2$ as their $be2$, our $Bh2$ as their $ph6$; transferred move g6g5 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to g5 (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



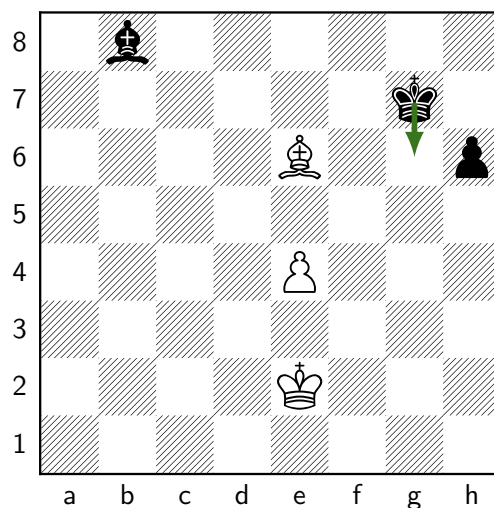
(a)



(b)

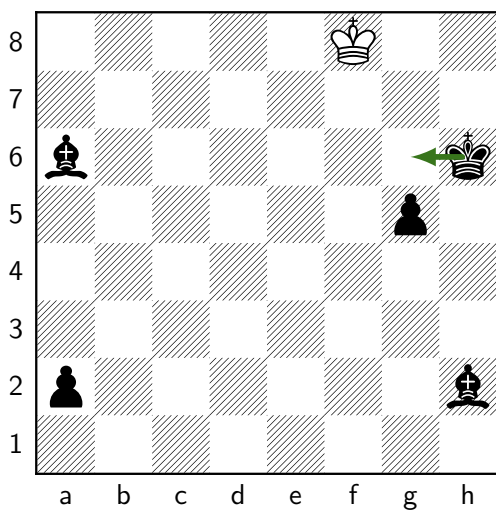


(c)

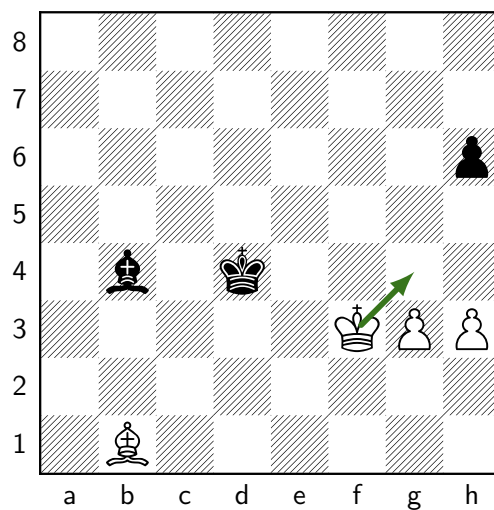


(d)

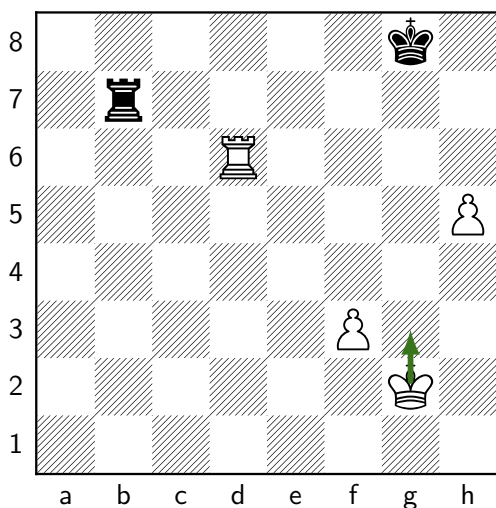
Figure 55: **Move 61, Black: Kh6**. (a) Our position; Melanie plays Kh6 (**h5h6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#103` (ply 112) [structural distance $d=6$, QAP obj=2]: the master played Kg3; under μ our Kh4 plays the role of that game's Kg2; cross-type roles: our Ba6 as their Ph5, our Bh2 as their Rd6; transferred move h5h6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2971` (ply 116) [structural distance $d=7$, QAP obj=2]: the master played Ke3; under μ our Kh4 plays the role of that game's Kf4; cross-type roles: our Ba6 as their pc3, our Bh2 as their nh8; transferred move h5h6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2764` (ply 117) [structural distance $d=4$, QAP obj=2]: the master played Kg6; under μ our Kh4 plays the role of that game's Kg2; cross-type roles: our Pa2 as their be6, our Bh2 as their pe4; transferred move h5h6 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Kh6 (summed QAP objective 24, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



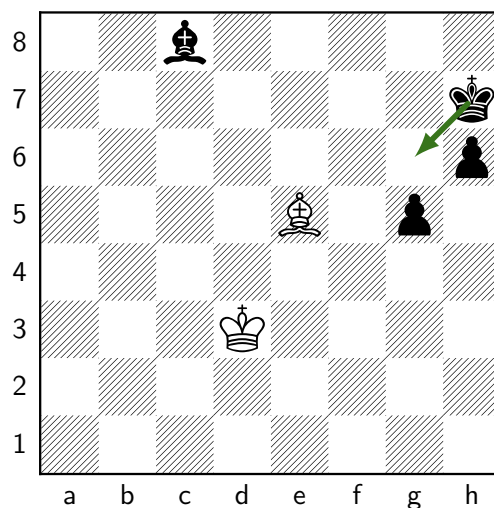
(a)



(b)

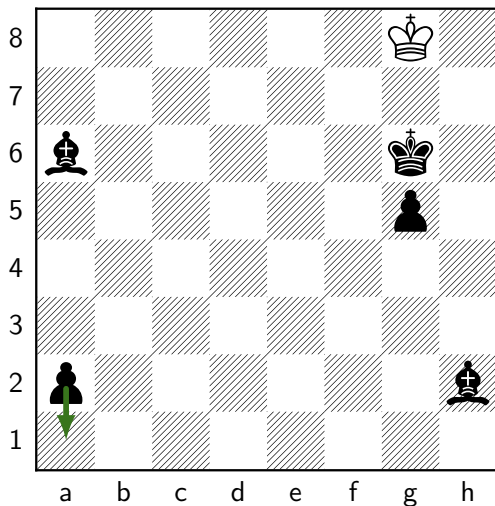


(c)

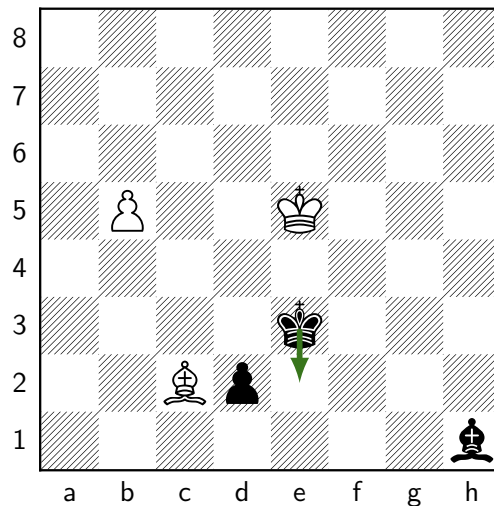


(d)

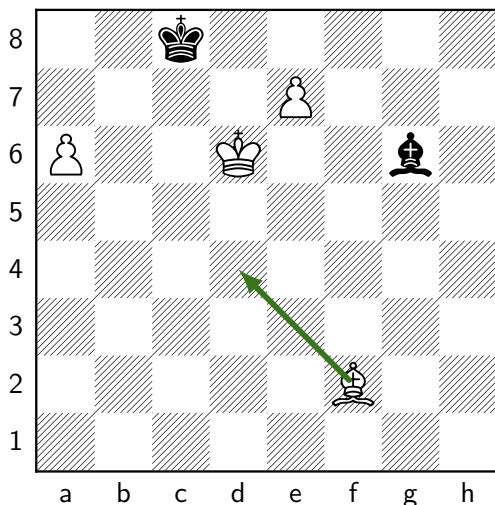
Figure 56: **Move 62, Black: Kg6**. (a) Our position; Melanie plays Kg6 (**h6g6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1894` (ply 112) [structural distance $d=4$, QAP obj=3]: the master played Kg4; under μ our *Kh3* plays the role of that game’s *Kf3*; cross-type roles: our *Bh2* as their *ph6*; transferred move **h6g6 (endorses this move)**. (c) Retrieved from `ccr1_top_p0003#103` (ply 112) [structural distance $d=6$, QAP obj=2]: the master played Kg3; under μ our *Kh3* plays the role of that game’s *Kg2*; cross-type roles: our *Ba6* as their *Ph5*, our *Pa2* as their *Rd6*; transferred move **h6g6 (endorses this move)**. (d) Retrieved from `ccr1_top_p0002#328` (ply 121) [structural distance $d=6$, QAP obj=2]: the master played Kg6; under μ our *Kh3* plays the role of that game’s *Kh2*; cross-type roles: our *Pa2* as their *be5*, our *Bh2* as their *Pg5*; transferred move **h6g6 (endorses this move)**. Synthesis: 12 retrieved analogs transferred to Kg6 (summed QAP objective 24, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



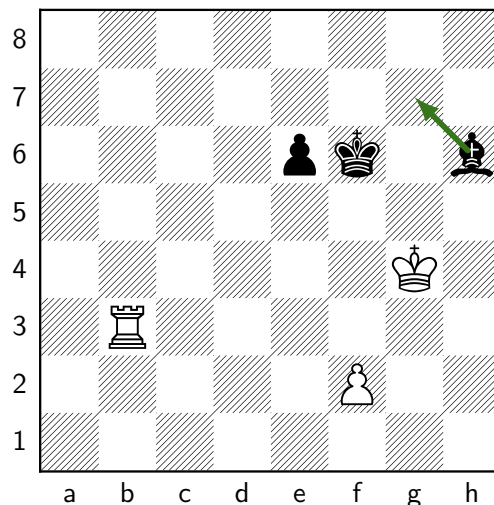
(a)



(b)

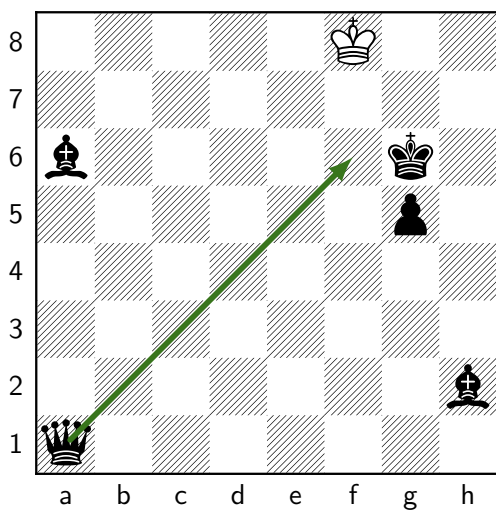


(c)

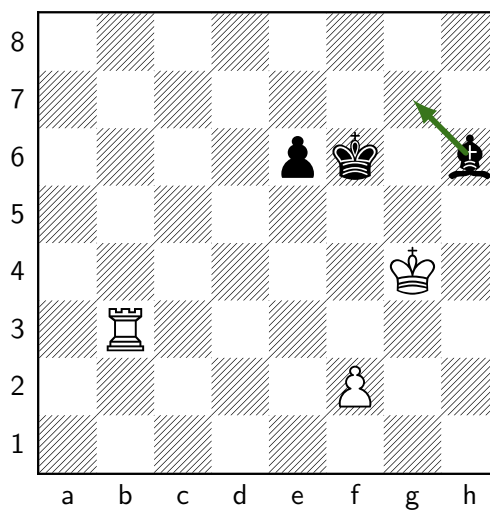


(d)

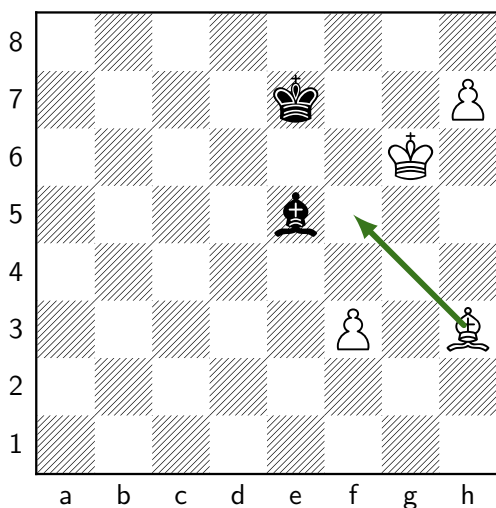
Figure 57: **Move 63, Black: a1=Q.** (a) Our position; Melanie plays **a1=Q** (**a2a1q**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#401` (ply 115) [structural distance $d=6$, QAP obj=2]: the master played **Ke2**; under μ our *Kg3* plays the role of that game’s *Ke6*; cross-type roles: our *Ba6* as their *pb5*, our *Pa2* as their *bc2*; transferred move **g6h5**. (c) Retrieved from `ccr1_top_p0001#376` (ply 128) [structural distance $d=4$, QAP obj=2]: the master played **Bd4**; under μ our *Ba3* plays the role of that game’s *Bf2*; transferred move **a6c4**. (d) Retrieved from `ccr1_top_p0003#994` (ply 115) [structural distance $d=5$, QAP obj=2]: the master played **Bg7**; under μ our *Ba3* plays the role of that game’s *Bh3*; cross-type roles: our *Bh2* as their *rb3*; transferred move **a6b7**. Synthesis: no retrieved analog transferred to **a1=Q** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



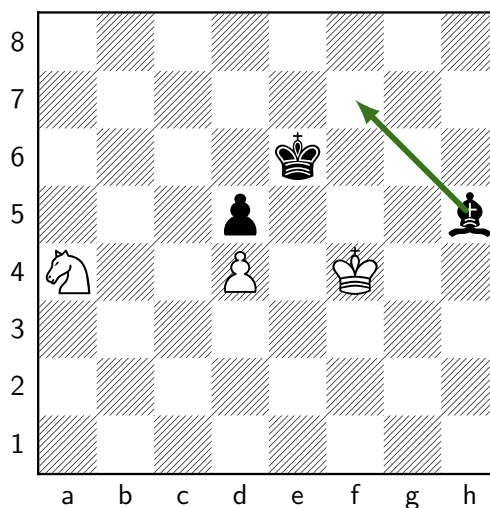
(a)



(b)

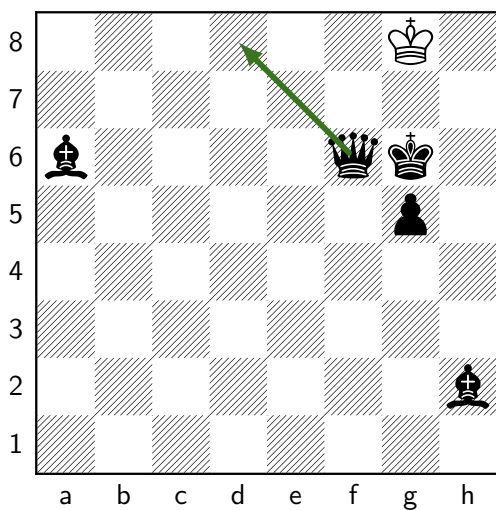


(c)

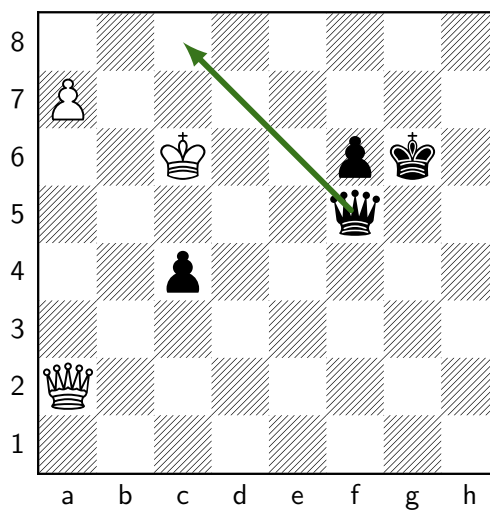


(d)

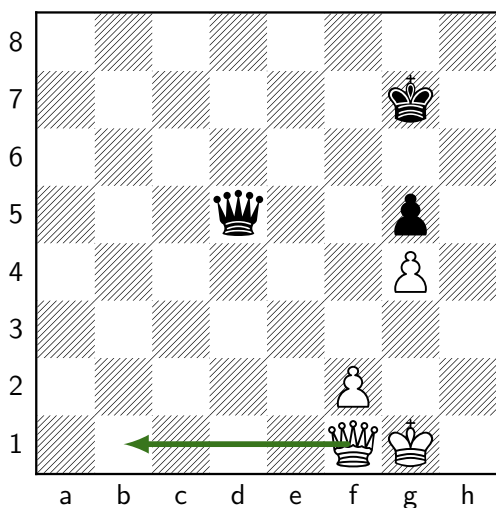
Figure 58: **Move 64, Black: Qf6+.** (a) Our position; Melanie plays Qf6+ (a1f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#994` (ply 115) [structural distance $d=9$, QAP obj=3]: the master played Bg7; under μ our *Qa8* plays the role of that game’s *Bh3* (a **cross-type** correspondence); cross-type roles: our *Ba6* as their *rb3*, our *Bh2* as their *pf2*; transferred move a1f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#109` (ply 132) [structural distance $d=10$, QAP obj=3]: the master played Bf5; under μ our *Qa8* plays the role of that game’s *Bh3* (a **cross-type** correspondence); cross-type roles: our *Ba6* as their *Pf3*; transferred move a1f6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2656` (ply 117) [structural distance $d=11$, QAP obj=2]: the master played Bf7; under μ our *Qa8* plays the role of that game’s *Bh4* (a **cross-type** correspondence); cross-type roles: our *Ba6* as their *na4*, our *Bh2* as their *pd4*; transferred move a1f6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qf6+ (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



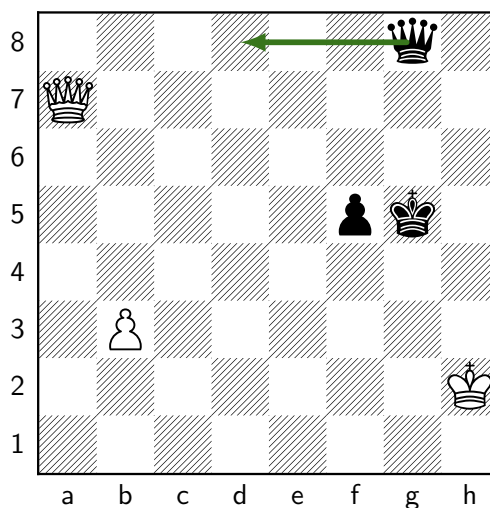
(a)



(b)



(c)



(d)

Figure 59: **Move 65, Black: Qd8# (f6d8)**. (a) Our position; Melanie plays Qd8# (f6d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#194` (ply 123) [structural distance $d=23$, QAP obj=2]: the master played Qc8+; under μ our Qf3 plays the role of that game’s Qf4; cross-type roles: our Ba6 as their pa7, our Bh2 as their qa2; transferred move f6d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2078` (ply 112) [structural distance $d=25$, QAP obj=2]: the master played Qb1; under μ our Qf3 plays the role of that game’s Qf1; cross-type roles: our Ba6 as their Pg4, our Bh2 as their qd5; transferred move f6d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1723` (ply 145) [structural distance $d=28$, QAP obj=1]: the master played Qd8; under μ our Qf3 plays the role of that game’s Qg1; cross-type roles: our Pg5 as their qa7, our Ba6 as their Pf5; transferred move f6d8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qd8# (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.